Conditional Latent-Prior Generative Compressed Sensing for Petrophysical Target-Log Reconstruction

Romulo Brito da Silva[1,3,*], Irineu de Azevedo Lima Neto[3]
Diego Passos[4], Herson Oliveira da Rocha[1,2,3]
Roseane Marchezi Missagia[1],
Marco Antonio Rodrigues de Ceia[1]

[1]LENEP, North Fluminense State University (UENF), Rodovia Amaral Peixoto, km 163, Imboassica, Macaé, RJ 27925-310, Brazil

[2]Polytechnic Institute (IPLOI), Campus Macaé, Federal University of Rio de Janeiro (UFRJ), Macaé, Brazil

[3]Scientific Computing Supporting Foundation (FACC), Petrópolis, Brazil

[4]Instituto Superior de Engenharia de Lisboa, Instituto Politécnico de Lisboa, Lisbon, Portugal

[*]Corresponding author at: LENEP/UENF, Rodovia Amaral Peixoto, km 163, Imboassica, Macaé, RJ 27925-310, Brazil. E-mail: romulobrito@lenep.uenf.br

Abstract

We reconstruct a continuous target-property profile when dense auxiliary wireline logs are available but the target is observed at only a few depths. Direct predictors can accept those observations as features without retaining them as explicit measurements during inference. Conditional Latent-Prior Compressed Sensing with Generative Models (CLP-CSGM) instead combines a learned target-window representation, a latent anchor $\boldsymbol{z}_0$ predicted from the dense logs, and test-time refinement against the sparse measurements. Its local recovery bound is conditional and does not establish separation of nearby windows by the trained decoder.

The cross-well evaluation uses three development wells and F03-4 as the held-out well. CLP-CSGM Ridge has lower mean root mean square error (RMSE) than a direct autoencoder in seven of nine configurations, especially at lower training support; it also improves on its conditional prior-only prediction in all nine. For the GR-only porosity case, ρ denotes the nominal fraction of window depths selected for observation. CLP-CSGM is lower at $\rho \in \{0.20, 0.30\}$, the

autoencoder is lower at $\rho \in \{0.40, 0.60\}$, and their errors are nearly equal at $\rho = 0.50$. At the reference noise standard deviation 0.01, depth interpolation has lower RMSE than CLP-CSGM in all nine cross-well configurations. The ordering becomes more favorable to CLP-CSGM under larger synthetic perturbations that were not calibrated to field uncertainties.

The F3 targets are processed curves: clay volume is Larionov-derived from gamma ray, porosity is density-derived, and sparse observations are subsamples rather than independent core or laboratory measurements. The results support conditional benefits of explicit latent refinement in the evaluated regimes, not general superiority, and motivate broader validation with independently calibrated targets.

Keywords: compressed sensing; generative model; inverse problem; well-log reconstruction; petrophysics; sparse assimilation

Computer Code Availability

CLP-CSGM (Conditional Latent-Prior Compressed Sensing with Generative Models) was developed by Romulo Brito da Silva, LENEP, North Fluminense State University, Rodovia Amaral Peixoto, km 163, Imboassica, Macae, RJ 27925-310, Brazil (contact: `romulobrito@lenep.uenf.br` ; telephone: +55 21 99101-3308). First publicly released in 2026, it is implemented in Python 3.10 using NumPy 1.26, pandas 2.3, Matplotlib 3.7, scikit-learn 1.6, PyTorch 2.6, and PyWavelets 1.4. The source repository occupies approximately 10 MB, excluding datasets and experiment outputs, and runs on a multi-core CPU with at least 8 GB of RAM; CUDA is optional.

Source code, installation instructions, dependencies, benchmark scripts, and documentation are available at `https://github.com/romulobrito/CLP--CSGM` under the MIT License. The repository covers the cross-well clay-volume and F03-4 porosity benchmarks, structural ablations, latent-prior sensitivity, and supplementary diagnostics. Processed well-log tables are not redistributed; the README specifies the required inputs and reproduction steps.

Data availability. The F3-derived benchmarks follow Zhang et al. [2025]; users must obtain the tabulated inputs from the cited releases or portals under the applicable licenses. The Lapa dataset follows Alvarez et al. [2025], whose underlying field data cannot be shared publicly. Supplementary analyses therefore require locally authorized inputs.

1 Introduction

Petrophysical target-log reconstruction builds a continuous property profile in a target well. Dense wireline logs run along the well, but they measure the target property only indirectly [Ellis and Singer, 2007, Asquith and Krygowski, 2004, Wang et al., 2024, Mondal and Singh, 2022, Sun et al., 2026]. A few depths already carry a value of that property. The task is to reconstruct the remaining depths while penalizing disagreement with those known values.

Supervised regressors learn a nonlinear map from logs to the target property. When the sparse observations are available, they are commonly joined to the dense logs and used as extra inputs [Bishop, 2006, Ahmadi and Chen, 2019, Wood, 2020, Pan et al., 2023, Mondal and Singh, 2022, Koray et al., 2024, Matinkia et al., 2023, Abdel Azim and Aljehani, 2022, Shi et al., 2024, He et al., 2025, Markus et al., 2026]. After training, that map is not required to reproduce the observations it was given. In this work those observations stay measurements: the reconstructed window is penalized for disagreeing with them, and that penalty does not force every observation to be met exactly.

Recent studies complete a missing log interval, translate a geomodel property map, or regress porosity and shale volume from the log suite [Koeshidayatullah et al., 2026, Al-Fakih et al., 2026, 2025, 2023]. Section 2 compares these approaches with the explicit measurement-fidelity formulation considered here.

An explicit measurement penalty still requires a useful representation of the complete target window. A fixed sparsity basis can be restrictive when petrophysical windows vary across wells [Donoho, 2006, Candes et al., 2006a,b, Bora et al., 2017]. Conditional Latent-Prior Compressed Sensing with Generative Models (CLP-CSGM) uses a learned decoder to define the candidate windows and a conditional map to predict the latent anchor $\boldsymbol{z}_0$ from the dense logs. Test-time optimization then updates the latent code using the sparse target measurements while penalizing departure from that anchor. Thus, contextual prediction and measurement assimilation remain distinct: the dense logs guide the search, whereas the sparse values directly assess candidate reconstructions. Ridge regression is the linear low-capacity reference for the conditional map, and an MLP variant is evaluated under the same channels and split.

The contributions are threefold. First, CLP-CSGM formulates target-log reconstruction as a

conditional latent inverse problem in which the measurement residual remains active at inference. Second, we use a classical maximum-a-posteriori (MAP)/Tikhonov interpretation to clarify the reconstruction objective and a conditional local error bound to identify the roles of decoder mismatch, latent-prior mismatch, measurement noise, and local measurement separability. Third, a controlled cross-well benchmark with three development wells and F03-4 as the single held-out well compares the formulation with direct predictors and structural ablations while varying training support, measurement availability, and latent-prior architecture.

CLP-CSGM Ridge has lower mean root mean square error (RMSE) than the direct autoencoder in seven of the nine cross-well cells, with the most consistent gains at steps 16 and 32, where the training pools are smaller. The full model also improves on the conditional prior-only reconstruction in all nine cells. At the reference noise level, depth interpolation remains lower in all nine configurations, showing that latent optimization is not always needed; larger synthetic perturbations yield a more favorable relative ordering for CLP-CSGM. These findings characterize the tested regimes rather than general cross-well or operational transferability. Likewise, the local recovery bound is conditional and does not certify the set-restricted eigenvalue condition (S-REC) for the trained decoder.

2 Related work

2.1 Petrophysical log prediction

Well logs provide dense but indirect measurements of rock and fluid properties, and data-driven methods are widely used to predict petrophysical targets [Ellis and Singer, 2007, Asquith and Krygowski, 2004, Ahmadi and Chen, 2019, Wood, 2020]. Recent studies address three neighboring tasks.

Log completion reconstructs missing intervals of a wireline curve from available observations and contextual information, including neighboring depths and other log channels. Sequence-based generative adversarial networks (GANs) generate synthetic logs and fill gaps in gamma ray, sonic, neutron porosity, deep resistivity, and bulk density [Al-Fakih et al., 2025]. A fine-tuned time-series foundation model forecasts the next interval of the same suite and flags anomalies, including a zero-shot check in a second basin [Koeshidayatullah et al., 2026]. The principal task is to predict missing

log responses using the learned temporal or spatial context, rather than to optimize a conditional latent code under an explicit target-measurement fidelity penalty at inference.

Geomodel translation reconstructs a reservoir-property image from another image. A Pix2Pix conditional GAN maps two-dimensional slices of facies, porosity, permeability, and water saturation, taking either a masked image of the same property or a different property as generator input [Al-Fakih et al., 2026]. Agreement with the supplied image is learned in training.

Direct property regression maps wireline logs to a petrophysical target. Artificial neural network (ANN) and random-forest models predict total porosity, effective porosity, and shale volume from logs that include neutron porosity [Al-Fakih et al., 2023]. Missing log values are treated in preprocessing. At inference the model returns the property from the logs, and target values act as training labels.

CLP-CSGM addresses target-property reconstruction from dense auxiliary logs and sparse observations of the target itself. Its inference formulation combines a conditional latent anchor, $\boldsymbol{z}_0 = h(\boldsymbol{u})$, with an explicit measurement-fidelity penalty applied during latent refinement. The distinction from the completion, translation, and direct-regression approaches discussed above concerns how the available observations enter inference, not whether those methods use known values. Because the target properties, observation patterns, and evaluation protocols differ, these comparisons establish methodological positioning rather than predictive superiority.

The experiments use the processed F3 block data of Zhang et al. [2025], who combine supervised prediction with Gardner and Larionov relations and Shapley additive explanations (SHAP). We use these logs to study reconstruction from dense context while penalizing disagreement with sparse target-well observations.

2.2 Classical and generative compressed sensing

Classical compressed sensing combines an explicit measurement operator with a sparsity or compressibility prior [Donoho, 2006, Candes et al., 2006a,b]. Here, M_Ω is a coordinate restriction selecting observed target depths. A fixed transform-domain prior may be restrictive for cross-well petrophysical windows, whose representation can vary across wells. Restriction operators are also common in seismic, ultrasonic, and sparse-sensor reconstruction [Herrmann and Hennenfent, 2008, Hennenfent and Herrmann, 2008, de Assis et al., 2022, Muir and Zhan, 2021, da Silva et al., 2025].

CSGM replaces fixed-basis sparsity with the range of a learned generator or decoder [Bora et al., 2017]. Reconstruction searches for a latent code $\boldsymbol{z}$ whose decoded signal reduces the measurement residual. Recovery properties and limitations of this learned-range prior have been studied under set-restricted conditions and related generative formulations [Bora et al., 2017, Kamath et al., 2020, Asim et al., 2020]. Missing-data reconstruction is also commonly formulated as an inverse problem. Generative-prior approaches have been investigated for inpainting as well as compressed sensing and denoising [Asim et al., 2020]. CLP-CSGM retains this decoder-range view and adds dense-log conditioning of the latent search.

2.3 Conditional generative inverse problems and positioning

Conditional approaches differ in where side information and measurements enter. Direct conditional predictors map side information, alone or with observations, to an output or latent code [Mirza and Osindero, 2014, Sohn et al., 2015], while measurement-conditional CSGM conditions the generator directly on compressed measurements and their sampling pattern [Kim et al., 2021]. Other formulations retain an explicit inverse problem: side information may regularize a measurement-consistent reconstruction [Ongie et al., 2020, Lucas et al., 2018], approximate or sample inverse posteriors [Adler and Öktem, 2018, Altekrüger et al., 2024, Bendel et al., 2024], or use score/diffusion models for inverse reconstruction [Dasgupta et al., 2024, Jacobsen et al., 2025].

CLP-CSGM belongs to this explicit conditional inverse-reconstruction setting. Dense auxiliary logs define the latent anchor $\boldsymbol{z}_0 = h(\boldsymbol{u})$, while sparse target observations enter the test-time measurement residual. The experiments distinguish this formulation from direct $[\boldsymbol{u}, \boldsymbol{b}]$ prediction, the decoded conditional prior $G(\boldsymbol{z}_0)$, and measurement-only CSGM. Measurement-conditional generators and posterior-sampling architectures remain outside the empirical comparison.

3 Method

CLP-CSGM separates representation, contextual prediction, and measurement assimilation. A decoder defines the candidate target windows, dense auxiliary logs determine a latent anchor $\boldsymbol{z}_0$, and test-time optimization refines that code against the sparse target observations. The quadratic penalty keeps the refined code near $\boldsymbol{z}_0$ without requiring an exact fit at the observed depths.

This section formalizes the measurement model and reconstruction objective, defines the structural ablations, and states the conditions used in the error interpretation.

3.1 Problem setup

Let $\boldsymbol{y} \in \mathbb{R}^N$ denote a target-property window of N depth samples and let $\boldsymbol{u} \in \mathbb{R}^p$ denote the vectorized dense-log channels over the same interval. Target values are observed at the ordered coordinate set $\Omega = \{\omega_1, \ldots, \omega_m\} \subset \{1, \ldots, N\}$, where $1 \leq m < N$. The realized observed fraction is m/N. In the experiments, ρ is the nominal measurement ratio used to choose m (Section 5). Rounding and a minimum of four observations can make m/N differ slightly from ρ.

The coordinate-restriction operator $M_\Omega : \mathbb{R}^N \to \mathbb{R}^m$ is represented by

$$M_\Omega = \begin{bmatrix} \boldsymbol{e}_{\omega_1}^\top \\ \vdots \\ \boldsymbol{e}_{\omega_m}^\top \end{bmatrix} \in \mathbb{R}^{m \times N}, \qquad M_\Omega \boldsymbol{y} = \begin{bmatrix} y_{\omega_1} \\ \vdots \\ y_{\omega_m} \end{bmatrix}, \tag{1}$$

where $\boldsymbol{e}_i$ is the i-th canonical basis vector of $\mathbb{R}^N$. Coordinate restriction is widely used to represent incomplete seismic, ultrasonic, and sparse-sensor observations [Herrmann and Hennenfent, 2008, de Assis et al., 2022, da Silva et al., 2025].

The sparse target measurements satisfy

$$\boldsymbol{b} = M_\Omega \boldsymbol{y} + \boldsymbol{\eta}, \qquad \boldsymbol{b}, \boldsymbol{\eta} \in \mathbb{R}^m, \tag{2}$$

where $\boldsymbol{\eta}$ is additive noise. Under the Gaussian model, $\boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{0}, \sigma_\eta^2 I_m)$, where I_m is the $m \times m$ identity; the noiseless case has $\boldsymbol{\eta} = \boldsymbol{0}$.

Although the application is naturally described as missing-data reconstruction, it also admits an underdetermined inverse-problem formulation. The forward model in Eq. (2) is coordinate restriction: M_Ω selects $m < N$ entries of a complete window. This operator is not injective on $\mathbb{R}^N$, so the sparse observations alone do not uniquely determine $\boldsymbol{y}$. Classical compressed sensing addresses such problems with a structural prior, typically sparsity, whereas CSGM uses the range of a learned generator [Bora et al., 2017]. CLP-CSGM follows that generative inverse-reconstruction view and adds a conditional latent prior from the dense logs. It does not invert a physical petrophysical

forward model, and it does not establish unique or exact recovery for the trained decoder. Recovery guarantees proved for other sensing operators, including certain random matrices, do not transfer to M_Ω.

The dense logs and the sparse measurements do different jobs. The dense-log vector $\boldsymbol{u}$ predicts the code $\boldsymbol{z}_0 = h(\boldsymbol{u})$, where h is the conditional map from dense logs to that code. The measurements $\boldsymbol{b}$ enter only as the penalized mismatch in Eq. (10). The first latent trajectory starts at $\boldsymbol{z}_0$, and the penalty center remains $\boldsymbol{z}_0$.

Standardization and modeling assumptions. All preprocessing statistics are computed from training windows only. Let $\boldsymbol{\mu}_y, \boldsymbol{s}_y$ and $\boldsymbol{\mu}_u, \boldsymbol{s}_u$ be the componentwise training means and standard deviations, with $D_y = \mathrm{diag}(\boldsymbol{s}_y)$ and $D_u = \mathrm{diag}(\boldsymbol{s}_u)$. These two matrices rescale each coordinate by its training standard deviation. They are not estimated covariance matrices of the profiles, and this rescaling leaves any correlation already present between coordinates unchanged. We use $\widetilde{\boldsymbol{y}} = D_y^{-1}(\boldsymbol{y} - \boldsymbol{\mu}_y)$ and $\widetilde{\boldsymbol{u}} = D_u^{-1}(\boldsymbol{u} - \boldsymbol{\mu}_u)$. Sparse measurements are standardized with $D_\Omega = M_\Omega D_y M_\Omega^\top$ and $\widetilde{\boldsymbol{b}} = D_\Omega^{-1}(\boldsymbol{b} - M_\Omega \boldsymbol{\mu}_y)$; near-zero standard deviations use the implementation safeguard. The decoder G returns a standardized window. The same window in the original target units is

$$G_{\mathrm{orig}}(\boldsymbol{z}) = \boldsymbol{\mu}_y + D_y G(\boldsymbol{z}), \tag{3}$$

and a standardized prediction returns to that scale by

$$\widehat{\boldsymbol{y}} = \boldsymbol{\mu}_y + D_y \widehat{\widetilde{\boldsymbol{y}}}. \tag{4}$$

For a fixed set of observed coordinates, Eq. (2) adds noise with $\boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{0}, \sigma_\eta^2 I_m)$. Independence across coordinates and a common variance are assumptions of this synthetic benchmark noise. They are not a demonstrated description of instrumental noise, and this manuscript does not identify them with the 30dB filename label. The standardized noise is $\widetilde{\boldsymbol{\eta}} = D_\Omega^{-1}\boldsymbol{\eta}$. Because D_Ω is diagonal,

$$\mathrm{Cov}(\widetilde{\boldsymbol{\eta}}) = \sigma_\eta^2 D_\Omega^{-1} D_\Omega^{-\top} = \sigma_\eta^2 D_\Omega^{-2}. \tag{5}$$

This diagonal transformation preserves the independence of the added Gaussian noise. When

the training standard deviations differ across coordinates, the standardized variances differ as well. For $\boldsymbol{r} = M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}}$,

$$\boldsymbol{r}^\top \operatorname{Cov}(\widetilde{\boldsymbol{\eta}})^{-1} \boldsymbol{r} = \frac{1}{\sigma_\eta^2} \|D_\Omega \boldsymbol{r}\|_2^2. \tag{6}$$

The right-hand side is the data term of Eq. (10) divided by σ_η^2. By Eq. (11), that data term equals $\|M_\Omega G_{\text{orig}}(\boldsymbol{z}) - \boldsymbol{b}\|_2^2$. Proposition 2 reads $\|M_\Omega G_{\text{orig}}(\boldsymbol{z}) - \boldsymbol{b}\|_2^2 + \lambda \|\boldsymbol{z} - \boldsymbol{z}_0\|_2^2$ as a MAP objective in which the data term has coefficient one and $\lambda = \sigma_\eta^2 / \sigma_z^2$ multiplies the latent penalty. The factor $\sigma_z^2 I_k$ in that proposition is the isotropic hypothesis behind the Euclidean latent penalty. It is not an estimated covariance of the residuals of h. Ridge standardizes the latent codes while fitting h and returns $\boldsymbol{z}_0$ to the encoder scale.

Where a later equation omits tildes, its coordinates are those written there. The refinement objective keeps the standardized residual and the diagonal weight D_Ω. The decoder $G : \mathbb{R}^k \to \mathbb{R}^N$, with $k < N$, supplies standardized windows, and G_{orig} supplies the corresponding windows in the original target units. We assume that target windows are adequately approximated by the windows G_{orig} can return, and that $h : \mathbb{R}^p \to \mathbb{R}^k$ provides a useful anchor $\boldsymbol{z}_0 = h(\boldsymbol{u})$. These assumptions are assessed empirically through reconstruction, ablations, and prior sensitivity.

3.2 Learned model components

Offline training fits an autoencoder on standardized target windows and a latent-prior predictor on paired dense-log windows and encoded targets. All learned components remain frozen during target-well reconstruction.

Generative decoder. Let $E_{\theta_E} : \mathbb{R}^N \to \mathbb{R}^k$ and $G_{\theta_G} : \mathbb{R}^k \to \mathbb{R}^N$, with $k < N$, denote the encoder and decoder. Given standardized training targets $\{\widetilde{\boldsymbol{y}}_i\}_{i=1}^{n_{\text{tr}}}$, their parameters are obtained from

$$\begin{aligned}(\widehat{\theta}_E, \widehat{\theta}_G) \in \underset{\theta_E, \theta_G}{\arg\min} \; & \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \|\widetilde{\boldsymbol{y}}_i - G_{\theta_G}\left(E_{\theta_E}\left(\widetilde{\boldsymbol{y}}_i\right)\right)\|_2^2 \\ & + \gamma_{\text{AE}} \left(\|\theta_E\|_2^2 + \|\theta_G\|_2^2\right),\end{aligned} \tag{7}$$

where n_{tr} is the number of training windows and $\gamma_{\text{AE}} \geq 0$ is the autoencoder regularization weight. After training, we write $E = E_{\widehat{\theta}_E}$ and $G = G_{\widehat{\theta}_G}$. The frozen decoder defines the learned

target-window classes

$$\mathcal{R}_G = \left\{ G(\boldsymbol{z}) : \boldsymbol{z} \in \mathbb{R}^k \right\} \subseteq \mathbb{R}^N, \qquad \mathcal{R}_{\text{orig}} = \left\{ G_{\text{orig}}(\boldsymbol{z}) : \boldsymbol{z} \in \mathbb{R}^k \right\} \subseteq \mathbb{R}^N. \tag{8}$$

$\mathcal{R}_G$ is the range of the standardized decoder. $\mathcal{R}_{\text{orig}}$ is that class in the original target units. M_Ω selects observed coordinates and is not either set. Target-well reconstruction therefore optimizes only $\boldsymbol{z} \in \mathbb{R}^k$ and produces $\widehat{\boldsymbol{y}} = G_{\text{orig}}(\widehat{\boldsymbol{z}}) \in \mathcal{R}_{\text{orig}}$; it does not update the network parameters or search over arbitrary vectors in $\mathbb{R}^N$. The encoder is used to train the decoder and to build the codes on which the dense-log map is fit. Inference does not encode the unknown window. In the reported model, $k = 16$ is the dimension of that code. A reconstruction $G(E(\boldsymbol{y}))$ that is given the full target window checks whether the decoder can represent that window. It uses target values that inference does not have, so it is not one of the compared predictors. Appendix B.12 reports that check. The later outputs are evaluated separately from this diagnostic. The prior-only ablation evaluates the decoded conditional prediction before sparse-measurement refinement, and the full CLP-CSGM results quantify reconstruction after refinement (Section 6.3). The measurement-only variant is a separate structural ablation, not an intermediate inference stage.

Log-conditioned latent prior. Each training target is encoded as $\boldsymbol{z}_i = E(\widetilde{\boldsymbol{y}}_i)$. Ridge regression learns $h : \mathbb{R}^p \to \mathbb{R}^k$ from the standardized auxiliary logs and these encoder codes, $\{(\widetilde{\boldsymbol{u}}_i, \boldsymbol{z}_i)\}_{i=1}^{n_{\text{tr}}}$. For that fit, $h(\widetilde{\boldsymbol{u}}) = W\widetilde{\boldsymbol{u}} + \boldsymbol{c}$, with parameters

$$(\widehat{W}, \widehat{\boldsymbol{c}}) \in \operatorname*{arg\,min}_{W,\boldsymbol{c}} \left\{ \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} \|\boldsymbol{z}_i - W\widetilde{\boldsymbol{u}}_i - \boldsymbol{c}\|_2^2 + \alpha \|W\|_F^2 \right\}, \tag{9}$$

where $\alpha \geq 0$ controls the ridge-regression penalty [Hoerl and Kennard, 1970]. After fitting, $h(\widetilde{\boldsymbol{u}}) = \widehat{W}\widetilde{\boldsymbol{u}} + \widehat{\boldsymbol{c}}$, and, suppressing tildes, the contextual anchor is $\boldsymbol{z}_0 = h(\boldsymbol{u}) \in \mathbb{R}^k$. At inference, $\boldsymbol{z}_0$ initializes the first trajectory and remains the center of the latent penalty. The decoder G turns latent codes into target windows. Ridge regression does not replace G and does not perform that latent optimization. CLP-CSGM is this scheme; the class of h is a configuration choice. It is the primary reference because it is a linear map from the dense logs to $\boldsymbol{z}_0$, not because it is the best map in every test. An MLP variant of h is compared under the same channels and the same split in Section 6.4. Direct ridge regression is a different comparator: it maps $[\boldsymbol{u}, \boldsymbol{b}]$ to the target window

(Appendix B.12).

Mapping the auxiliary logs to a latent code serves the limited-information regimes this reconstruction is designed to address. The target property is observed only at sparse depths, and the cross-well protocol includes configurations with limited training support. The conditional map estimates $\boldsymbol{z}_0$ in the latent space of G. Sparse observations then update that code through the measurement-fidelity term, and the decoder returns the window, so the same coordinates carry learned window structure, dense-log context, and sparse-measurement refinement. In the reported model the search adjusts the $k = 16$ code of each length-64 window; under the published sampling rule, a window can contain fewer observations than latent coordinates. The decoder range and the penalty around $\boldsymbol{z}_0$ provide the structural information used in that case. Direct predictors may instead map the logs, or the logs together with the sparse values, directly to the target window. AE $[\boldsymbol{u}, \boldsymbol{b}]$ also predicts a latent code, so the empirical comparison in Section 6 concerns how that code is conditioned and refined. Reconstruction accuracy ultimately depends on the adequacy of G, the conditional anchor, and the observations available at inference.

3.3 Test-time reconstruction

At inference time, G and h remain fixed, and only the latent code of the current target window is optimized. Given standardized dense logs $\widetilde{\boldsymbol{u}} \in \mathbb{R}^p$ and sparse target observations $\widetilde{\boldsymbol{b}} \in \mathbb{R}^m$, define $\boldsymbol{z}_0 = h(\widetilde{\boldsymbol{u}})$. CLP-CSGM computes

$$\begin{aligned} \widehat{\boldsymbol{z}}_{\text{num}} &\in \underset{\boldsymbol{z}\in\mathbb{R}^k}{\arg\min} \left\{ \left\| D_\Omega \left(M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}} \right) \right\|_2^2 + \lambda \left\| \boldsymbol{z} - \boldsymbol{z}_0 \right\|_2^2 \right\}, \\ \widehat{\boldsymbol{y}} &= G_{\text{orig}} \left(\widehat{\boldsymbol{z}}_{\text{num}} \right). \end{aligned} \tag{10}$$

The first term penalizes disagreement with the sparse observations and the second, weighted by $\lambda \geq 0$, penalizes deviation from $\boldsymbol{z}_0$. Their sum does not force the observations to be met exactly, so a residual at the observed depths can remain. Because each row of M_Ω selects one window coordinate,

$$\left\| D_\Omega \left(M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}} \right) \right\|_2^2 = \left\| M_\Omega G_{\text{orig}}(\boldsymbol{z}) - \boldsymbol{b} \right\|_2^2 . \tag{11}$$

The left-hand side is the data term in Eq. (10). The right-hand side is the squared residual in the

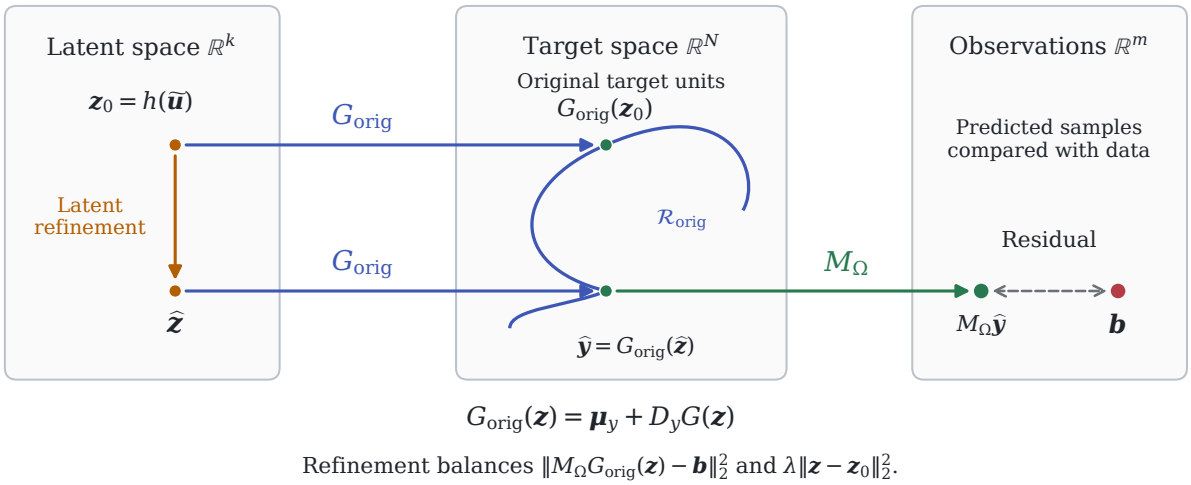


Figure 1: Geometry of the conditional latent code, in the original target units. Dense logs determine $z_0 = h(u)$. That code initializes the first trajectory and remains the center of the quadratic penalty. The vertical arrow is latent refinement from z_0 to $\widehat{z}$. The drawing is schematic and does not show a measured numerical trajectory. The two horizontal arrows are decoding by G_{orig}: one maps z_0 to $G_{\text{orig}}(z_0)$, and the other maps $\widehat{z}$ to $\widehat{y} = G_{\text{orig}}(\widehat{z})$. The blue curve illustrates $\mathcal{R}_{\text{orig}}$, the set of decoder outputs in the original target units. M_Ω selects the observed coordinates and is not that set. The measurement residual is the gap between $M_\Omega\widehat{y}$ and b. Refinement balances that residual against deviation from z_0, and the observations need not be met exactly. This schematic figure was prepared with assistance from OpenAI's ChatGPT (model GPT-5.5). The tool was used to support the initial visual organization of the conceptual diagram. The authors reviewed and edited the final figure and remain fully responsible for its scientific content and interpretation.

original target units, which is the residual evaluated by the implementation. The diagonal entries of D_y are the training standard deviations of the window coordinates. On the cross-well V_c training windows those entries are not equal: the ratio of the largest to the smallest is about 1.13 at step 8, 1.23 at step 16, and 1.40 at step 32. A single factor therefore does not convert a penalty weight between the unweighted standardized residual and this objective. The implementation also divides both terms by the number of windows. Multiplying the entire objective by the same positive factor leaves its minimizers unchanged. $\lambda \geq 0$ is selected on validation windows for each measurement ratio and seed. Proposition 2 identifies λ with σ_η^2/σ_z^2 when the likelihood is an isotropic Gaussian in those original target units. That is the scale of Eq. (11). The same ratio does not apply to the unweighted standardized residual $\|M_\Omega G(z) - \widetilde{b}\|_2^2$. The reported experiments do not compute λ from those variances. For each seed and each measurement ratio, validation full-window RMSE selects one candidate from $\{10^{-4}, 3\cdot10^{-4}, 10^{-3}, 3\cdot10^{-3}, 10^{-2}, 3\cdot10^{-2}, 10^{-1}\}$, and that choice is

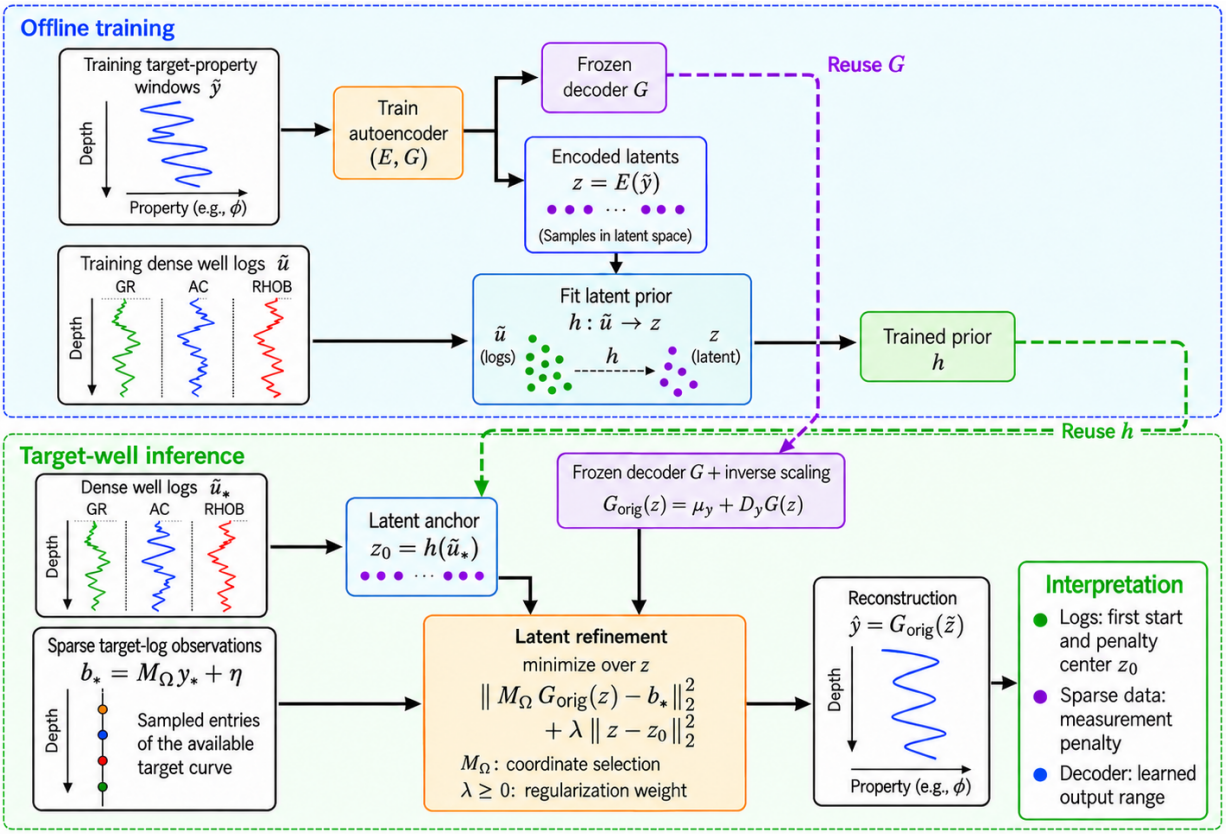


Figure 2: CLP-CSGM workflow. Offline, standardized target windows train the autoencoder (E, G), and paired dense-log windows fit the latent-prior predictor h. At inference, $\boldsymbol{u}_*$ determines $\boldsymbol{z}_0 = h(\boldsymbol{u}_*)$, sparse observations $\boldsymbol{b}_*$ enter latent refinement as a measurement penalty through M_Ω, and the decoded result is returned to the original target scale. The displayed log labels are illustrative; the input channels differ by benchmark, and gamma ray is excluded from the main cross-well clay-volume benchmark. OpenAI's ChatGPT, using the GPT-5.5 model, was used to assist in generating the initial explanatory schematic and visual layout. The final figure was reviewed, edited, and validated by the authors.

fixed before the test score is computed. The selected λ is the validation choice for the original-scale objective. Because G is nonlinear, the objective is generally nonconvex and is solved with Adam [Kingma and Ba, 2015]. The first run starts from $\boldsymbol{z}_0$. Each additional restart starts at $\boldsymbol{z}_0$ plus an independent draw from $\mathcal{N}(\mathbf{0}, 0.05^2 I)$, with the same shape as $\boldsymbol{z}_0$. The penalty center remains $\boldsymbol{z}_0$ on every restart and every iteration, and the lowest-objective candidate is retained. Thus $\widehat{\boldsymbol{z}}_{\text{num}}$ is a numerical solution, not a certified global minimizer.

Ablation variants. The *latent-prior-only* variant suppresses refinement and uses $\widehat{\boldsymbol{z}}_{\text{prior}} = \boldsymbol{z}_0$, followed by $\widehat{\boldsymbol{y}}_{\text{prior}} = G_{\text{orig}}(\widehat{\boldsymbol{z}}_{\text{prior}})$. The *measurement-only CSGM* variant sets $\lambda = 0$ and computes

$$\widehat{\boldsymbol{z}}_{\text{meas}} \in \arg\min_{\boldsymbol{z}\in\mathbb{R}^k} \left\| D_\Omega \left(M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}} \right) \right\|_2^2. \tag{12}$$

The latent-prior-only variant returns the decoded prediction of the conditional prior, $G_{\text{orig}}(\boldsymbol{z}_0)$, with no latent search. In that variant $\boldsymbol{z}_0$ is not an initialization, because no trajectory is optimized. The measurement-only variant uses only the sparse measurements and recovers measurement-only generative compressed sensing [Bora et al., 2017]. Comparison with the full objective in Eq. (10) shows what each piece contributes. Results are reported in Section 6.3.

3.4 Interpretation and theoretical context

The results below answer three questions about the method. How should the two terms of the objective be interpreted? Under what conditions does the problem have a minimizer? Which factors condition recovery of the full window from the sparse observations? The MAP/Tikhonov reading and the existence argument are classical. They are used to interpret the formulation and are not presented as new theoretical contributions. Complete proofs appear in Appendix A.

Under the Gaussian hypotheses in Proposition 2, the objective admits a MAP interpretation. The quadratic penalty is centered at the conditioned code $\boldsymbol{z}_0 = h(\boldsymbol{u})$, not at the origin [Tikhonov, 1963, Burger and Lucka, 2014]. In that reading the data term has coefficient one, and $\lambda = \sigma_\eta^2/\sigma_z^2$ multiplies the latent penalty. The data term is the original-scale residual, which Eq. (11) identifies with the standardized residual weighted by D_Ω. The reported experiments do not estimate this variance ratio. Validation selects λ, as stated in Section 3.3. The MAP reading explains the form of the penalty. It does not demonstrate empirical superiority.

Remark 1 (Relation to classical CSGM). Setting $\lambda = 0$ recovers the measurement-only CSGM problem in Eq. (12) [Bora et al., 2017]. Increasing λ shifts the reconstruction toward the decoded prediction of the conditional prior, $G_{\text{orig}}(\boldsymbol{z}_0)$, by penalizing departures from $\boldsymbol{z}_0 = h(\boldsymbol{u})$ more strongly. That shift is not a claim that a larger λ improves the reconstruction.

Proposition 3 is a classical existence result. If the decoder is continuous and $\lambda > 0$, the objective is continuous and coercive, so it attains a minimum. The result does not give uniqueness, convexity, or convergence of Adam to a global minimizer. It is stated for $\lambda > 0$ and does not extend automatically to the measurement-only case $\lambda = 0$. For any global minimizer $\widehat{\boldsymbol{z}}$,

$\mathcal{J}(\widehat{\boldsymbol{z}}) \leq \mathcal{J}(\boldsymbol{z}_0) = \left\| D_\Omega \left(M_\Omega G(\boldsymbol{z}_0) - \widetilde{\boldsymbol{b}} \right) \right\|_2^2$. Exact minimization therefore cannot worsen the optimized objective relative to the latent-prior-only candidate. It does not guarantee lower error with respect to the unknown target. For the numerical solution $\widehat{\boldsymbol{z}}_{\text{num}}$, this comparison holds only when the computed values satisfy $\mathcal{J}(\widehat{\boldsymbol{z}}_{\text{num}}) \leq \mathcal{J}(\boldsymbol{z}_0)$. $\widehat{\boldsymbol{z}}_{\text{num}}$ remains a numerical solution, not a certified global minimizer.

Remark 2 (Representation error and decoder mismatch). Every CLP-CSGM reconstruction in the original target units satisfies $\widehat{\boldsymbol{y}} = G_{\text{orig}}(\widehat{\boldsymbol{z}}) \in \mathcal{R}_{\text{orig}}$. Consequently, for an unknown target window $\boldsymbol{y}^\star \in \mathbb{R}^N$, the error of any such reconstruction is bounded below by the oracle representation error

$$\operatorname{dist}\left(\boldsymbol{y}^\star, \mathcal{R}_{\text{orig}}\right) := \inf_{\boldsymbol{y}' \in \mathcal{R}_{\text{orig}}} \left\| \boldsymbol{y}^\star - \boldsymbol{y}' \right\|_2 = \inf_{\boldsymbol{z} \in \mathbb{R}^k} \left\| \boldsymbol{y}^\star - G_{\text{orig}}(\boldsymbol{z}) \right\|_2 . \tag{13}$$

Thus, even with noiseless observations and exact global optimization, latent refinement cannot remove mismatch between $\boldsymbol{y}^\star$ and $\mathcal{R}_{\text{orig}}$. This structural error is represented by $\boldsymbol{r}^\dagger$ in Proposition 1 and is distinct from the latent-prior mismatch $\|\boldsymbol{z}^\dagger - \boldsymbol{z}_0\|_2$.

An inaccurate code $\boldsymbol{z}_0 = h(\boldsymbol{u})$ may bias the search toward a suboptimal latent region, because it initializes the first trajectory and remains the center of the penalty. The data-fidelity term can partially correct this bias when the sparse observations are informative and λ does not dominate the objective. The decoder range determines what can be represented; the conditional code $\boldsymbol{z}_0$ determines where the latent penalty is centered.

Four factors enter the bound in Proposition 1. Representation error is the difference between the true window and a window the decoder can represent. Noise and observed discrepancy combine the measurement perturbations with the part of that representation error present at the observed coordinates. Deviation from the conditioned code is the distance between a reference code $\boldsymbol{z}^\dagger$ and the penalty center $\boldsymbol{z}_0$. Local separability is the ability of the observed coordinates to distinguish different windows produced by the decoder in the region under consideration. The statement below assigns symbols to these factors.

Proposition 1 (Conditional local recovery interpretation)**.** *Let $\boldsymbol{y}^\star \in \mathbb{R}^N$ satisfy $\boldsymbol{b} = M_\Omega \boldsymbol{y}^\star + \boldsymbol{\eta}$, let $\boldsymbol{z}_0 = h(\boldsymbol{u})$, $\lambda > 0$, and let $\widehat{\boldsymbol{z}}$ be any global minimizer of Eq.* (10)*. The residual $\|M_\Omega G_{\text{orig}}(\boldsymbol{z}) - \boldsymbol{b}\|_2^2$ is the right-hand side of Eq.* (11)*. For any reference code $\boldsymbol{z}^\dagger \in \mathbb{R}^k$, define*

$$
\begin{aligned}
\boldsymbol{r}^\dagger &= \boldsymbol{y}^\star - G_{\text{orig}}(\boldsymbol{z}^\dagger), & \boldsymbol{e}^\dagger &= M_\Omega \boldsymbol{r}^\dagger + \boldsymbol{\eta}, \\
\Delta^\dagger &= \left(\|\boldsymbol{e}^\dagger\|_2^2 + \lambda \|\boldsymbol{z}^\dagger - \boldsymbol{z}_0\|_2^2 \right)^{1/2}, & R^\dagger &= \frac{\Delta^\dagger}{\sqrt{\lambda}}.
\end{aligned}
\tag{14}
$$

Let $\mathcal{R}_{\text{orig},R^\dagger} = G_{\text{orig}}(B(\boldsymbol{z}_0, R^\dagger))$*, where* $B(\boldsymbol{z}_0, R^\dagger) = \left\{ \boldsymbol{z} \in \mathbb{R}^k : \|\boldsymbol{z} - \boldsymbol{z}_0\|_2 \le R^\dagger \right\}$*. The constants* $\gamma_R > 0$ *and* $\delta_R \ge 0$ *below refer to pairs of windows in* $\mathcal{R}_{\text{orig},R^\dagger}$*. They are not constants of the standardized decoder* G*. If*

$$
\|M_\Omega(\boldsymbol{y}_1 - \boldsymbol{y}_2)\|_2 \ge \gamma_R \|\boldsymbol{y}_1 - \boldsymbol{y}_2\|_2 - \delta_R, \qquad \forall\, \boldsymbol{y}_1, \boldsymbol{y}_2 \in \mathcal{R}_{\text{orig},R^\dagger},
\tag{15}
$$

then

$$
\|G_{\text{orig}}(\widehat{\boldsymbol{z}}) - \boldsymbol{y}^\star\|_2 \le \|\boldsymbol{r}^\dagger\|_2 + \frac{\Delta^\dagger + \|\boldsymbol{e}^\dagger\|_2 + \delta_R}{\gamma_R}.
\tag{16}
$$

Proposition 1 organizes factors that can contribute to the reconstruction error. It is a sufficient conditional interpretation, not a verified guarantee for the trained decoder. The latent penalty localizes the S-REC condition around $\boldsymbol{z}_0$, and the bound follows the decoder-range view of generative compressed sensing [Candes et al., 2006b, Bora et al., 2017]. The bound depends on the local separability condition stated above. Coordinate restriction is not proved to satisfy S-REC on the trained decoder range; this depends on local decoder secants. The empirical separability diagnostics in Appendix B.1 do not certify that condition for every pair of windows. The statement above is for a global minimizer. The extension to an inexact numerical optimizer is a separate argument in Appendix A.3. The result does not demonstrate superiority over interpolation, linear regression, or the other comparators. Increasing λ does not necessarily tighten the bound or improve the reconstruction, because an unsuitable center $\boldsymbol{z}_0$ can also introduce bias.

4 Data and exploratory analysis

We evaluate two F3 settings. In the first, clay volume is reconstructed in a held-out well. In the second, porosity in one well is reconstructed from gamma ray alone.

4.1 Field settings and benchmark provenance

The experiments use the processed F3 well-log curves of Zhang et al. [2025]; no new field data were acquired. We use the FS2–FS6 intervals of F02-1, F03-2, F03-4, and F06-1. Sonic and gamma ray are available in all wells; density is measured in F02-1/F03-2 and predicted in F03-4/F06-1. The processed porosity target is density-derived, whereas V_c denotes the dimensionless clay-volume fraction estimated from gamma ray using the Larionov relation; further geological and processing details are given in the source study.

The cross-well benchmark reconstructs V_c using processed F3 well-log tables associated with the dataset of Zhang et al. [2025], with F02-1, F03-2, and F06-1 used for development and F03-4 held out for testing. Zhang et al. report an experimental condition with an SNR of 30 but do not specify its mathematical definition or explicitly state whether it is expressed in decibels. We retain the designation `30dB` used in the input filenames for consistency and reproducibility, without interpreting it as an independently verified signal-to-noise ratio. The single-well benchmark reconstructs F03-4 porosity using gamma ray as the only dense input. In both benchmarks, sparse target observations are generated by coordinate subsampling of the corresponding processed target curve. Appendix D provides a limited-sample Lapa permeability case based on Alvarez et al. [2025].

Table 1: Summary of the two F3 benchmarks. The cross-well target V_c is derived from gamma ray through the Larionov relation, whereas the single-well target is processed porosity.

dataset	well	target	n_rows	depth_min	depth_max	target_mean	target_std	target_min	target_max
cross_train	F02-1	V_c	165	1149.7056	1174.6992	0.1215	0.0407	0.0513	0.1934
cross_train	F03-2	V_c	811	951.9000	1073.4000	0.1184	0.0490	0.0393	0.2688
cross_train	F06-1	V_c	122	1133.1000	1151.2500	0.1223	0.0437	0.0427	0.2175
cross_test	F03-4	V_c	832	946.5000	1071.1500	0.0824	0.0341	0.0112	0.1699
real_f03_gr	F03-4	Porosity	1435	856.0500	1071.1500	0.2789	0.0142	0.2471	0.3264

Table 1 summarizes the two settings. Held-out F03-4 has mean $V_c = 0.0824$, versus about 0.12 in the training wells, and Fig. 3 shows a lower-valued but overlapping distribution. Larger sliding-window steps further reduce training support, whereas CLP-CSGM can still refine the held-out-well estimate using $\boldsymbol{b}$ at inference.

Channel-target correlations and input design. Table 2 reports descriptive Pearson correlations used only to guide input design. Clay volume in this benchmark is computed from gamma ray through the Larionov relation [Zhang et al., 2025]; including gamma ray would therefore expose

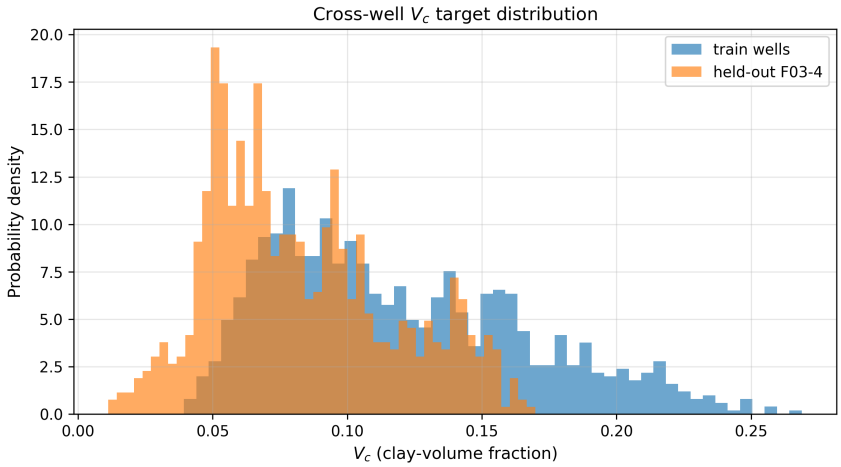


Figure 3: Empirical cross-well V_c distributions for the pooled training wells and the held-out F03-4 well. The held-out distribution is shifted toward lower clay-volume values while retaining partial overlap with the training distribution.

the model to the log from which the target was derived. The main cross-well input instead uses sonic, density, acoustic impedance, and V_p. The gamma-ray index is linear between the clean and shale endpoints, whereas the Larionov transformation from that index to clay volume is nonlinear. In a fully logged well the relation can be evaluated directly, so the proposed model is not intended to replace it. Gamma ray is restored only in Appendix B.7 to assess the effect of that channel.

Table 2: Pearson correlations between the available input channels and the target properties in the cross-well V_c and single-well F03-4 porosity benchmarks.

dataset	channel	target	pearson_r
cross_train	Sonic	V_c	-0.1642
cross_train	RHOB	V_c	0.0807
cross_train	GR	V_c	0.9733
cross_train	AI	V_c	0.1672
cross_train	V_p	V_c	0.1819
cross_test	Sonic	V_c	-0.1606
cross_test	RHOB	V_c	-0.0061
cross_test	GR	V_c	0.9772
cross_test	AI	V_c	0.1330
cross_test	V_p	V_c	0.1077
real_f03	AC	Porosity	0.9996
real_f03	GR	Porosity	-0.1759

In F03-4, sonic travel time follows the processed porosity almost one to one, whereas gamma ray is only weakly related to it. Sonic is left out for that reason. The density curve behind this porosity was itself predicted from sonic and gamma ray. The test that remains uses gamma ray as the only dense log and keeps a few values of the porosity curve as measurements. It is not a porosity

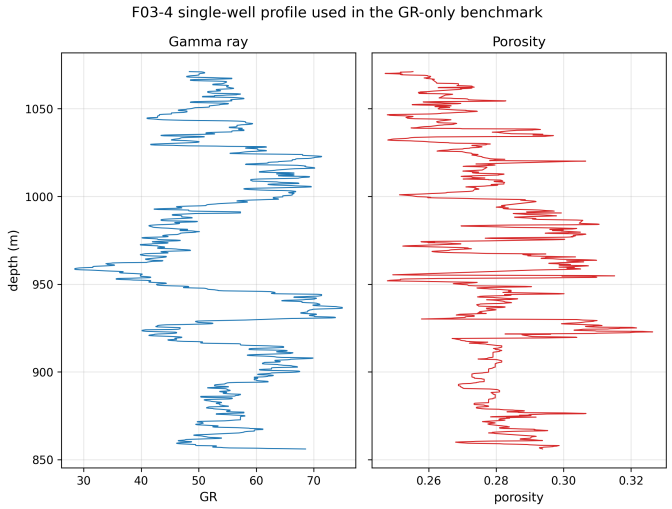


Figure 4: F03-4 depth profiles used in the gamma-ray-only (GR-only) porosity benchmark. Gamma ray is the only dense input, while porosity is reconstructed from the log-conditioned latent prior and sparse target observations.

interpretation from density, neutron, and sonic together (Fig. 4).

5 Experimental setup

The evaluation uses paired protocols for the cross-well low-data V_c benchmark and the single-well F03-4 GR-only porosity benchmark. Within each cell, all methods share the same partitions, measurement ratios, random seeds, sampled coordinates, and noise realizations.

5.1 Cross-well clay-volume (V_c) low-data benchmark

The cross-well benchmark reconstructs length-$N = 64$ processed V_c windows in the held-out F03-4 well. F02-1, F03-2, and F06-1 provide the model-development data, and the dense input $\boldsymbol{u}$ concatenates sonic, bulk density, acoustic impedance, and V_p. Reported cross-well performance refers to this single held-out well. As justified in Section 4, gamma ray is left out because clay volume was computed from that log [Zhang et al., 2025]; the benchmark therefore evaluates reconstruction from alternative dense logs when the source channel is withheld.

Sparse observations follow $\boldsymbol{b} = M_\Omega \boldsymbol{y} + \boldsymbol{\eta}$ at $\rho \in \{0.05, 0.10, 0.20\}$. Sliding-window steps $\{8, 16, 32\}$ control training support, and each step–ρ cell uses three paired seed realizations. This is a controlled sparse-assimilation benchmark on a processed target curve, not recovery against

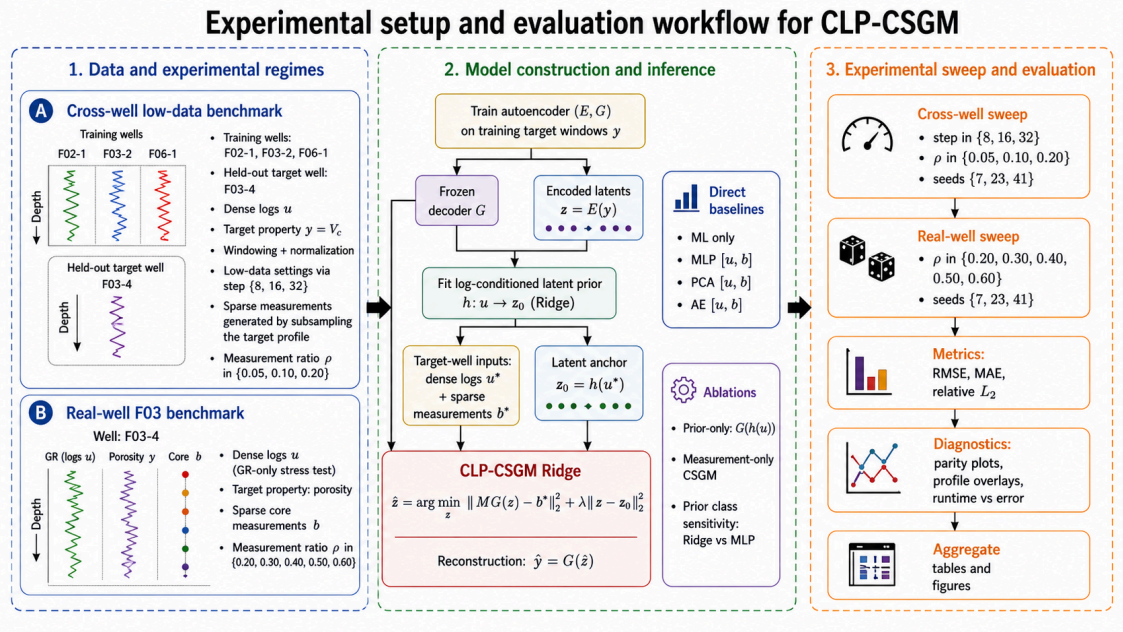


Figure 5: Experimental protocol. Offline training fits the decoder and the map from dense logs to z_0. At test time the first trajectory starts at that code and the penalty remains centered there. Partitions, measurement ratios, seeds, and sparse-observation draws are shared across methods, which are evaluated using RMSE, mean absolute error (MAE), and relative ℓ_2 error. OpenAI's ChatGPT, using the GPT-5.5 model, was used to assist in generating the initial explanatory schematic and visual layout. The final figure was reviewed, edited, and validated by the authors.

an independently acquired measurement. Split and sampling details are given in Section 5.3; GR inclusion and the embargo-style boundary check are reported in Appendix B.7.

5.2 Single-well F03-4 GR-only porosity benchmark

The second benchmark reconstructs length-$N = 64$ windows of the processed density-derived F03-4 porosity curve using gamma ray as the only dense input. Acoustic travel time is excluded because it is nearly collinear with the target (Pearson $r \approx 0.9996$) and would provide a near-direct proxy. The target follows the processing of Zhang et al. [2025], in which the density curve underlying F03-4 porosity is itself predicted from sonic and gamma ray; the high sonic correlation is therefore a property of that processed curve, not validation against independent laboratory porosity. This design leaves gamma ray as the only dense log and evaluates a restricted-input reconstruction rather than a replacement for density- or sonic-porosity relations when their required logs are available.

Sparse observations are sampled at $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$. Depth is partitioned before window construction, preventing overlap among training, validation, and test windows. Its errors

therefore characterize this GR-only protocol and do not transfer directly to a suite that also includes density, neutron, and sonic.

The smaller Lapa pre-salt permeability case [Alvarez et al., 2025] is retained as supplementary evidence in Appendix D. Appendix E reports a separate single-well check on laboratory porosity with depth-held-out blocks and within-window scoring of coordinates not supplied as measurements in each evaluated window. This check is not part of the main grid; its results are summarized in Section 6.

5.3 Training, validation, and test protocol

All preprocessing statistics and model choices are determined without test-target leakage. Training statistics are reused unchanged on validation and test windows. The training split fits the direct baselines, autoencoder (E, G), and latent-prior predictor h. Validation targets select λ and baseline-specific hyperparameters, and test targets are used only for final evaluation.

In the cross-well benchmark, F02-1, F03-2, and F06-1 form the development pool and F03-4 is held out entirely for testing. Steps $\{8, 16, 32\}$ produce $\{115, 58, 30\}$ development windows. Reserving 10% of each development well for validation leaves $\{104, 52, 27\}$ training windows and $\{11, 6, 3\}$ validation windows. At step 32 that pool has 30 windows; Appendix B.8 redraws it ten times with the measurement pattern held fixed. Because the validation tail is selected after window construction, boundary windows may overlap when step $< N$; Appendix B.7 reports an embargo-style check that removes these overlap-prone windows.

For F03-4 porosity, depth samples are divided into contiguous training, validation, and test blocks using fractions 60%, 20%, and 20% before windows are constructed independently. This depth-first partition prevents overlap and preserves contiguous-depth generalization.

For each seed and nominal measurement ratio ρ, the number of observed coordinates is

$$m = \max\{4, \mathrm{round}(\rho N)\}.$$

We use nearest-integer rounding, with half-integer ties resolved to the nearest even integer. The realized fraction is m/N, which can differ slightly from ρ. One coordinate set $\Omega = \{\omega_1, \ldots, \omega_m\}$ is sampled for each (seed, ρ) cell and held fixed across all training, validation, and test windows in

that cell. The set defines

$$\boldsymbol{b} = \begin{bmatrix} y_{\omega_1} & \cdots & y_{\omega_m} \end{bmatrix}^\top + \boldsymbol{\eta}. \tag{17}$$

Within each (seed, ρ) cell, all methods share the partitions, the fixed sampled coordinates, their ordering, and the noise realization. Direct baselines receive $[\boldsymbol{u}, \boldsymbol{b}] \in \mathbb{R}^{p+m}$, with only the measured target values in the ordering of M_Ω. Because Ω is fixed within the cell, the j-th entry of $\boldsymbol{b}$ consistently corresponds to coordinate ω_j across windows, making coordinate identity implicit in the stable input ordering. No zero-padded target, binary mask, or measurement matrix is added. CLP-CSGM uses M_Ω explicitly in the measurement term of Eq. (10). The protocol therefore compares direct fusion with operator-based assimilation under a fixed sampling pattern, giving the direct models a stable coordinate-to-input mapping within each cell. Appendix B.10 gives one direct AE the same measurements on the window coordinates plus a binary mask that remains unscaled. This comparator is not part of the main grid.

Appendix B.11 redraws the coordinate set and the measurement noise five times, while the model-seed identities stay fixed. Section 6 summarizes the mask-aware and measurement-redraw comparisons against the main-grid ordering.

Appendix B.12 adds direct linear and ridge maps from $[\boldsymbol{u}, \boldsymbol{b}]$ to the window, depth interpolation of the sparse measurements, and a second-difference smoother whose penalty is chosen on validation, on this same cross-well grid. It also raises the measurement-noise standard deviation above the published value 0.01. In that sweep the decoder and the conditional map stay fixed for each step and seed, and λ is selected again on validation at each noise level. Section 6 summarizes that noise dependence for interpolation, second-difference smoothing, and CLP-CSGM Ridge. The linear maps and the per-seed tables are not merged into the submitted nine-cell grid.

Noise is applied after coordinate subsampling. Each observed coordinate receives an independent standard-normal draw, scaled to a fixed standard deviation and added to the subsampled target value. That standard deviation is $\sigma_\eta = 0.01$ for cross-well V_c and $\sigma_\eta = 0.02$ for F03-4 porosity, in the units of the target curve. It is not a percentage of the curve. The held-out V_c curve has standard deviation about 0.035, so 0.01 is about 0.29 times that deviation. These two levels were adopted for the controlled benchmarks. These levels define controlled synthetic perturbations and

are not interpreted as estimates of instrumental uncertainty. Gaussian contamination has precedent as a synthetic stress test in well-log inversion [Abdelrahman and Szabó, 2024], but this precedent does not calibrate the amplitudes used here. Because the benchmark target is derived from gamma ray through the nonlinear Larionov relation, an instrument-aware analysis would require estimating gamma-ray and endpoint uncertainties and propagating them through that transformation [Adjei et al., 2019]. Such propagation is outside the scope of this benchmark. The input files carry the filename designation 30dB (Section 4.1). That designation is not this added standard deviation, and it is not a verified signal-to-noise ratio. Within one run, every method uses the same noisy observations. Direct $[\boldsymbol{u}, \boldsymbol{b}]$ models standardize the concatenated features after the noise has been added.

5.4 Implementation details

Table 3 reports the architectures, optimization settings, and validation grids. The compact backbone uses latent dimension $k = 16$ and a decoder that remains frozen during target-well reconstruction. The scientific code was originally developed by the authors. During subsequent development, xAI's Grok 4.7 was used to assist with refactoring the existing implementation. This assistance concerned code restructuring and editing rather than the original development of the research methodology. The authors reviewed the resulting changes and retain responsibility for the correctness, validation, and reproducibility of the computational implementation.

For CLP-CSGM, the first latent-refinement run starts from $\boldsymbol{z}_0 = h(\boldsymbol{u})$. Each additional run starts at $\boldsymbol{z}_0$ plus an independent draw from $\mathcal{N}(\mathbf{0}, 0.05^2 I)$. The penalty center remains $\boldsymbol{z}_0$ on every run and every iteration, and the candidate with the lowest objective is retained. The measurement-only ablation sets $\lambda = 0$ and starts from zero, whereas the prior-only variant returns the decoded prediction of the conditional prior, $G(h(\boldsymbol{u}))$, with no latent search. Predictions are returned to the original scale using Eq. (4).

5.5 Baselines and evaluation metrics

The direct baselines are ML only, which uses only $\boldsymbol{u}$; MLP $[\boldsymbol{u}, \boldsymbol{b}]$, which predicts the target directly; principal component analysis (PCA) $[\boldsymbol{u}, \boldsymbol{b}]$, which predicts target principal-component scores; and AE $[\boldsymbol{u}, \boldsymbol{b}]$, which predicts codes of a separately trained target autoencoder. CLP-CSGM Ridge is

Table 3: Implementation settings for offline training, target-well latent refinement, and direct supervised baselines.

Phase	Module	Implementation
Offline training	Standardization	Training-set mean and standard deviation for u and y; reused unchanged on validation and test windows.
	Autoencoder (E, G)	MLP with one hidden layer (width 128) in encoder and decoder, latent dimension $k = 16$, ReLU activations, linear output; 200 epochs, Adam [Kingma and Ba, 2015] (10^{-3}), weight decay 10^{-5}, batch size 64.
	Latent prior $h(u)$	Ridge regressor with $\alpha = 1.0$ on standardized (u, z) pairs (main experiments). Sensitivity variant: MLP with two hidden layers (width 128), L_2 penalty 10^{-4}, learning rate 10^{-3}, max. 500 iterations, early stopping.
Target-well inference	Latent refinement	Adam [Kingma and Ba, 2015] for 400 iterations, learning rate 0.05, three restarts (first start at $z_0 = h(u)$; two further starts at z_0 plus $\mathcal{N}(0, 0.05^2 I)$; penalty center remains z_0); λ selected by validation RMSE from $\{10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}, 3{\cdot}10^{-2}, 10^{-1}\}$ for the original-scale residual, not computed as σ_η^2/σ_z^2.
	Reconstruction	$\hat{y} = G(\hat{z})$ after inverse standardization to the original target scale.
Direct baselines	$[u, b]$ regressors	ML only and MLP $[u, b]$: sklearn MLP with two hidden layers (width 128), max. 500 iterations, learning rate 10^{-3}, L_2 penalty 10^{-4}, early stopping. PCA $[u, b]$: PCA score grid $\{8, 16, 24, 32, 48, 64\}$ with the same MLP head. AE $[u, b]$: separate MLP autoencoder ($k = 32$, hidden width 128, 120 epochs) plus MLP from $[u, b]$ to latent codes.

the proposed reference configuration. Prior-only Ridge and measurement-only CSGM are structural ablations, while the MLP latent prior is used only for sensitivity analysis. AE $[\boldsymbol{u}, \boldsymbol{b}]$ is the direct comparator emphasized in the main cross-well comparison because it has the lowest unweighted mean RMSE among MLP, PCA, and AE on that grid. That role does not make it the strongest comparator overall. Depth interpolation, second-difference smoothing, and direct linear and ridge regression from $[\boldsymbol{u}, \boldsymbol{b}]$ are additional comparators in Section 6 and Appendix B.12.

Metrics are computed after inverse standardization on the original target scale. RMSE is the primary metric and selects λ on validation data; MAE and relative ℓ_2 error provide complementary summaries. Results are computed under the paired protocol for each seed and then summarized across seeds, with lower values indicating better reconstruction.

Uncertainty summaries. For the main nine-cell benchmark, all seed-level intervals use the three fixed seeds $\{7, 23, 41\}$. Reported 95% half-widths are Student-t intervals with df $= n_{\text{seeds}} - 1 = 2$ (critical value $t_{2,0.975} \approx 4.303$), not normal $z = 1.96$ intervals. Where CLP-CSGM Ridge is compared

with AE $[\boldsymbol{u}, \boldsymbol{b}]$, we also report the paired seed-wise difference

$$\Delta\mathrm{RMSE} = \mathrm{RMSE}_{\mathrm{AE}} - \mathrm{RMSE}_{\mathrm{CLP}},$$

together with its seed-level standard error and Student-t half-width. Positive ΔRMSE favors CLP-CSGM Ridge. With only three seeds, these intervals describe seed-level reproducibility within the paired protocol; their interpretation remains limited by the small seed panel. Two supplements widen that evidence and leave the three-seed tables as the main result. Appendix B.9 repeats the comparison on ten seeds. Each extra seed refits the models and draws a new measurement realization, so training randomness, observation coordinates, and measurement noise change together. Those intervals describe seed-to-seed variability on the single held-out well F03-4 rather than ten independent geological tests; the nine per-configuration intervals are not adjusted as one simultaneous test. Appendix B.11 holds the same three model seeds and redraws observation coordinates together with a new measurement-noise realization. That check assesses sensitivity to the measurement realization without isolating coordinate placement from noise. Results from both supplements are reported in Section 6; overlapping windows inside a well are not counted as further independent samples.

6 Results

All comparisons follow the paired protocol of Section 5: within each (seed, ρ) cell, the methods share partitions, sampled coordinates, and noisy sparse observations. Method orderings in this section refer to the reported mean metrics across the paired seeds.

6.1 Cross-well low-data results

Three questions organize the results. Where does conditional latent refinement improve on the strongest direct autoencoder in the main grid, and does that ordering persist across additional seeds and measurement realizations? What do the structural ablations reveal about the conditional prior and sparse-measurement refinement? When do interpolation, smoothing, and their lower computational cost make latent optimization unnecessary?

Table 4 compares CLP-CSGM Ridge with AE $[\boldsymbol{u}, \boldsymbol{b}]$, the direct baseline with the lowest unweighted mean RMSE across the grid. CLP-CSGM Ridge has lower mean RMSE in seven of the nine cells; the exceptions are step 8 with $\rho \in \{0.05, 0.10\}$. It has lower mean RMSE at every tested ratio for steps 16 and 32. Across the three tested step sizes, the descriptive RMSE gap shifts progressively in favor of CLP-CSGM as training support decreases. At step 32, the RMSE reductions relative to AE are approximately 32% at $\rho = 0.10$ and 57% at $\rho = 0.20$.

Across the complete grid, the unweighted mean of the nine cell-level mean RMSE values decreases from 0.03496 for AE $[\boldsymbol{u}, \boldsymbol{b}]$ to 0.02862 for CLP-CSGM Ridge, a reduction of approximately 18.1%. The ordering still depends on the amount of training data: the direct AE has lower mean RMSE in the two cells with the largest training pool, step 8 at $\rho = 0.05$ and $\rho = 0.10$.

Table 4: Cross-well V_c comparison between CLP-CSGM Ridge and AE $[\boldsymbol{u}, \boldsymbol{b}]$. Values are RMSE mean and standard deviation across seeds; a negative gap favors CLP-CSGM.

step	n_train_approx	rho	clp_csgm_rmse_mean	clp_csgm_rmse_std	ae_ub_rmse_mean	ae_ub_rmse_std	gap_vs_ae_pct
8	104	0.05000	0.04483	0.00524	0.03567	0.00100	25.69014
8	104	0.10000	0.03646	0.00194	0.03195	0.00208	14.10258
8	104	0.20000	0.02267	0.00381	0.02600	0.00164	-12.83374
16	52	0.05000	0.03344	0.00151	0.03712	0.00098	-9.91538
16	52	0.10000	0.02689	0.00235	0.03420	0.00291	-21.35910
16	52	0.20000	0.01771	0.00138	0.03214	0.00742	-44.89074
32	27	0.05000	0.03217	0.00268	0.03901	0.00259	-17.53826
32	27	0.10000	0.02640	0.00342	0.03894	0.00228	-32.21019
32	27	0.20000	0.01705	0.00119	0.03959	0.00234	-56.94200

Table 5 reports the paired seed-wise difference $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ with Student-t half-widths (df = 2). At steps 16 and 32, where training support is smallest, the mean paired gap is positive at every ρ and CLP-CSGM Ridge wins all three seeds; at step 8, where the development pool is largest, AE $[\boldsymbol{u}, \boldsymbol{b}]$ wins the two smallest ρ cells, matching the mean-RMSE ordering in Table 4. Several of these three-seed intervals include zero. Appendix B.9 repeats the same pairwise comparison on ten seeds. None of the nine per-configuration intervals includes zero, and the cell ordering remains unchanged: CLP-CSGM is lower at steps 16 and 32, whereas AE remains lower at step 8 for $\rho = 0.05$ and $\rho = 0.10$. These intervals describe seed-to-seed variability on F03-4, not independent geological tests, and they are not adjusted as one simultaneous nine-cell test.

Two further checks preserve this pattern. A direct AE supplied with placed measurements and an unscaled binary mask remains higher than CLP-CSGM in seven of nine cells, including all six cells at steps 16 and 32 (Appendix B.10). When the sparse coordinates and measurement noise are redrawn five times with model seeds fixed, the published winner is unchanged in every cell

(Appendix B.11). CLP-CSGM is lower in every draw at steps 16 and 32, while step 8 retains the mixed ordering of the main grid. Because these checks change several sources of variation together and add no held-out wells, they support the within-benchmark ordering without isolating its cause or extending its geological scope. Overlapping windows are not counted as additional independent samples.

Appendix B.12 compares those same nine cells with ordinary least squares and ridge regression from $[\boldsymbol{u}, \boldsymbol{b}]$, and with linear interpolation of the sparse measurements along depth. The linear maps are scored at the published noise standard deviation 0.01. At that reference level, interpolation has the lowest full-window RMSE in all nine cells and in all 27 seed comparisons. CLP-CSGM Ridge is lower than both direct linear maps at step 16, $\rho = 0.20$, and at step 32 for every ρ.

Figure 6 and Table 20 keep the evaluation windows and observation coordinates fixed, while varying the added Gaussian noise and reselecting the regularization weight by validation at each noise level. At the reference noise level, $\sigma_\eta = 0.01$, depth interpolation achieves lower mean RMSE than CLP-CSGM in all nine configurations. As the simulated perturbation increases, the relative RMSE comparison becomes more favorable to CLP-CSGM: it has lower RMSE than interpolation in seven of the nine configurations at $\sigma_\eta = 0.05$ and in all nine at $\sigma_\eta = 0.10$ and 0.20. The same progression appears against the validation-selected second-difference smoother in Appendix B.12. Smoothing is lower in eight of nine cells at the reference level, whereas CLP-CSGM is lower in five cells at $\sigma_\eta = 0.05$, eight at 0.10, and all nine at 0.20. The smoother selects the largest weight in 62 of 108 seed rows; its affine infinite-penalty limit is reported separately.

This behavior is consistent with the role of regularization in balancing agreement with noisy observations against prior information Hansen [2010]. Interpolation passes through the perturbed observations, whereas CLP-CSGM allows measurement residuals, restricts reconstruction to the decoder range, and penalizes departures from the conditional latent prediction. These mechanisms can limit the response to measurement perturbations, but can also introduce bias when the learned representation or conditional prediction does not describe the target accurately. Related generative compressed-sensing theory explicitly accounts for representation error, measurement noise, and optimization error under suitable assumptions Bora et al. [2017]. That theory provides context for the observed relative robustness; the sweep itself does not separate the effects of decoder restriction, conditional penalization, and validation-based parameter selection.

The held-out target curve has a standard deviation of approximately 0.035, so the added levels 0.05, 0.10, and 0.20 are about 1.4, 2.9, and 5.7 times that value. They define a controlled synthetic sensitivity analysis, not calibrated field uncertainty, and the added noise is unrelated to the 30dB filename label. AE $[\boldsymbol{u}, \boldsymbol{b}]$, ordinary least squares, and direct ridge were not rerun across this sweep. The comparison therefore establishes a change in relative full-window RMSE against interpolation and smoothing within this protocol, not a general noise threshold or recovery of local peaks.

The decoder and the dense-log map are fit once per step and seed. The penalty weight is then selected on validation for each measurement fraction and each noise level, so the curves are not a fixed-penalty reconstruction. The sensitivity sweep uses a different, wider grid. It omits the factor-of-three candidates of the published grid and adds 1, 10, and 100. The weight selected at 0.01 and at 0.05 also belongs to the published grid. That membership does not mean the two searches were the same. At noise 0.10, step 16 and $\rho = 0.10$, one seed selects the weight 1, above the published grid and still inside the sweep. At noise 0.20, some cells select 100, the largest candidate in the sweep grid. That choice is the best value among the candidates that were tried, not an optimum outside the grid. The points at noise 0.01 belong to this same refit. The ordering between interpolation and CLP-CSGM Ridge matches the published comparison at that noise, while a cell mean can differ slightly from the published CLP-CSGM value. At step 16 and $\rho = 0.20$, the published mean is 0.01771 and the refit mean is 0.0159. These runs are not merged into the submitted tables.

The accuracy gains must also be weighed against iterative inference. On this cross-well grid, Table 22 reports 0.23517 s per test window for CLP-CSGM Ridge and 0.03156 s for AE $[\boldsymbol{u}, \boldsymbol{b}]$; the separately normalized runtime diagnostic is 9.79 and is not the quotient of those averages. The extra optimization is most relevant where its accuracy gain exceeds what can be obtained from the simpler comparator available for the same observations.

Figure 7 places the pairwise result among the five direct and latent methods of the main grid. The upper row keeps MLP and PCA on the same axis as the lower-RMSE methods. The lower row enlarges ML only, AE $[\boldsymbol{u}, \boldsymbol{b}]$, and CLP-CSGM Ridge. That enlargement is not a ranking against interpolation or smoothing. AE has the lowest unweighted mean RMSE among the direct baselines MLP, PCA, and AE. Table 4 therefore focuses on the most competitive direct comparator in this five-method grid.

Figure 8 shows one held-out F03-4 profile for step 16, seed 7, and $\rho = 0.20$. The four panels share

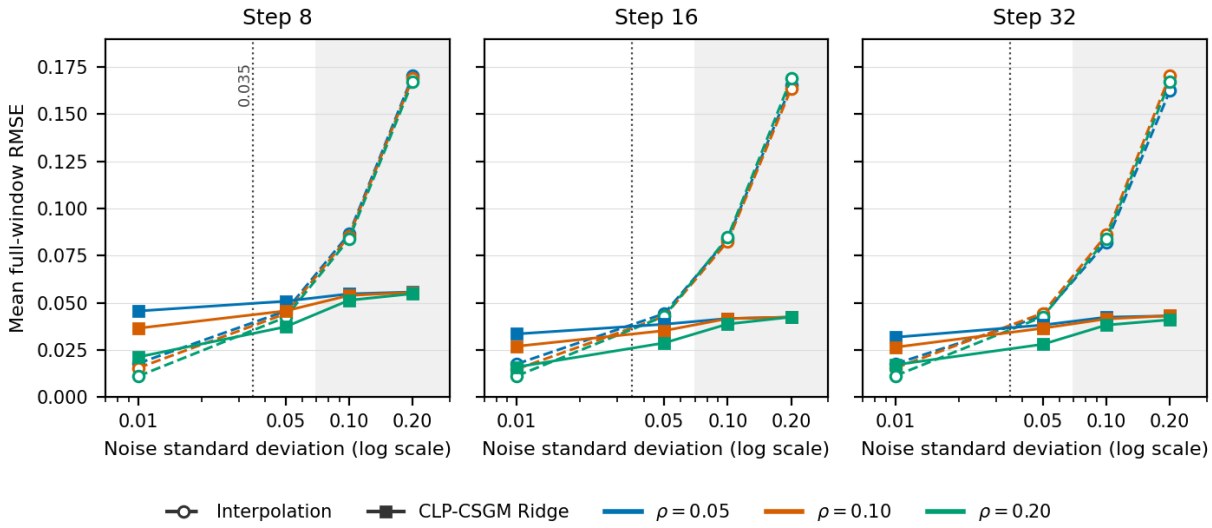


Figure 6: Mean full-window RMSE of depth interpolation and CLP-CSGM Ridge. The horizontal axis is logarithmic. Each panel is one step and each measurement fraction is a separate curve. Open circles are interpolation and squares are CLP-CSGM Ridge. Values are means over seeds $\{7, 23, 41\}$. The dotted line is the held-out V_c standard deviation 0.035, on the same scale as the noise. Noise 0.05 is about 1.43 times that deviation. The shaded region marks 0.10 and 0.20, about 2.86 and 5.71 times that deviation. These noise levels were used for sensitivity analysis and were not calibrated to field measurement uncertainties. The decoder and the dense-log map are fit once per step and seed. The penalty weight is selected on validation for each measurement fraction and each noise level. The points at noise 0.01 belong to this refit. Cell means and seed-level win counts are reported in Table 20.

Table 5: Paired seed-level $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on the cross-well V_c grid ($n = 3$ seeds). SE is the seed-level standard error; the 95% half-width uses Student-t with df = 2. Positive mean ΔRMSE favors CLP-CSGM Ridge. Wins count seeds with strictly lower RMSE for each method.

step	ρ	mean ΔRMSE	SE	t_2 95% half-width	CLP wins	AE wins
8	0.05	-0.00916	0.00350	0.01507	0	3
8	0.10	-0.00451	0.00231	0.00992	1	2
8	0.20	0.00334	0.00200	0.00860	2	1
16	0.05	0.00368	0.00123	0.00531	3	0
16	0.10	0.00730	0.00187	0.00803	3	0
16	0.20	0.01443	0.00362	0.01556	3	0
32	0.05	0.00684	0.00120	0.00515	3	0
32	0.10	0.01254	0.00216	0.00930	3	0
32	0.20	0.02254	0.00202	0.00869	3	0

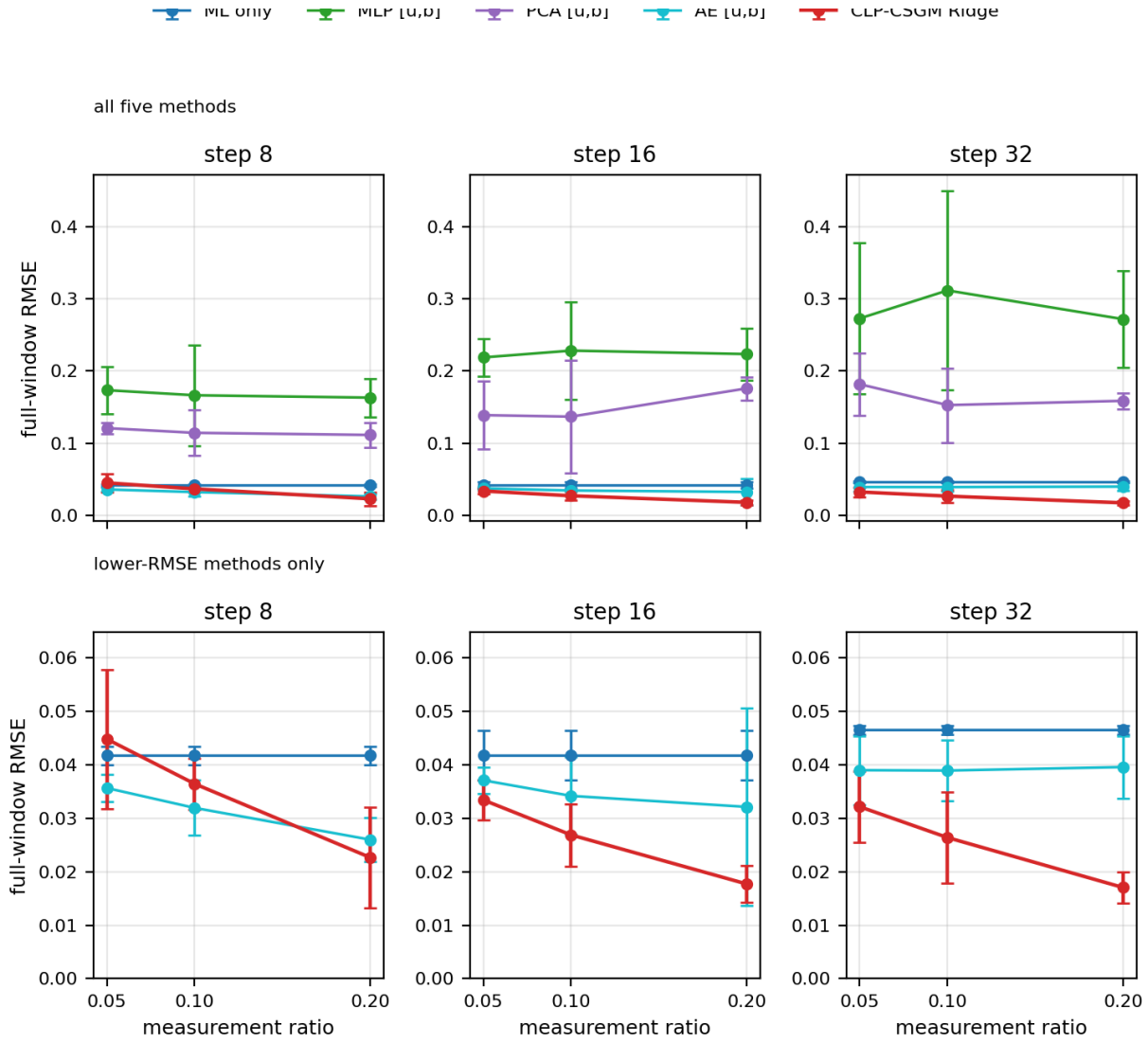


Figure 7: Cross-well V_c full-window RMSE for five methods. The upper row shows ML only, MLP $[\boldsymbol{u}, \boldsymbol{b}]$, PCA $[\boldsymbol{u}, \boldsymbol{b}]$, AE $[\boldsymbol{u}, \boldsymbol{b}]$, and CLP-CSGM Ridge. The lower row uses a smaller vertical range for ML only, AE, and CLP-CSGM Ridge. Error bars are Student-t 95% half-widths across seeds (df = 2). This figure does not include depth interpolation or second-difference smoothing.

one V_c scale and one depth axis, increasing downward. In this displayed example, CLP-CSGM Ridge follows the broad variations of the processed reference more closely than AE $[\boldsymbol{u}, \boldsymbol{b}]$. ML only stays in a narrower band. MLP and PCA are omitted here because their range is much wider; Fig. 24 shows those two predictions on their full horizontal range for the same interval. The individual window in Fig. 15 is a different configuration: step 16, seed 7, $\rho = 0.10$, and window index 0. There, CLP-CSGM Ridge is closer to the reference, the autoencoder prediction is positively offset, and local deviations remain. These examples illustrate differences in reconstruction behavior within the displayed configurations and are consistent with their aggregate RMSE. Their role is qualitative; the paired aggregate metrics, rather than individual windows, determine the benchmark ordering.

The paired step-16 checks in Appendix B.7 (Table 14) show that including gamma ray reduces absolute error and that the embargo changes training support, but neither modification reverses the ordering of CLP-CSGM Ridge against AE $[\boldsymbol{u}, \boldsymbol{b}]$ and measurement-only CSGM at the tested ratios.

6.2 Single-well F03 GR-only results

The F03-4 benchmark uses gamma ray as the only dense log. Table 6 shows that CLP-CSGM Ridge achieves mean RMSE 0.01390 across the tested grid, compared with 0.01434 for AE $[\boldsymbol{u}, \boldsymbol{b}]$. CLP-CSGM Ridge has lower mean RMSE at $\rho \in \{0.20, 0.30\}$, the methods are nearly tied at $\rho = 0.50$, and AE has lower mean RMSE at $\rho \in \{0.40, 0.60\}$. In this controlled benchmark, the largest observed RMSE advantage of CLP-CSGM Ridge over the direct autoencoder occurs at the sparsest ratios. Direct AE fusion becomes competitive as more entries of $\boldsymbol{b}$ are available.

Table 6: Single-well F03-4 GR-only porosity benchmark: mean RMSE by measurement ratio ρ. The prior-only and measurement-only rows are structural reference variants.

method	$\rho = 0.20$	$\rho = 0.30$	$\rho = 0.40$	$\rho = 0.50$	$\rho = 0.60$
CLP-CSGM Ridge	0.01596	0.01475	0.01377	0.01281	0.01220
AE [u,b]	0.01850	0.01554	0.01304	0.01286	0.01178
Prior-only Ridge	0.01958	0.01958	0.01958	0.01958	0.01958
Measurement-only CSGM	0.02519	0.02285	0.02033	0.01622	0.01494
ML only	0.02359	0.02359	0.02359	0.02359	0.02359
PCA [u,b]	0.02149	0.02187	0.02261	0.02409	0.02426
MLP [u,b]	0.03022	0.02997	0.03712	0.04199	0.04627

Figure 9 displays the complete comparison with seed-level Student-t 95% half-widths (df = 2). Table 7 reports the paired $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on the same three seeds: the mean gap is positive at $\rho = 0.20$ and $\rho = 0.30$ (CLP wins all seeds), near zero at $\rho = 0.50$, and negative

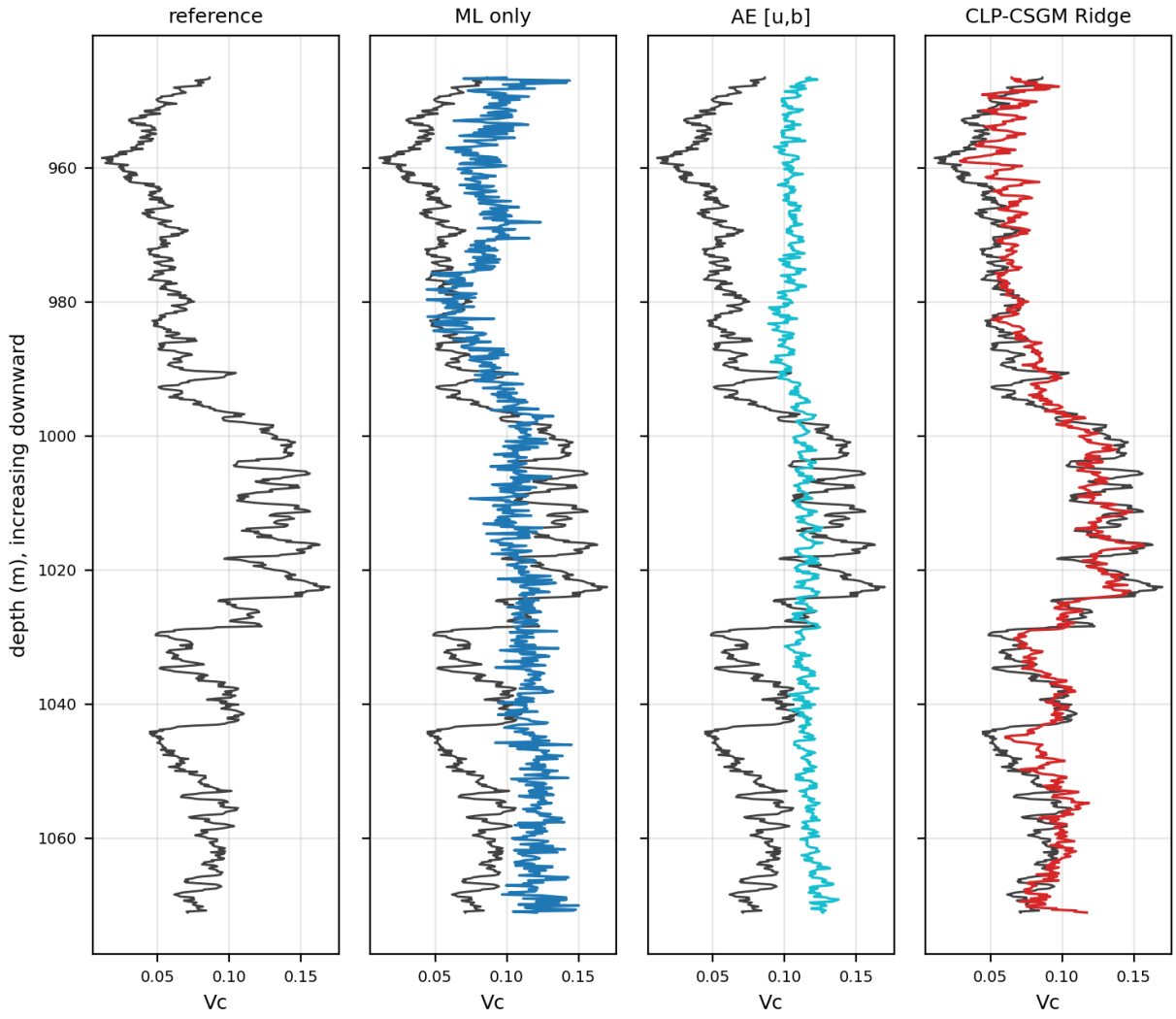


Figure 8: Held-out F03-4 V_c reconstruction at step 16, seed 7, and $\rho = 0.20$. The target is the processed Larionov-derived curve, and the sparse observations are coordinate subsamples of that curve. All four panels use the same horizontal V_c scale and the same depth axis, increasing downward. MLP and PCA are shown separately in Fig. 24.

at $\rho \in \{0.40, 0.60\}$ (AE wins all seeds). With $n = 3$, several paired intervals include zero. The F03 ordering therefore changes with how many depths are measured, and the three-seed intervals are not a large-sample significance claim. RMSE variability, MAE, and relative ℓ_2 error are reported in Appendix B.2 (Table 13) and Appendix B.3 (Figs. 12–13); these metrics retain the same regime-dependent interpretation.

Table 7: Paired seed-level $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on the F03-4 GR-only porosity benchmark ($n = 3$ seeds). SE is the seed-level standard error; the 95% half-width uses Student-t with df = 2. Positive mean ΔRMSE favors CLP-CSGM Ridge.

ρ	mean ΔRMSE	SE	t_2 95% half-width	CLP wins	AE wins
0.20	0.00254	0.00042	0.00180	3	0
0.30	0.00079	0.00032	0.00139	3	0
0.40	-0.00073	0.00035	0.00151	0	3
0.50	0.00006	0.00033	0.00144	2	1
0.60	-0.00041	0.00024	0.00103	0	3

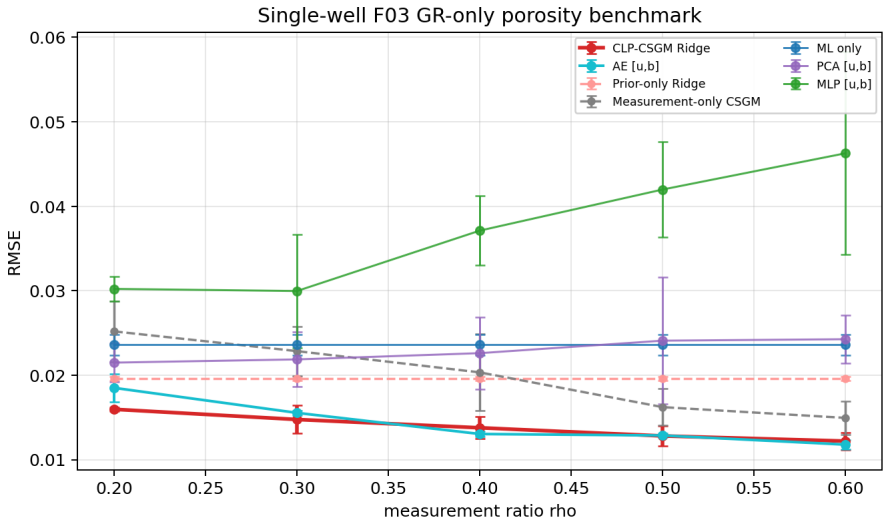


Figure 9: F03-4 GR-only porosity RMSE by measurement ratio. CLP-CSGM Ridge has the lowest mean RMSE at the sparsest ratios, while AE $[\boldsymbol{u}, \boldsymbol{b}]$ becomes competitive as more target observations are available. Error bars are Student-t 95% half-widths across seeds (df = 2).

The stress tests in Appendix C (Table 23) extend the sparse grid to $\rho \in \{0.05, 0.10, 0.20\}$. CLP-CSGM Ridge retains lower mean RMSE than AE under increased noise and a smaller training pool, while the ordering becomes less stable with a coarse sliding step and fewer test windows.

Figure 10 shows the depth-profile reconstruction at $\rho = 0.20$, the main-grid ratio at which CLP-CSGM Ridge has its largest mean RMSE advantage over the direct AE baseline.

The porosity example is more mixed than the clay-volume profile. In this displayed profile, CLP-

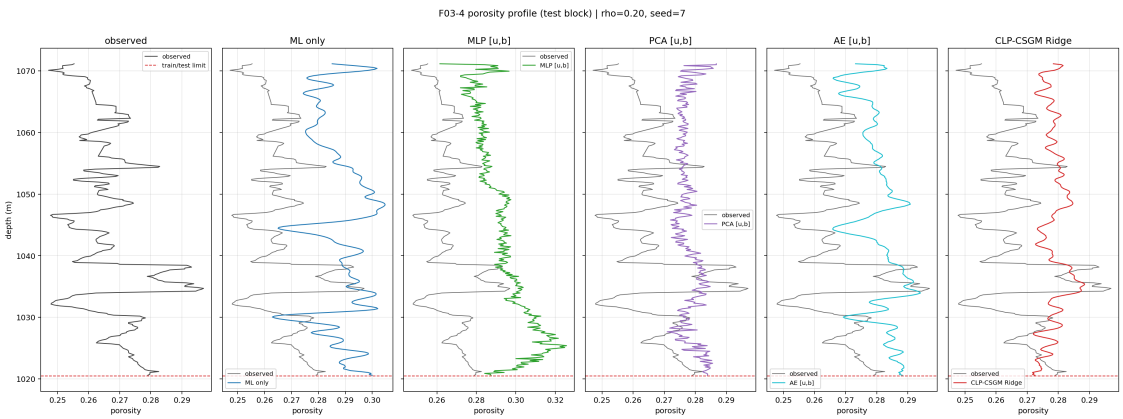


Figure 10: F03-4 depth-profile reconstruction at $\rho = 0.20$. The target is the processed density-derived porosity curve, and the sparse observations are coordinate subsamples of that curve. In this displayed profile, CLP-CSGM Ridge follows parts of the broad trend and smooths local variations. The panels do not share one horizontal scale.

CSGM Ridge follows parts of the broad trend and smooths local variations. Because the profile combines predictions from overlapping windows, the view is qualitative rather than an independent pointwise test. The displayed window in Fig. 14 uses the corresponding configuration seed 7, $\rho = 0.20$, and window index 0. There, both CLP-CSGM Ridge and AE $[\boldsymbol{u}, \boldsymbol{b}]$ miss the central depression of the reference. The pooled parity plot in Fig. 17 prints a correlation of 0.4632 for CLP-CSGM Ridge and 0.3720 for the autoencoder. Those values describe the points drawn on that figure, and overlapping-window predictions are not independent observations. With $m = 13$ observations per length-64 window, CLP-CSGM has mean RMSE of approximately 0.016 at $\rho = 0.20$, consistent with Table 6. Across the aggregate results, its advantage over the direct autoencoder remains concentrated at the sparse end of this controlled GR-only benchmark.

The limited-sample Lapa permeability case in Appendix D provides supplementary evidence. The prior-guided CLP-CSGM variants have lower mean RMSE than the direct baselines that use $[\boldsymbol{u}, \boldsymbol{b}]$ across the tested grid. Their comparison with measurement-only CSGM is ratio-dependent.

Appendix E scores laboratory porosity on depth-held-out blocks of one cored interval, using coordinates not supplied as measurements within each evaluated window. CLP-CSGM is not superior on the mean: the direct AE has the lower RMSE at $\rho = 0.25$ and $\rho = 0.50$ (0.0537 and 0.0523, against 0.0602 and 0.0580 for CLP-CSGM Ridge). CLP-CSGM Ridge has lower RMSE than the log-only anchor in every block–seed pair and has the lower block-mean RMSE on the two shallower blocks. On the deepest block the auxiliary logs change, the decoded prediction $G(\boldsymbol{z}_0)$ differs sub-

Table 8: Cross-well V_c ablation results. Gains are relative RMSE reductions of CLP-CSGM Ridge with respect to the corresponding structural ablation; negative gain indicates that measurement-only CSGM has lower mean RMSE.

Step	ρ	CLP-CSGM	Prior-only	Meas.-only	AE $[u, b]$	Gain vs prior (%)	Gain vs meas. (%)
8	0.0500	0.0448	0.0564	0.0368	0.0357	20.4913	-21.7274
8	0.1000	0.0365	0.0564	0.0297	0.0320	35.3433	-22.7163
8	0.2000	0.0227	0.0564	0.0251	0.0260	59.8016	9.5581
16	0.0500	0.0334	0.0426	0.0362	0.0371	21.5310	7.6727
16	0.1000	0.0269	0.0426	0.0295	0.0342	36.8933	8.9585
16	0.2000	0.0177	0.0426	0.0213	0.0321	58.4391	16.7449
32	0.0500	0.0322	0.0432	0.0341	0.0390	25.5774	5.6452
32	0.1000	0.0264	0.0432	0.0271	0.0389	38.9248	2.4363
32	0.2000	0.0170	0.0432	0.0219	0.0396	60.5584	22.0073

stantially from the target windows, and the direct AE remains lower. These block differences are compatible with sensitivity to prior mismatch, but that mechanism is not causally isolated here. The block RMSE, the log summaries, and the split counts are in Appendix E.

6.3 Ablation analysis

The structural ablations drop one term at a time. Prior-only returns the decoded prediction of the conditional prior, $G(h(\boldsymbol{u}))$, with no latent search. Measurement-only sets $\lambda = 0$ and searches the decoder range using only the sparse measurements [Bora et al., 2017]. The comparison shows what each term contributes (Remark 1).

In the cross-well benchmark, CLP-CSGM Ridge has lower mean RMSE than prior-only in all nine cells, with reductions from 20.5% to 60.6% (Table 8). It also has lower mean RMSE than measurement-only CSGM in seven cells. The exceptions are step 8 with $\rho \in \{0.05, 0.10\}$, where measurement-only CSGM has lower mean RMSE and the Ridge code $\boldsymbol{z}_0$ is not beneficial. At steps 16 and 32, the full model has lower mean RMSE than both ablations at every tested ratio.

In the F03-4 GR-only benchmark, CLP-CSGM Ridge has lower mean RMSE than both ablations at every tested ratio (Table 9). Its reduction relative to prior-only increases from approximately 18.5% at $\rho = 0.20$ to 37.7% at $\rho = 0.60$, while its largest gains over measurement-only occur at the sparsest ratios. Both terms contribute. The cross-well exceptions show that the conditional code $\boldsymbol{z}_0$ is not helpful in every cell.

Table 9: Single-well F03-4 GR-only ablation results. Gains are relative RMSE reductions of CLP-CSGM Ridge with respect to the corresponding structural ablation.

ρ	CLP-CSGM	Prior-only	Meas.-only	AE $[u, b]$	Gain vs prior (%)	Gain vs meas. (%)
0.2000	0.0160	0.0196	0.0252	0.0185	18.4646	36.6254
0.3000	0.0148	0.0196	0.0228	0.0155	24.6433	35.4375
0.4000	0.0138	0.0196	0.0203	0.0130	29.6514	32.2641
0.5000	0.0128	0.0196	0.0162	0.0129	34.5823	21.0300
0.6000	0.0122	0.0196	0.0149	0.0118	37.6999	18.3817

Table 10: Cross-well V_c latent-prior sensitivity. Positive signed Ridge gain favors Ridge, whereas negative values favor MLP.

Step	ρ	CLP-CSGM MLP	CLP-CSGM Ridge	Ridge gain (%)
8	0.0500	0.0338	0.0448	-32.7769
8	0.1000	0.0286	0.0365	-27.4065
8	0.2000	0.0198	0.0227	-14.4546
16	0.0500	0.0296	0.0334	-12.9037
16	0.1000	0.0247	0.0269	-8.8053
16	0.2000	0.0162	0.0177	-9.2911
32	0.0500	0.0309	0.0322	-3.9736
32	0.1000	0.0270	0.0264	2.1090
32	0.2000	0.0178	0.0170	4.3457

6.4 Latent-prior model sensitivity

Tables 10 and 11 compare Ridge and MLP implementations of h. This sensitivity analysis is limited to these two prior classes. Signed Ridge gain is defined as $100(\mathrm{RMSE}_{\mathrm{MLP}} - \mathrm{RMSE}_{\mathrm{Ridge}})/\mathrm{RMSE}_{\mathrm{MLP}}$; positive values favor Ridge.

In the controlled cross-well comparison, both prior architectures use the same auxiliary channels, excluding gamma ray, and the same training, validation, test, and sparse-observation protocol. The MLP prior achieves lower mean RMSE in seven of the nine configurations. Ridge has lower mean RMSE at step 32 for measurement ratios 0.10 and 0.20. The unweighted mean RMSE across the nine configurations decreases from 0.02862 for Ridge to 0.02538 for MLP, a relative reduction of 11.3%. The MLP advantage becomes smaller as training support decreases and is nearly absent at step 32. Ridge remains the headline comparator because it is the fixed linear reference defined in Section 5.5, not because it dominates MLP on this grid.

In the controlled F03-4 GR-only comparison, both prior architectures use the same depth-first partition and the same 224 test windows. Ridge achieves lower mean RMSE at all five measurement ratios, with relative reductions of approximately 15.6% to 19.7% compared with MLP. The prior

Table 11: Single-well F03-4 GR-only latent-prior sensitivity. Signed Ridge gain follows the definition used in Table 10.

ρ	CLP-CSGM MLP	CLP-CSGM Ridge	Ridge gain (%)
0.2000	0.0190	0.0160	15.9713
0.3000	0.0177	0.0148	16.4356
0.4000	0.0163	0.0138	15.5648
0.5000	0.0160	0.0128	19.7190
0.6000	0.0151	0.0122	19.4423

architecture therefore affects reconstruction accuracy, but the preferred configuration depends on the evaluated data regime.

Taken together, the rankings show that the preferred map depends on the dataset and measurement regime. Its class should be selected on validation data for the intended test rather than after observing the held-out result.

Supplementary diagnostics. Appendix B.6 audits validation, selected λ, and per-window errors; Appendix B.4 shows realized sparse-measurement windows; Appendix B.1 assesses empirical decoder-range separation; and Appendix B.14 reports inference cost. These analyses serve interpretive and reproducibility purposes and are separate from the benchmark comparisons.

7 Interpretation, scope, and limitations

The methodological distinction of CLP-CSGM is not the latent representation alone, but the way dense context and sparse target measurements enter inference. Dense logs determine the anchor $\boldsymbol{z}_0 = h(\boldsymbol{u})$, while the sparse values remain in an explicit measurement residual during latent refinement. This separates the method from direct $[\boldsymbol{u}, \boldsymbol{b}]$ prediction and from approaches that complete a missing log interval, translate one property image to another, or regress a target from the available logs [Koeshidayatullah et al., 2026, Al-Fakih et al., 2026, 2025, 2023]. The decoder range determines what can be represented; the anchor guides the search within that range; and the observations can correct the conditioned prediction without being fitted exactly.

The ablations provide the clearest internal evidence for this design. In the cross-well grid, the full Ridge model has lower mean RMSE than prior-only in all nine cells and lower mean RMSE than measurement-only CSGM in seven. At steps 16 and 32 it is lower than both ablations at every tested ratio, whereas measurement-only inversion is lower in the two step-8 exceptions. On the F03-

4 GR-only grid, the full model is lower than both ablations at all five ratios. Sparse-measurement refinement therefore improves the conditioned prediction consistently in these experiments, while the value of the conditional anchor depends on the configuration.

The external comparisons identify where that internal gain translates into better prediction. CLP-CSGM Ridge is lower than the direct AE in seven of nine cross-well cells, including all cells at steps 16 and 32. The additional seed and measurement-realization checks preserve that ordering within the same held-out well. Because the sliding step also changes window sampling and overlap, the result supports the complete procedure in those limited-support configurations rather than a causal effect of training-set size alone. The depth profiles complement this aggregate evidence: the multi-channel model follows the broad V_c variation in the displayed cross-well interval, while the GR-only porosity reconstruction is smoother and misses some local structure. Attainable detail depends jointly on the dense-log information, the decoder range, and the number and placement of sparse observations.

The class of the conditional map also matters. Under matched channels and splits, the MLP prior is lower than Ridge in seven of nine cross-well cells and reduces the unweighted mean RMSE from 0.02862 to 0.02538 (11.3%). Its advantage narrows as training support decreases, and Ridge is lower at step 32 for $\rho = 0.10$ and $\rho = 0.20$. The ordering reverses on the controlled F03-4 grid, where Ridge is lower at all five ratios. Ridge remains the primary reference because it supplies one fixed linear low-capacity map across the main grid, not because it is uniformly preferable. The class of h, latent dimension, and regularization should consequently be chosen on validation data for the intended measurements; the experiments do not isolate model capacity as the cause of the observed ranking.

Simple reconstruction provides an important counterpoint. At the reference noise standard deviation 0.01, depth interpolation is lower than CLP-CSGM in all nine cross-well cells, and the validation-selected second-difference smoother is lower in eight. In this regime the sparse observations already support lower-RMSE depth-domain reconstruction in this benchmark, so iterative latent refinement adds cost without improving RMSE. As the synthetic perturbation increases, CLP-CSGM becomes lower than interpolation in seven cells at 0.05 and all nine at 0.10 and 0.20; against smoothing, the corresponding counts are five, eight, and nine. This pattern is relative robustness of the complete procedure, with λ reselected by validation at each level. Decoder restriction, condi-

tional penalization, and parameter reselection are all compatible with the behavior, but the sweep does not isolate their individual effects.

The larger perturbations are sensitivity levels rather than estimates of field uncertainty. An instrument-aware analysis would require gamma-ray and endpoint uncertainties to be propagated through the nonlinear Larionov relation. Moreover, AE, ordinary least squares, and direct ridge were not rerun across the sweep. The result therefore concerns the reported comparisons with interpolation and smoothing, not a universal noise threshold or general superiority under noisy measurements. Computational cost reinforces this conditional interpretation: CLP-CSGM uses iterative inference, whereas direct predictors require one forward pass. Its additional cost is most relevant when explicit sparse-measurement refinement yields a meaningful accuracy gain; faster solvers, adaptive stopping, and amortized refinement remain useful directions.

The benchmark provenance defines a second boundary on interpretation. Cross-well V_c is a processed Larionov-derived curve, so gamma ray is withheld from the main input to avoid supplying the source of the target. When gamma ray is available, the conventional relation can be evaluated directly. Restoring that channel lowers absolute error without changing the step-16 ordering among CLP-CSGM Ridge, direct AE, and measurement-only CSGM. The F03-4 porosity target is likewise processed and density-derived; sonic is excluded because it follows that curve almost one to one, and gamma ray is the only dense input. In both F3 settings, sparse observations are subsamples of the processed target rather than independently acquired property measurements. The experiments therefore test reconstruction under deliberately restricted information, not replacement of the Larionov or porosity relations in a fully logged well.

A field analogue would combine a missing or unreliable source log with a few independently acquired target measurements, but the present benchmarks reproduce only its restricted-input and sparse-observation structure. The Lapa permeability case [Alvarez et al., 2025] extends the evaluation to another lithology and target property with a small sample. The laboratory porosity supplement uses a target not deterministically derived from the auxiliary logs by this pipeline: CLP-CSGM improves on the decoded conditional prior in every block–seed pair, while direct AE has the lower aggregate mean RMSE at both ratios. One cored interval, within-window withheld coordinates, and overlapping training windows limit the scope of that evidence. Broader transfer assessment, such as the cross-basin setting studied by Koeshidayatullah et al. [2026], will require

additional wells, log suites, independently calibrated targets, and geological labels.

The available metrics are not stratified by lithofacies, texture, or formal stratigraphic interval because independent labels are unavailable at each depth sample. RMSE, MAE, and relative ℓ_2 error consequently aggregate heterogeneous windows, while the depth profiles provide qualitative rather than facies-resolved audits. Such resolution would require independently interpreted or core-calibrated interval labels, particularly because decoder adequacy and prior accuracy may vary across geological regimes.

Finally, the theoretical statements retain their conditional scope. Proposition 1 separates decoder mismatch, latent-prior mismatch, measurement noise, optimization quality, and local separability, but assumes rather than proves a local S-REC condition for the trained decoder. The sampled diagnostics in Appendix B.1 therefore support interpretation of the decoder outputs without providing an operator-level recovery guarantee.

8 Conclusion

CLP-CSGM combines a learned target-window representation with two sources of information that retain distinct roles at inference: dense auxiliary logs predict a latent anchor, and sparse target observations refine that prediction through an explicit measurement residual. The resulting MAP/Tikhonov interpretation clarifies the balance between the two terms, while the local error bound separates decoder mismatch, prior mismatch, measurement noise, and local separability under stated conditions.

In the cross-well F3 benchmark, CLP-CSGM Ridge achieves lower mean RMSE than the direct autoencoder baseline AE $[\boldsymbol{u}, \boldsymbol{b}]$ in seven of nine configurations, including every tested ratio at steps 16 and 32. Across the grid, the unweighted mean decreases from 0.03496 to 0.02862, an 18.1% reduction. Because the sliding step also changes window sampling and overlap, this result characterizes the complete procedure in the evaluated limited-support configurations rather than a causal effect of training-set size alone. The ablations show more directly what refinement contributes: the full model is lower than prior-only in all nine cross-well cells and lower than measurement-only CSGM in seven. On the F03-4 GR-only grid, it is lower than both ablations at every ratio, while its ordering against the direct AE changes with the observed fraction. The opposite Ridge–MLP

rankings across the two benchmarks further show that the appropriate conditional map depends on the data regime.

Simple methods define the other side of that result. At the reference noise standard deviation 0.01, depth interpolation is lower than CLP-CSGM in all nine cross-well cells and the validation-selected smoother is lower in eight. When sparse observations already support lower-RMSE depth-domain reconstruction, the additional latent optimization is not justified by RMSE. Under larger synthetic perturbations, CLP-CSGM becomes lower than interpolation in seven cells at 0.05 and all nine at 0.10 and 0.20; against smoothing, the corresponding counts are five, eight, and nine. This is relative robustness of the complete validation-selected procedure, not an instrument-calibrated noise threshold or an isolated effect of the latent penalty.

These findings establish a controlled reconstruction result rather than operational transfer. Cross-well V_c is Larionov-derived from gamma ray, F03-4 porosity is density-derived, and the sparse observations are subsamples of those processed curves. The cross-well comparison uses one held-out well and does not establish operational cross-well or multi-field transferability. In the separate laboratory-porosity supplement, CLP-CSGM improves on its decoded conditional prior in every block–seed pair, but direct AE retains the lower aggregate mean RMSE. The main empirical lesson is therefore conditional: explicit sparse-measurement refinement can improve a dense-log-conditioned latent prediction, especially in the evaluated limited-support configurations, while accurate low-noise observations may favor substantially simpler reconstruction. Testing that lesson operationally will require larger multi-well datasets, independently calibrated targets, geological labels, uncertainty estimates, and faster refinement strategies.

Author Contributions

Romulo Brito da Silva: Conceptualization, Methodology, Software, Formal Analysis, Data Curation, Validation, Visualization, Writing—Original Draft, Writing—Review & Editing.

Diego Passos: Visualization, including figure organization and standardization, Writing—Review & Editing.

Irineu de Azevedo Lima Neto: Visualization, including figure organization and standardization, Writing—Review & Editing.

Herson Oliveira da Rocha: Visualization, including figure organization and standardization,

Writing—Review & Editing.

Roseane Marchezi Missagia: Senior Review, Advisory Support, Writing—Review & Editing.

Marco Antonio Rodrigues de Ceia: Senior Review, Advisory Support, Writing—Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI's ChatGPT to assist with translating author-written text into English, improving linguistic clarity and organization, reducing manuscript length to comply with journal requirements, and preparing selected explanatory schematic figures, as identified in their respective captions.

The authors additionally used xAI's Grok 4.7 to assist with refactoring existing scientific code. The research implementation was not originally generated by AI.

All AI-assisted content and code modifications remained subject to human review and verification. The authors retain full responsibility for the scientific methodology, computational implementation, reproducibility, reported results, and conclusions. Generative AI was not used to fabricate or alter research data or quantitative results, or to determine the scientific conclusions.

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A Supplementary Theoretical Details

This section states the MAP/Tikhonov and existence results together with their proofs, and it gives the conditional local-recovery argument omitted from the main text. That argument separates four contributions to the error: how far the true window is from anything the decoder can produce, how far $\boldsymbol{z}_0$ is from a good code, measurement noise, and whether the measurements can tell nearby windows apart.

A.1 MAP and Tikhonov interpretation

Proposition 2 (MAP and Tikhonov interpretation)**.** *Assume* $p(\boldsymbol{b} \mid \boldsymbol{z}, \boldsymbol{u}) = p(\boldsymbol{b} \mid \boldsymbol{z})$ *and* $\boldsymbol{b} \mid \boldsymbol{z} \sim \mathcal{N}\left(M_\Omega G_{\text{orig}}(\boldsymbol{z}), \sigma_\eta^2 I_m\right)$*, while* $\boldsymbol{z} \mid \boldsymbol{u} \sim \mathcal{N}\left(\boldsymbol{z}_0, \sigma_z^2 I_k\right)$*, where* $\boldsymbol{z}_0 = h(\boldsymbol{u})$*,* $\sigma_\eta > 0$*, and* $\sigma_z > 0$*. Then every MAP estimator satisfies*

$$\widehat{\boldsymbol{z}}_{\text{MAP}} \in \arg\min_{\boldsymbol{z} \in \mathbb{R}^k} \left\{ \left\| D_\Omega \left(M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}} \right) \right\|_2^2 + \lambda \left\| \boldsymbol{z} - \boldsymbol{z}_0 \right\|_2^2 \right\}, \qquad \lambda = \frac{\sigma_\eta^2}{\sigma_z^2}. \tag{18}$$

By Eq. (11)*, this data term is the original-scale residual. The ratio* σ_η^2/σ_z^2 *is the MAP weight on the latent penalty in that objective. The data term keeps coefficient one. The ratio is not the weight of the unweighted standardized residual, because the diagonal entries of* D_y *vary by coordinate. Thus, CLP-CSGM is a Tikhonov-regularized latent inverse problem. Its center is the log-conditioned anchor, not the origin [Tikhonov, 1963, Burger and Lucka, 2014]. The likelihood induces the data-fidelity term, while the conditional prior induces the latent penalty.*

Proof of Proposition 2. The MAP estimate is obtained by maximizing the posterior density

$$p(z \mid b, u).$$

By Bayes' rule,

$$p(z \mid b, u) = \frac{p(b \mid z, u) p(z \mid u)}{p(b \mid u)}.$$

Using the conditional independence assumption $p(b \mid z, u) = p(b \mid z)$, and noting that $p(b \mid u)$ does not depend on z, we obtain

$$p(z \mid b, u) \propto p(b \mid z) p(z \mid u).$$

Therefore,

$$\hat{z}_{\mathrm{MAP}} \in \arg\min_z \left[-\log p(b \mid z) - \log p(z \mid u) \right].$$

Since

$$b \mid z \sim \mathcal{N}\big(M_\Omega(\mu_y + D_y G(z)), \sigma_\eta^2 I_m\big),$$

we have, up to constants independent of z,

$$-\log p(b \mid z) = \frac{1}{2\sigma_\eta^2} \|M_\Omega(\mu_y + D_y G(z)) - b\|_2^2 + \text{const.}$$

Similarly, since

$$z \mid u \sim \mathcal{N}(z_0(u), \sigma_z^2 I_k),$$

we have, up to constants independent of z,

$$-\log p(z \mid u) = \frac{1}{2\sigma_z^2} \|z - z_0(u)\|_2^2 + \text{const.}$$

Hence the MAP estimate minimizes

$$\frac{1}{2\sigma_\eta^2} \|M_\Omega(\mu_y + D_y G(z)) - b\|_2^2 + \frac{1}{2\sigma_z^2} \|z - z_0(u)\|_2^2.$$

Multiplying this objective by the positive constant $2\sigma_\eta^2$ leaves its minimizers unchanged. Thus,

$$\hat{z}_{\mathrm{MAP}} \in \arg\min_z \|M_\Omega(\mu_y + D_y G(z)) - b\|_2^2 + \frac{\sigma_\eta^2}{\sigma_z^2} \|z - z_0(u)\|_2^2.$$

By the identity in Eq. (11), the data term is the weighted standardized residual minimized in Eq. (10). Therefore the CLP-CSGM objective is recovered on the original measurement scale with

$$\lambda = \frac{\sigma_\eta^2}{\sigma_z^2}.$$

The second term is a Tikhonov regularization term centered at the log-conditioned latent anchor $z_0(u)$, not at the origin. □

A.2 Existence of a minimizer

Proposition 3 (Existence of a minimizer)**.** *Assume that $G : \mathbb{R}^k \to \mathbb{R}^N$ is continuous and $\lambda > 0$. For fixed M_Ω, D_Ω, $\widetilde{\boldsymbol{b}} \in \mathbb{R}^m$, and $\boldsymbol{z}_0 \in \mathbb{R}^k$, the objective*

$$\mathcal{J}(\boldsymbol{z}) = \left\| D_\Omega \left(M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}} \right) \right\|_2^2 + \lambda \|\boldsymbol{z} - \boldsymbol{z}_0\|_2^2$$

is continuous and coercive. Consequently,

$$\arg\min_{\boldsymbol{z} \in \mathbb{R}^k} \mathcal{J}(\boldsymbol{z}) \neq \varnothing. \tag{19}$$

Proof of Proposition 3. Since G is continuous, M_Ω is linear, and D_Ω does not depend on z, the map

$$z \mapsto \|D_\Omega(M_\Omega G(z) - \tilde{b})\|_2^2$$

is continuous. The quadratic term

$$z \mapsto \lambda \|z - z_0\|_2^2$$

is also continuous. Hence

$$J(z) = \|D_\Omega(M_\Omega G(z) - \tilde{b})\|_2^2 + \lambda \|z - z_0\|_2^2$$

is continuous on $\mathbb{R}^k$.

Moreover, since $\lambda > 0$,

$$J(z) \geq \lambda \|z - z_0\|_2^2.$$

As $\|z\|_2 \to \infty$, we also have $\|z - z_0\|_2 \to \infty$. Therefore,

$$J(z) \to +\infty \qquad \text{as} \qquad \|z\|_2 \to \infty.$$

Thus J is coercive.

Choose the reference level

$$c = J(z_0),$$

and define the sublevel set

$$S_c = \{z \in \mathbb{R}^k : J(z) \leq c\}.$$

The set S_c is nonempty because $z_0 \in S_c$. It is closed since

$$S_c = J^{-1}((-\infty, c]),$$

J is continuous, and $(-\infty, c]$ is closed in $\mathbb{R}$.

Since J is coercive, there exists $R > 0$ such that

$$\|z\|_2 > R \quad \Longrightarrow \quad J(z) > c.$$

Therefore no point with $\|z\|_2 > R$ belongs to S_c, and hence

$$S_c \subset \overline{B}_R = \{z \in \mathbb{R}^k : \|z\|_2 \leq R\}.$$

Thus S_c is bounded. Since S_c is closed and bounded in the finite-dimensional space $\mathbb{R}^k$, it is compact.

By the Weierstrass theorem, the continuous function J attains its minimum on S_c. Hence there exists $\hat{z} \in S_c$ such that

$$J(\hat{z}) = \min_{z \in S_c} J(z).$$

It remains only to note that this minimum is also global. Indeed, if $z \in S_c$, then

$$J(\hat{z}) \leq J(z)$$

by the definition of $\hat{z}$. If $z \notin S_c$, then $J(z) > c$, while $\hat{z} \in S_c$ implies

$$J(\hat{z}) \leq c.$$

Therefore,

$$J(\hat{z}) \leq c < J(z)$$

for every $z \notin S_c$. Hence no point outside S_c can have objective value smaller than $J(\hat{z})$. Conse-

quently,

$$J(\hat{z}) = \min_{z \in \mathbb{R}^k} J(z),$$

and J admits at least one global minimizer in $\mathbb{R}^k$. □

A.3 Conditional local recovery interpretation

Proposition 1 gives the conditional local recovery interpretation summarized in the main text. The full proof is followed by the corresponding extension for a numerical optimizer.

Proof of Proposition 1. The residual $\|M_\Omega G_{\text{orig}}(z) - b\|_2^2$ is the right-hand side of Eq. (11), so it is the data term of $\mathcal{J}$. By global optimality of $\hat{z}$,

$$\mathcal{J}(\hat{z}) \leq \mathcal{J}(z^\dagger).$$

Since $y^\star = G_{\text{orig}}(z^\dagger) + r^\dagger$, we have

$$b = M_\Omega y^\star + \eta = M_\Omega G_{\text{orig}}(z^\dagger) + M_\Omega r^\dagger + \eta = M_\Omega G_{\text{orig}}(z^\dagger) + e^\dagger.$$

Therefore,

$$\mathcal{J}(z^\dagger) = \|M_\Omega G_{\text{orig}}(z^\dagger) - b\|_2^2 + \lambda \|z^\dagger - z_0\|_2^2 = \|e^\dagger\|_2^2 + \lambda \|z^\dagger - z_0\|_2^2 = (\Delta^\dagger)^2.$$

Hence $\mathcal{J}(\hat{z}) \leq (\Delta^\dagger)^2$. Since each term in $\mathcal{J}(\hat{z})$ is nonnegative,

$$\lambda \|\hat{z} - z_0\|_2^2 \leq \mathcal{J}(\hat{z}) \leq (\Delta^\dagger)^2,$$

which yields

$$\|\hat{z} - z_0\|_2 \leq \frac{\Delta^\dagger}{\sqrt{\lambda}}.$$

Also, by the definition of $\Delta^\dagger$,

$$\Delta^\dagger \geq \sqrt{\lambda} \|z^\dagger - z_0\|_2,$$

so

$$\|z^\dagger - z_0\|_2 \le \frac{\Delta^\dagger}{\sqrt{\lambda}}.$$

Thus both $\hat{z}$ and $z^\dagger$ belong to $B(z_0, R^\dagger)$.

The local S-REC on $\mathcal{R}_{\mathrm{orig},R^\dagger}$ therefore applies to $G_{\mathrm{orig}}(\hat{z})$ and $G_{\mathrm{orig}}(z^\dagger)$, giving

$$\gamma_R \|G_{\mathrm{orig}}(\hat{z}) - G_{\mathrm{orig}}(z^\dagger)\|_2 - \delta_R \le \|M_\Omega(G_{\mathrm{orig}}(\hat{z}) - G_{\mathrm{orig}}(z^\dagger))\|_2.$$

By the triangle inequality,

$$\|M_\Omega(G_{\mathrm{orig}}(\hat{z}) - G_{\mathrm{orig}}(z^\dagger))\|_2 \le \|M_\Omega G_{\mathrm{orig}}(\hat{z}) - b\|_2 + \|b - M_\Omega G_{\mathrm{orig}}(z^\dagger)\|_2.$$

The first term satisfies

$$\|M_\Omega G_{\mathrm{orig}}(\hat{z}) - b\|_2^2 \le \mathcal{J}(\hat{z}) \le (\Delta^\dagger)^2,$$

so $\|M_\Omega G_{\mathrm{orig}}(\hat{z}) - b\|_2 \le \Delta^\dagger$. The second term is $\|e^\dagger\|_2$. Hence

$$\|M_\Omega(G_{\mathrm{orig}}(\hat{z}) - G_{\mathrm{orig}}(z^\dagger))\|_2 \le \Delta^\dagger + \|e^\dagger\|_2.$$

Combining the last two displays gives

$$\|G_{\mathrm{orig}}(\hat{z}) - G_{\mathrm{orig}}(z^\dagger)\|_2 \le \frac{\Delta^\dagger + \|e^\dagger\|_2 + \delta_R}{\gamma_R}.$$

Finally, since $y^\star = G_{\mathrm{orig}}(z^\dagger) + r^\dagger$, a triangle inequality yields

$$\|G_{\mathrm{orig}}(\hat{z}) - y^\star\|_2 \le \|G_{\mathrm{orig}}(\hat{z}) - G_{\mathrm{orig}}(z^\dagger)\|_2 + \|r^\dagger\|_2 \le \|r^\dagger\|_2 + \frac{\Delta^\dagger + \|e^\dagger\|_2 + \delta_R}{\gamma_R}.$$

□

The same reasoning extends to a numerical solution $\hat{z}_{\mathrm{num}}$ produced by an iterative optimizer such as Adam. If

$$\mathcal{J}(\hat{z}_{\mathrm{num}}) \le \mathcal{J}(z^\dagger) + \tau,$$

then one replaces $(\Delta^\dagger)^2$ by $(\Delta^\dagger)^2 + \tau$ throughout the argument, so the bounds hold up to an

additional optimization tolerance.

Coordinate restriction and incoherent secants. The local S-REC assumption above is not automatic for a coordinate-restriction operator. If $A_\Omega = \sqrt{N/m}\, M_\Omega$ denotes the rescaled restriction operator and Ω is sampled uniformly, then for any fixed $v \in \mathbb{R}^N$,

$$\mathbb{E}\|A_\Omega v\|_2^2 = \|v\|_2^2.$$

Thus, random coordinate restriction preserves the energy of a fixed vector in expectation. However, concentration around this expectation requires that the energy of v is not concentrated on a small number of coordinates. A standard sufficient condition is an incoherence bound of the form

$$\|v\|_\infty^2 \le \frac{\mu}{N}\|v\|_2^2.$$

Under such a condition, random coordinate samples concentrate $\|A_\Omega v\|_2^2$ around $\|v\|_2^2$. For a finite set of candidate signals, a union bound extends this concentration to all pairwise secants, provided the number of sampled coordinates is large enough relative to the incoherence level and the logarithm of the number of pairs. For the continuous decoder range, the analogous requirement would need to hold uniformly over the normalized secants

$$\frac{G_{\mathrm{orig}}(z_1) - G_{\mathrm{orig}}(z_2)}{\|G_{\mathrm{orig}}(z_1) - G_{\mathrm{orig}}(z_2)\|_2}, \qquad z_1, z_2 \in B(z_0, R^\dagger).$$

This strong assumption is not verified analytically for the trained decoder. We therefore do not claim that M_Ω satisfies S-REC over the learned decoder range. The local S-REC serves only as a sufficient theoretical lens; Appendix B.1 empirically assesses the practical separability of decoder-generated candidates under coordinate subsampling.

B Supplementary Diagnostics

B.1 Measurement Separability

Table 12 and Fig. 11 provide an empirical check of the separation property that is relevant for generative compressed sensing. For pairs of decoder-generated training windows, each returned to the original target units, we compute

$$r_{ij} = \frac{\sqrt{N/m}\,\|M_\Omega(y_i - y_j)\|_2}{\|y_i - y_j\|_2}.$$

Values close to one indicate that coordinate subsampling preserves pairwise distances after energy normalization. This diagnostic does not prove S-REC over the learned decoder range. It instead examines the corresponding practical question of whether sparse observations distinguish plausible generated windows. The ratios describe separation of G_{orig} windows. They are not the constants γ_R and δ_R. The median ratio remains close to one in all tested cells. Lower percentiles are generally stronger in denser settings, despite some non-monotonic variation at intermediate ρ values. The weakest case is the sparsest cross-well setting, where only four entries are observed per window, consistent with the larger ill-posedness expected at very low measurement ratios.

Table 12: Empirical S-REC-style separation diagnostics on decoder-generated windows. Values are percentiles of the energy-normalized ratio $\sqrt{N/m}\|M_\Omega(y_i - y_j)\|_2/\|y_i - y_j\|_2$.

Dataset	ρ	m	P1	P5	P10	Median
Cross-well V_c	0.05	4	0.320	0.483	0.567	0.963
Cross-well V_c	0.10	6	0.291	0.527	0.646	1.004
Cross-well V_c	0.20	13	0.770	0.882	0.930	1.039
F03-4 GR-only	0.20	13	0.641	0.753	0.815	1.000
F03-4 GR-only	0.40	26	0.600	0.767	0.833	0.999
F03-4 GR-only	0.60	38	0.856	0.902	0.927	0.998

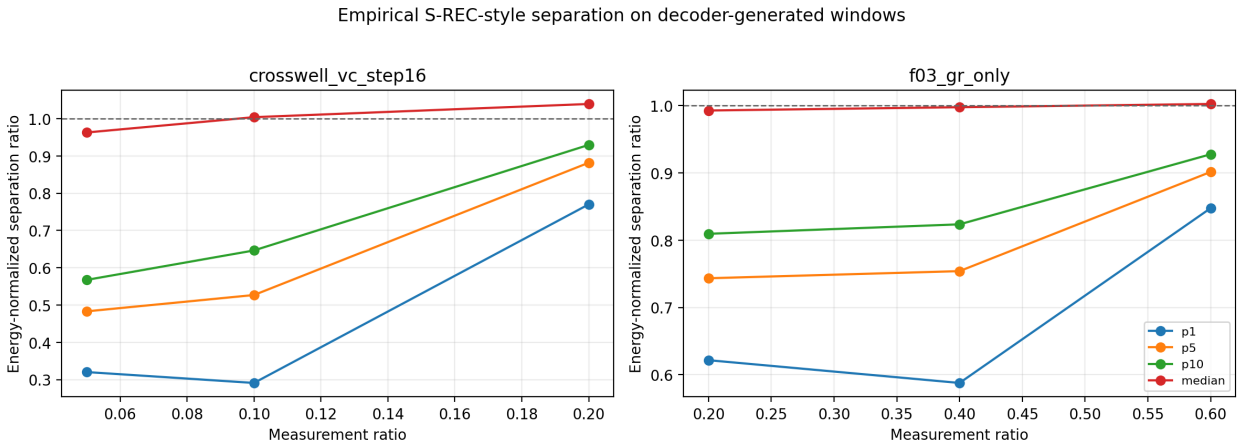


Figure 11: Empirical S-REC-style separation diagnostics. The curves show lower percentiles and medians of the energy-normalized pairwise separation ratio on decoder-generated windows returned to the original target units. Higher lower-percentile values indicate stronger separation of plausible generated candidates under coordinate subsampling.

B.2 Full F03-4 benchmark table

Table 13 reports the complete single-well F03-4 GR-only porosity benchmark used in Section 6.2. In addition to mean RMSE, it includes seed-level RMSE variability (standard deviation and Student-t 95% half-width with df $= 2$), MAE, and relative ℓ_2 error for all seven methods at each tested measurement ratio.

B.3 F03-4 Metric Robustness Views

The F03-4 GR-only discussion in Section 6.2 uses RMSE because the protocol selects and reports models on that scale. Figures 12 and 13 evaluate the same interpretation under alternative error summaries. MAE downweights isolated large deviations relative to RMSE. Relative ℓ_2 error indicates whether the ranking mainly reflects a scale effect. Both plots retain the sparse-regime pattern: CLP-CSGM Ridge is lowest or close to lowest at lower measurement ratios, and AE $[u, b]$ becomes more competitive as more entries of b are available. The degradation of direct MLP $[u, b]$ also appears in more than one metric, consistent with sparse measurements acting as high-variance features when the measurement operator is not handled explicitly.

Table 13: Full single-well F03-4 GR-only porosity benchmark by measurement ratio, including direct baselines, structural reference variants, and multi-metric summaries across seeds. The column rmse_t95_half is the Student-t 95% half-width for the seed mean (df = 2), not a normal $z = 1.96$ interval.

rho	method	rmse	rmse_std	rmse_t95_half	mae	relative_l2
0.20000	CLP-CSGM Ridge	0.01596	0.00010	0.00024	0.01409	0.06000
0.20000	AE [u,b]	0.01850	0.00067	0.00166	0.01619	0.06951
0.20000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.20000	PCA [u,b]	0.02149	0.00093	0.00230	0.01796	0.08073
0.20000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.20000	Measurement-only CSGM	0.02519	0.00145	0.00360	0.01993	0.09459
0.20000	MLP [u,b]	0.03022	0.00059	0.00146	0.02588	0.11356
0.30000	CLP-CSGM Ridge	0.01475	0.00067	0.00166	0.01294	0.05544
0.30000	AE [u,b]	0.01554	0.00011	0.00027	0.01351	0.05837
0.30000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.30000	PCA [u,b]	0.02187	0.00130	0.00323	0.01799	0.08214
0.30000	Measurement-only CSGM	0.02285	0.00117	0.00290	0.01786	0.08578
0.30000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.30000	MLP [u,b]	0.02997	0.00270	0.00670	0.02555	0.11253
0.40000	AE [u,b]	0.01304	0.00013	0.00033	0.01116	0.04897
0.40000	CLP-CSGM Ridge	0.01377	0.00052	0.00129	0.01197	0.05176
0.40000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.40000	Measurement-only CSGM	0.02033	0.00182	0.00452	0.01560	0.07635
0.40000	PCA [u,b]	0.02261	0.00172	0.00427	0.01839	0.08490
0.40000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.40000	MLP [u,b]	0.03712	0.00166	0.00412	0.03244	0.13932
0.50000	CLP-CSGM Ridge	0.01281	0.00048	0.00119	0.01110	0.04813
0.50000	AE [u,b]	0.01286	0.00010	0.00025	0.01107	0.04832
0.50000	Measurement-only CSGM	0.01622	0.00087	0.00217	0.01281	0.06089
0.50000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.50000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.50000	PCA [u,b]	0.02409	0.00301	0.00748	0.01952	0.09050
0.50000	MLP [u,b]	0.04199	0.00227	0.00564	0.03757	0.15764
0.60000	AE [u,b]	0.01178	0.00023	0.00058	0.01003	0.04425
0.60000	CLP-CSGM Ridge	0.01220	0.00041	0.00102	0.01051	0.04583
0.60000	Measurement-only CSGM	0.01494	0.00081	0.00200	0.01176	0.05611
0.60000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.60000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.60000	PCA [u,b]	0.02426	0.00114	0.00284	0.01976	0.09109
0.60000	MLP [u,b]	0.04627	0.00483	0.01199	0.04076	0.17381

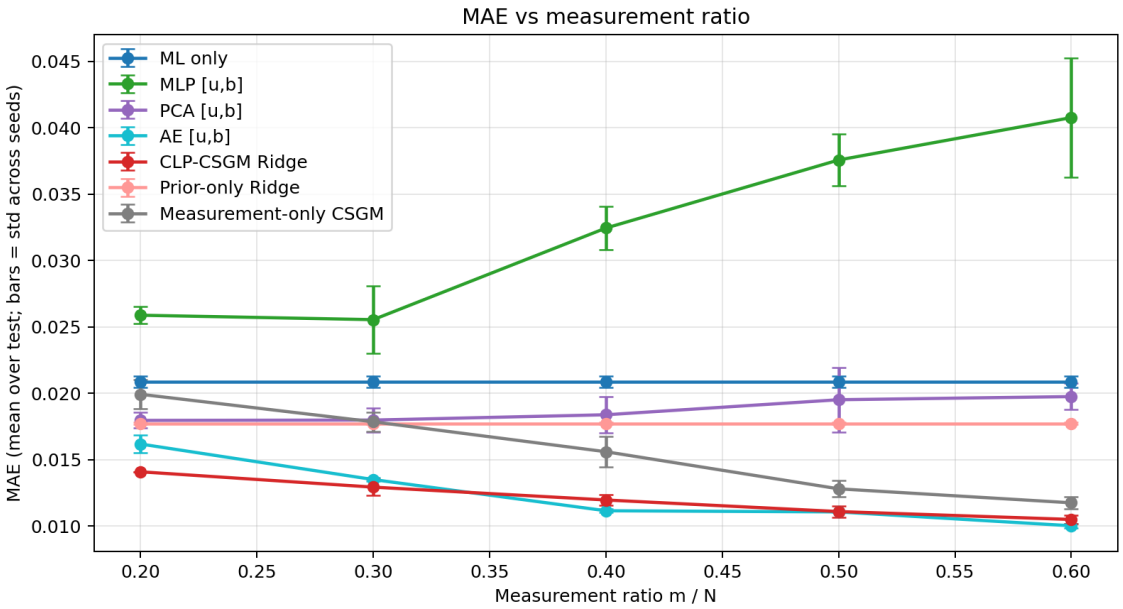


Figure 12: Single-well F03-4 GR-only MAE by measurement ratio. The ranking is consistent with the RMSE view in the main text: CLP-CSGM Ridge has the lowest MAE in the sparse regime and remains competitive as the measurement ratio increases.

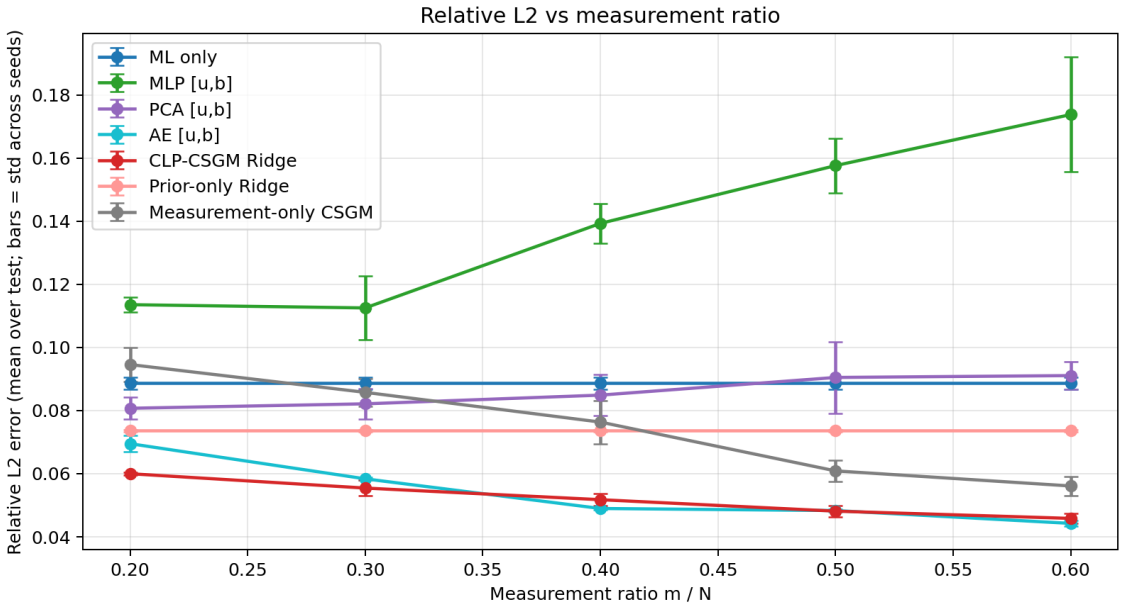


Figure 13: Relative ℓ_2 error on the single-well F03-4 GR-only benchmark. The relative-error view preserves the same conclusion as RMSE and MAE, indicating that the gain is not caused by scale effects alone.

B.4 Illustrative sparse-measurement windows

The main tables and curves in Section 6 pool scalar errors over many test windows. The examples in Figs. 14 and 15 display the target profile, the coordinates where b is observed, and the full-window predictions generated under the paired (seed, ρ) design used in the main experiments, thereby complementing the aggregate comparisons with window-level views.

The horizontal axis indexes coordinates within one length-L window ($L = 64$). The blue curve is the target window y. Black markers show the realized noisy measurements b at the sampled coordinates. The red curve is CLP-CSGM Ridge and the green curve is AE $[u, b]$, when shown. These markers are the actual sparse inputs generated under the benchmark noise model.

Each figure fixes one test-window index, one seed, and one ρ. These audit examples show that the markers alone underdetermine the reconstruction and that the methods interpolate between sparse constraints using different inductive biases. In the displayed porosity window, both CLP-CSGM Ridge and AE $[u, b]$ miss the central depression of the reference. In the displayed clay-volume window, the corresponding configuration is step 16, seed 7, $\rho = 0.10$, and window index 0. CLP-CSGM Ridge is closer to the reference, the autoencoder prediction is positively offset, and local deviations remain. These windows do not establish a uniform advantage. Aggregate tables and the main-text diagnostics provide the overall performance comparisons.

B.5 F03-4 Qualitative Reconstruction Views

This subsection complements the depth-profile reconstruction in the main text (Fig. 10). Because the depth profile follows the contiguous test interval, it provides the clearest qualitative view. The figures below present window-level and pointwise audits.

Figure 16 shows representative F03-4 test windows and reveals local shape behavior compressed by scalar metrics. Direct MLP $[u, b]$ can produce oscillatory predictions, whereas PCA, AE, and CLP-CSGM Ridge are smoother. In these displayed windows, CLP-CSGM Ridge and the autoencoder miss part of the central depression of the reference. Aggregate metrics determine the window-level rankings. Figure 17 pools pointwise predictions against ground truth over the F03-4 test windows. The figure prints a correlation of 0.4632 for CLP-CSGM Ridge and 0.3720 for AE $[u, b]$. The scatter remains broad. This comparison describes the pooled points on the plot. It

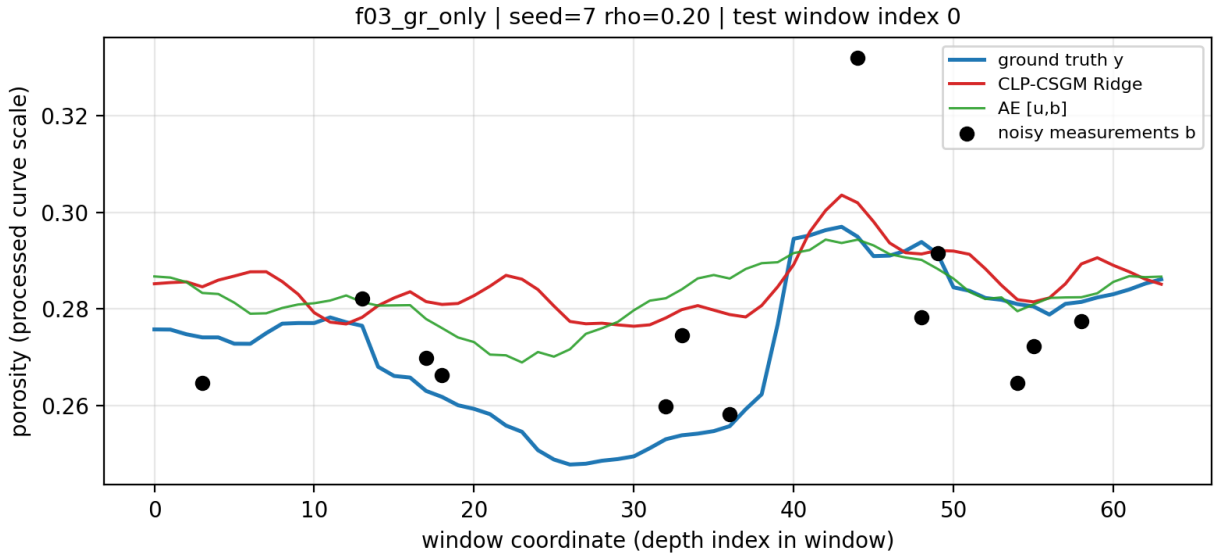


Figure 14: F03-4 GR-only porosity example (test window index 0, seed 7, $\rho = 0.20$): ground truth, sparse noisy measurements b, CLP-CSGM Ridge, and AE $[u, b]$. In this displayed window, both reconstructions miss the central depression of the reference.

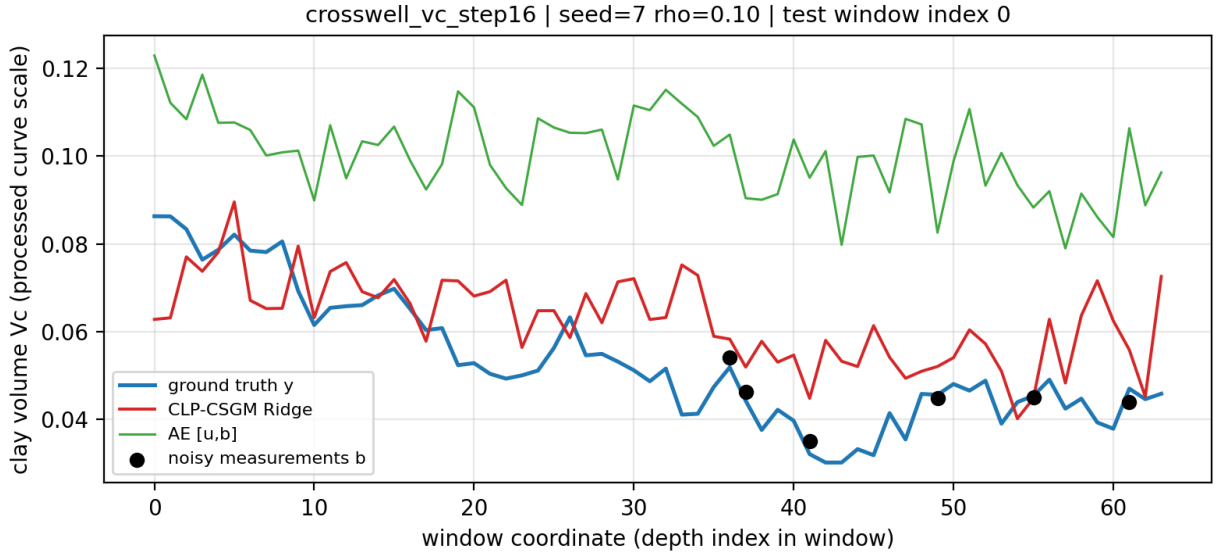


Figure 15: Cross-well V_c example (held-out F03-4 test window index 0, step 16, seed 7, $\rho = 0.10$): ground truth, sparse noisy measurements b, CLP-CSGM Ridge, and AE $[u, b]$. In this displayed window, CLP-CSGM Ridge is closer to the reference. The autoencoder prediction is positively offset, and local deviations remain.

does not establish statistical significance, and overlapping-window predictions are not independent observations.

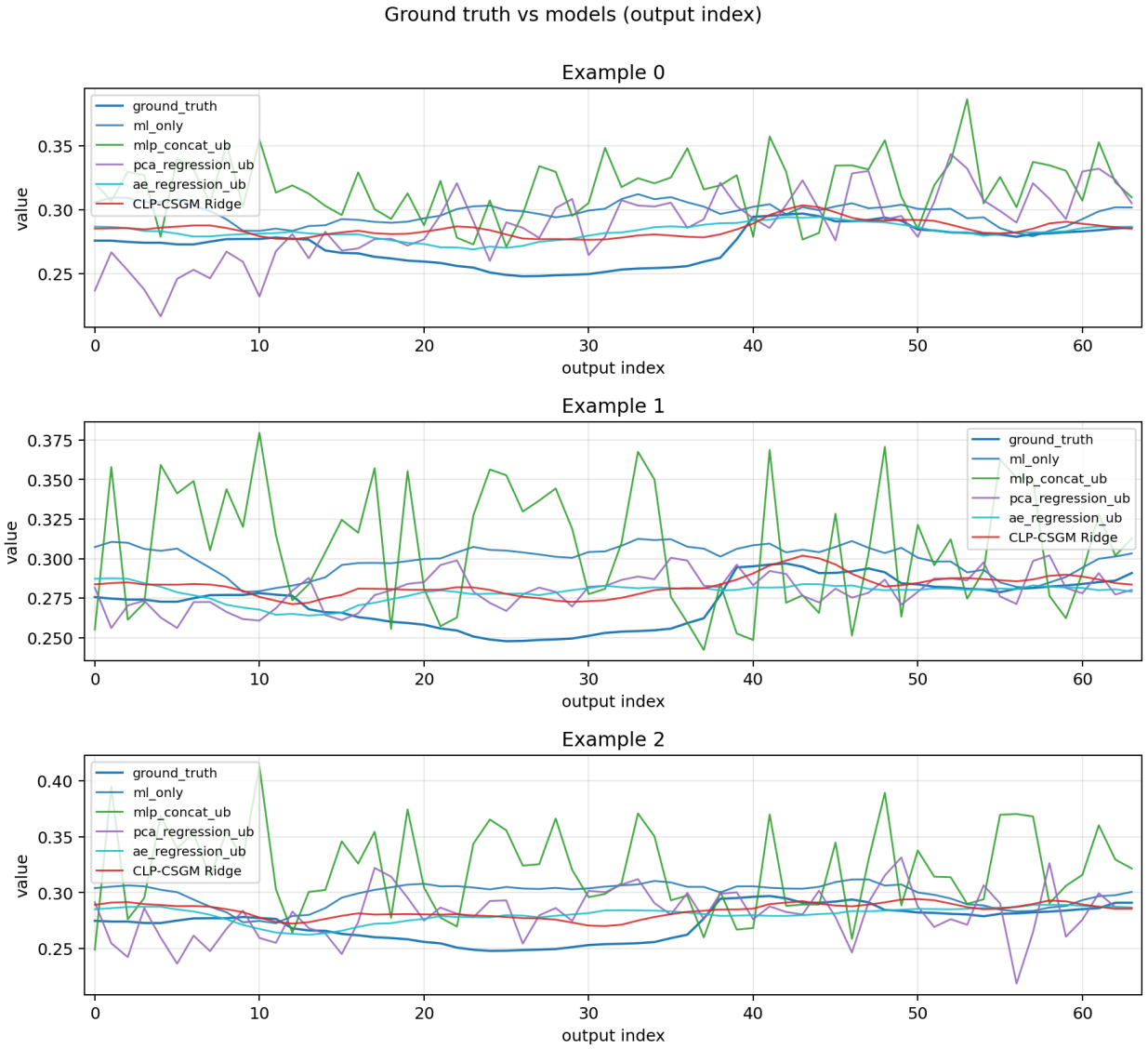


Figure 16: Representative F03-4 test windows. Direct MLP $[u, b]$ is visibly more oscillatory. CLP-CSGM Ridge and AE $[u, b]$ are smoother and, in these displayed windows, miss part of the central depression of the reference.

B.6 Validation and Generalization Diagnostics

Figures 18 and 19 audit quantities used by the experimental protocol: validation RMSE for selecting λ, selected regularization values, per-window test RMSE distributions, and the relation between autoencoder training reconstruction error and final test RMSE. These diagnostics support

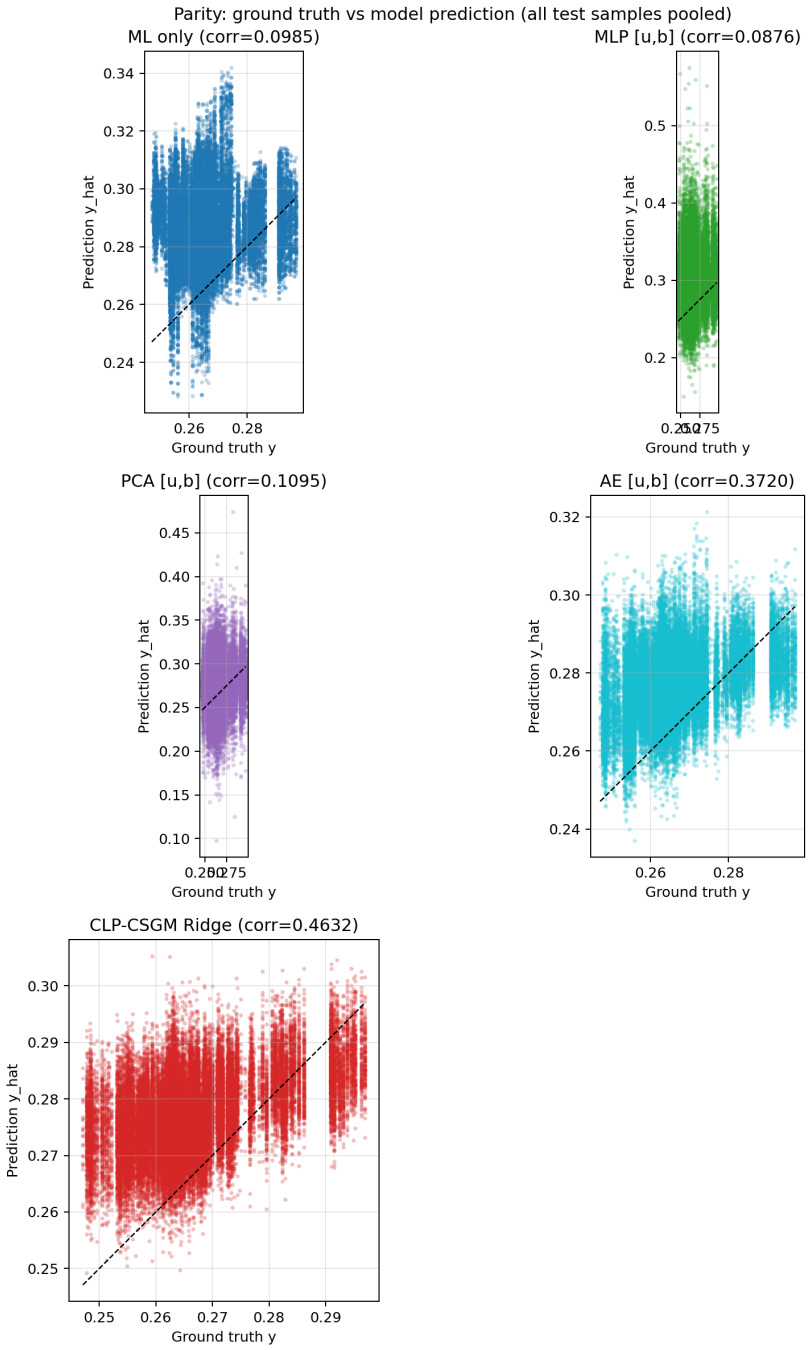


Figure 17: Pooled parity plots for F03-4 test windows. The displayed correlations are 0.4632 for CLP-CSGM Ridge and 0.3720 for AE $[u, b]$. Broad scatter remains. These values describe the pooled points on the figure and are not a significance test.

the reliability and interpretation of the main rankings.

Figure 18 summarizes the cross-well V_c case. The validation and test RMSE values remain of similar magnitude across the low-data grid, with no sign that lower validation error is obtained only by shifting error to the test well. The selected λ values are small but nonzero, consistent with sparse target-log observations providing the main target-well information while the log-conditioned latent prior acts as a stabilizing regularizer. The per-window RMSE boxplots reinforce the ablation result: CLP-CSGM Ridge has a lower median error than the latent-prior-only decoder and a substantially tighter distribution than measurement-only CSGM.

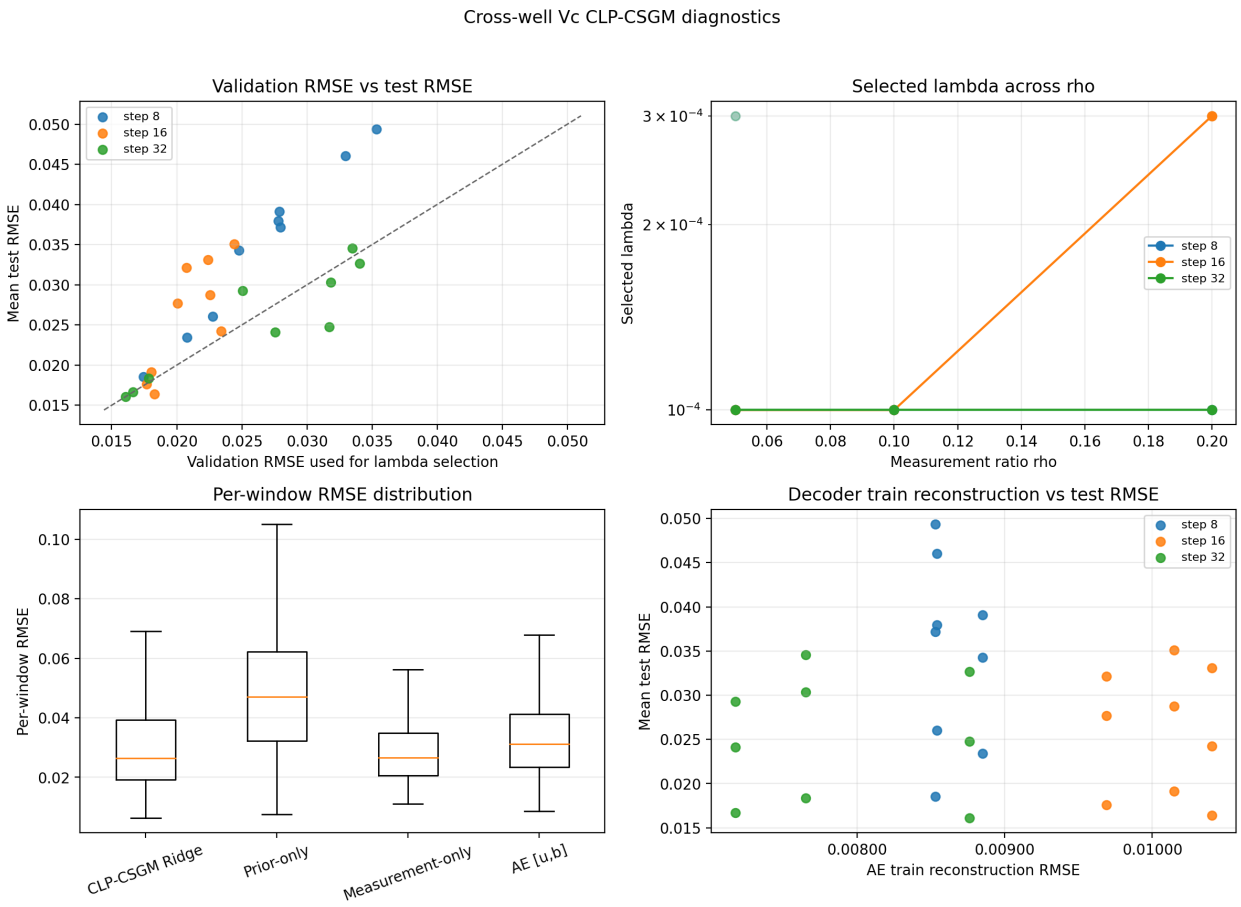


Figure 18: Cross-well V_c diagnostics for CLP-CSGM Ridge summarizing the experiments in Section 6.1. The panels compare validation RMSE with mean test RMSE for CLP-CSGM Ridge, show the selected λ values across measurement ratios, summarize per-window RMSE distributions, and relate autoencoder training reconstruction error to final test RMSE.

Figure 19 gives the analogous diagnostic view for the F03-4 porosity benchmark with GR-only dense input. The validation–test relation is less diagonal than in the cross-well case, consistent

with the harder stress test: the dense input is restricted to GR, and the test block is a contiguous depth interval, not a random split. The trend across measurement ratios remains coherent. Larger ρ generally reduces the full-model test error, and the selected λ remains stable across ratios. The per-window RMSE distributions show that CLP-CSGM Ridge improves the central error relative to both ablated variants while remaining close to AE $[u, b]$, matching the main F03 regime-dependent result.

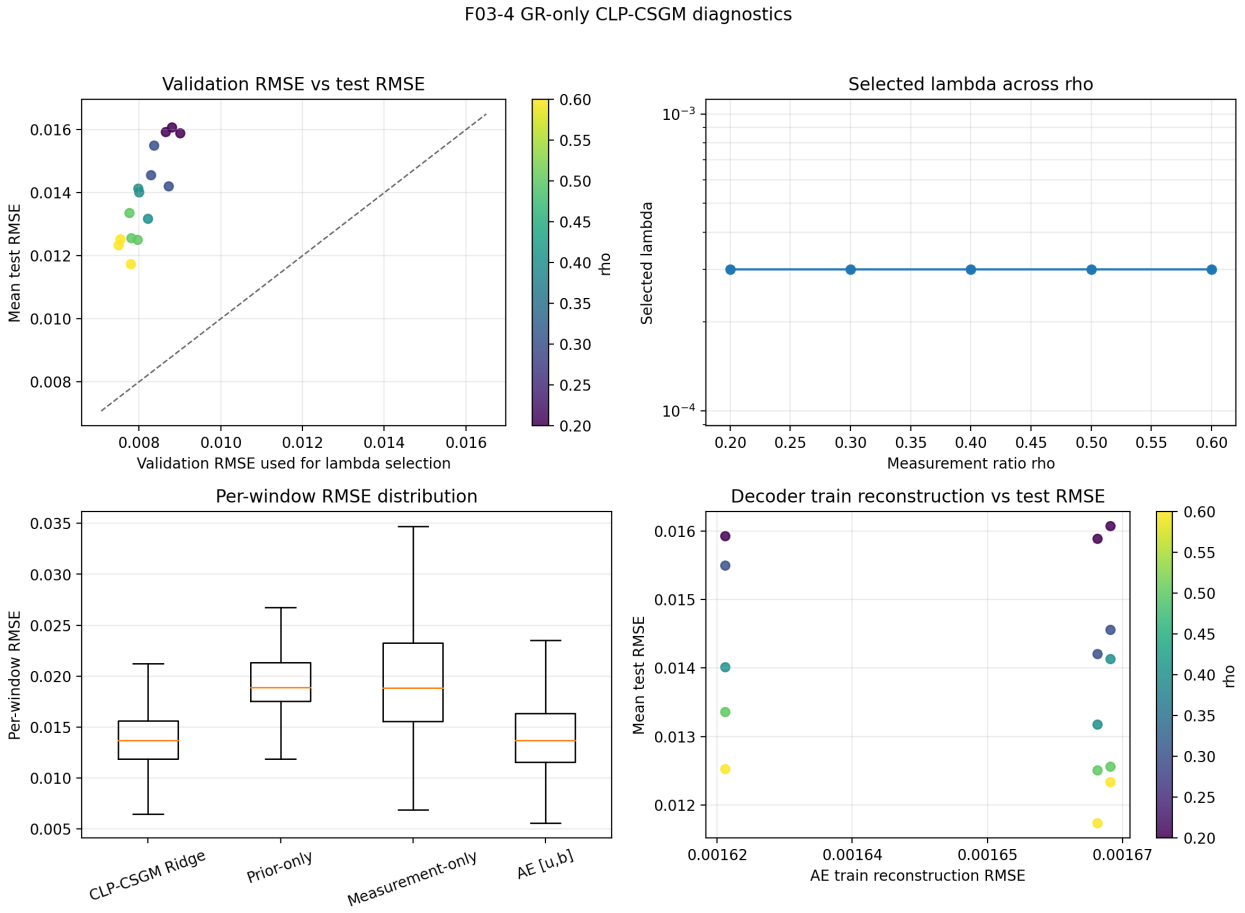


Figure 19: F03-4 GR-only diagnostics for CLP-CSGM Ridge summarizing the experiments in Section 6.2. The panels report validation RMSE versus mean test RMSE, selected λ, per-window RMSE distributions, and the relation between autoencoder training reconstruction error and final test RMSE.

B.7 Cross-well Channel-Ablation and Embargo-Style Boundary Check

This subsection reports a paired robustness check for the cross-well V_c step-16 setting. It tests two possible sensitivities: reintroducing GR into u, relative to the main no-GR protocol in Section 5, and

removing overlap-prone windows near the train/validation boundary. The latter is an embargo-style boundary check: a small band of windows adjacent to the boundary is removed before validation to create a gap between training and validation windows. We compare four scenarios under the same seeds and measurement-ratio grid, $\rho \in \{0.05, 0.10, 0.20\}$:

- No boundary check + GR in u (stronger dense context reference).
- Embargo-style removal (4 boundary windows) + GR in u.
- No boundary check + GR excluded from u (matches the main cross-well u).
- Embargo-style removal (4 boundary windows) + GR excluded from u.

Operationally, the embargo-style removal drops four overlap-prone boundary windows before the validation tail in each training well. In this setup, n_{train} decreases from 52 to 41, while n_{val} and n_{test} stay fixed.

Table 14: Cross-well step-16 robustness check across GR ablation and embargo-style boundary-check scenarios. Ridge RMSE is reported as mean ± std over three seeds. Gains are relative RMSE reductions of CLP-CSGM Ridge against AE $[u, b]$ and measurement-only CSGM.

scenario	rho	ridge rmse	ae rmse	meas-only rmse	ridge gain vs ae (%)	ridge gain vs meas-only (%)
Emb4+GR	0.05	0.01487 ± 0.00035	0.02056	0.03647	27.7	59.2
Emb4+GR	0.10	0.01406 ± 0.00032	0.01953	0.02849	28.0	50.7
Emb4+GR	0.20	0.01282 ± 0.00026	0.01932	0.02167	33.6	40.8
Emb4-noGR	0.05	0.03117 ± 0.00020	0.03781	0.03647	17.6	14.5
Emb4-noGR	0.10	0.02549 ± 0.00268	0.03439	0.02849	25.9	10.6
Emb4-noGR	0.20	0.01625 ± 0.00139	0.03771	0.02167	56.9	25.0
NoEmb+GR	0.05	0.01403 ± 0.00064	0.01893	0.03622	25.9	61.3
NoEmb+GR	0.10	0.01380 ± 0.00107	0.01802	0.02954	23.4	53.3
NoEmb+GR	0.20	0.01289 ± 0.00052	0.01822	0.02127	29.3	39.4
NoEmb-noGR	0.05	0.03344 ± 0.00151	0.03712	0.03622	9.9	7.7
NoEmb-noGR	0.10	0.02689 ± 0.00235	0.03420	0.02954	21.4	9.0
NoEmb-noGR	0.20	0.01771 ± 0.00138	0.03214	0.02127	44.9	16.7

Table 14 shows three stable patterns. Within each boundary protocol, including GR in u lowers absolute RMSE, as expected because V_c is Larionov-derived from GR. Under the no-GR protocol matching the main benchmark, CLP-CSGM Ridge has the lowest RMSE relative to AE $[u, b]$ and measurement-only CSGM throughout the tested grid. The embargo-style boundary check changes the absolute values without reversing this ordering. These checks support the main interpretation: the cross-well V_c benchmark tests sparse target-log reconstruction under distribution shift without a GR shortcut in u, and the reported gains are not solely an artifact of train/validation boundary overlap.

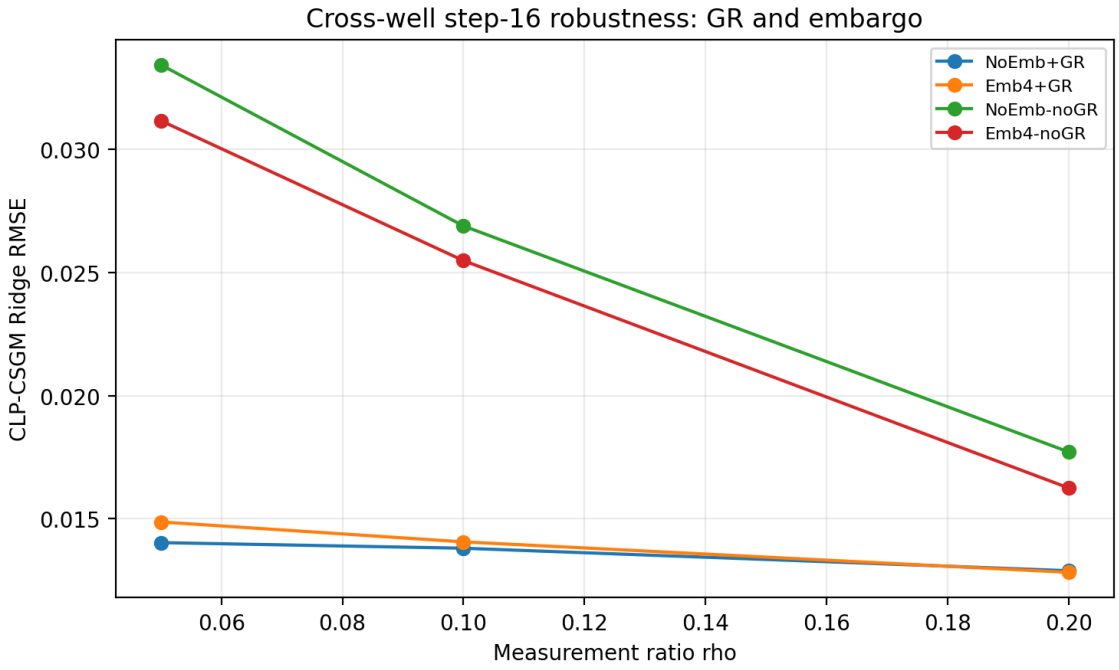


Figure 20: Cross-well step-16 RMSE sensitivity for CLP-CSGM Ridge across GR inclusion in u and embargo-style boundary-check scenarios. Omitting GR from u relative to GR-inclusive rows increases error; the boundary check shifts absolute values without overturning the ranking trends.

B.8 Step-32 development-split resampling

The main cross-well grid uses a depth-tail validation split. At step 32 the development pool is small ($n_{\text{train}} = 27$, $n_{\text{val}} = 3$ after the split). To test whether the CLP-CSGM Ridge versus AE $[u, b]$ gap in that cell depends on the particular validation windows, we redrew the same validation budget ten times inside each development well (`random_per_well`), keeping F03-4 as the held-out test well and freezing the sparse coordinates Ω to the paper seed 7. Prior-only and measurement-only ablations were not rerun.

Table 15: Cross-well step-32 resampling of the development train/validation split ($n = 10$). RMSE is the mean ± standard deviation across splits. Positive mean ΔRMSE (AE − CLP) favors CLP-CSGM Ridge. CLP win frac is the fraction of splits with lower CLP RMSE.

ρ	n_{splits}	n_{train}	n_{val}	CLP RMSE	AE RMSE	mean ΔRMSE	CLP win frac
0.05	10	27	3	0.0342 ± 0.0024	0.0407 ± 0.0025	0.0064	1.00
0.10	10	27	3	0.0305 ± 0.0032	0.0388 ± 0.0015	0.0083	1.00
0.20	10	27	3	0.0188 ± 0.0010	0.0362 ± 0.0064	0.0174	1.00

Table 15 shows that CLP-CSGM Ridge has lower mean RMSE than AE $[u, b]$ at every tested ρ, and lower RMSE in all ten splits. This check does not add a second held-out well and does not vary

Ω; it only resamples the small development split. Adjacent windows have length 64 and step 32, so they share 32 depth samples. This random-per-well redraw does not apply a boundary embargo. In each of the ten splits, all three validation windows share at least one depth sample with a training window of the same well (30 of 30 validation windows). One validation window comes from F02-1 and two from F03-2; F06-1 has only two windows and stays in the training set. The ten splits test sensitivity to that development partition. They do not represent independent validation without shared samples, and they are not ten independent wells.

B.9 Ten-seed CLP versus AE panel

The main cross-well intervals use three seeds (t critical value with df = 2). Several of those paired ΔRMSE intervals include zero, and the CLP-CSGM Ridge versus AE $[u, b]$ ranking changes between step 8 and steps 16–32. We therefore reran that pairwise comparison with ten seeds $\{7, 23, 41, 13, 29, 31, 37, 43, 47, 53\}$, keeping the submitted tail validation split, the no-GR input, and F03-4 as the held-out well. Direct MLP and PCA, structural ablations, and the F03-4 grid were not rerun.

Table 16: Cross-well paired $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on ten seeds. SE is the seed-level standard error; the 95% half-width uses Student-t with df = 9. Positive mean ΔRMSE favors CLP-CSGM Ridge. Wins count seeds with strictly lower RMSE.

step	ρ	n	mean ΔRMSE	SE	t 95% half-width	CLP wins	AE wins	CI includes 0
8	0.05	10	-0.00826	0.00121	0.00273	0	10	no
8	0.10	10	-0.00318	0.00092	0.00209	2	8	no
8	0.20	10	0.00360	0.00129	0.00293	7	3	no
16	0.05	10	0.00434	0.00159	0.00360	9	1	no
16	0.10	10	0.00677	0.00134	0.00304	10	0	no
16	0.20	10	0.01382	0.00145	0.00328	10	0	no
32	0.05	10	0.00865	0.00129	0.00292	10	0	no
32	0.10	10	0.01217	0.00118	0.00266	10	0	no
32	0.20	10	0.01934	0.00185	0.00418	10	0	no

Table 16 shows that none of the nine ten-seed intervals include zero. The CLP-CSGM Ridge advantage is concentrated where training support is smallest (steps 16 and 32, 52 and 27 training windows): lower RMSE at every ρ. At step 8, where the training set is largest (104 training windows), AE $[u, b]$ has lower RMSE at $\rho = 0.05$ and $\rho = 0.10$; CLP-CSGM Ridge has lower RMSE only at $\rho = 0.20$. This is a low-training-support pattern, not a uniform gain at every small measurement ratio. The submitted three-seed tables are unchanged. Each interval describes

variability across the ten seeds on the single held-out well F03-4. It is not evidence of generalization across wells, and the ten seeds are not ten independent geological tests. The nine intervals are separate cell summaries, not a simultaneous multiplicity-corrected test, and overlapping windows are not counted as extra independent samples. Each of these seeds also draws its own coordinates and measurement noise, so the panel does not separate training randomness from the observation realization. Appendix B.11 holds the model seeds fixed and redraws those coordinates together with the measurement noise, without isolating placement from noise.

B.10 Explicit-mask direct AE

The main direct baselines receive $[\boldsymbol{u}, \boldsymbol{b}]$ with positional identity implicit in the column order. CLP-CSGM instead applies M_Ω to the length-N window. To test whether the CLP-CSGM gain is only that coordinate placement, we fit one additional direct AE on the same nine cells, seeds, and noise. Its input is $[\boldsymbol{u}, \boldsymbol{b}_\Omega, \text{mask}]$: $\boldsymbol{b}_\Omega \in \mathbb{R}^N$ holds the noisy measurement at each observed coordinate and zero elsewhere, and the mask is the binary indicator of Ω. Numeric channels are standardized from the training windows only. The trailing binary mask is left unscaled, so its entries remain in $\{0, 1\}$. The mask-aware AE uses the same base training seed as the conventional AE. Both direct models and CLP-CSGM Ridge share the same partitions, observation coordinates, and noisy target measurements within each cell. MLP, PCA, ablations, and the F03-4 grid were not rerun, and this method is not added to the main tables.

An earlier historical run scaled the full concatenated vector and trained the masked AE at seed offset 100. That preprocessing zeroed the constant mask columns. The difference between that historical variant and the corrected protocol cannot be attributed to either change alone.

The corrected explicit-mask AE preserves the binary mask without centering or standardization and uses the same base training seed as the conventional AE. Both direct models and CLP-CSGM Ridge are evaluated using the same data partitions, observation coordinates, and noisy target measurements within each cell.

CLP-CSGM Ridge achieves lower mean RMSE than the mask-aware AE in seven of the nine cross-well configurations. At steps 16 and 32, corresponding to the smaller training sets, CLP-CSGM Ridge has lower mean RMSE in all six configurations and lower RMSE in 17 of the 18 paired seed comparisons. At step 8, the mask-aware AE remains better at measurement ratios 0.05

Table 17: Cross-well CLP-CSGM Ridge versus AE $[\boldsymbol{u}, \boldsymbol{b}]$ and an AE that also receives an explicit window mask ($n = 3$ seeds). Δ is the other method's RMSE minus CLP RMSE. Positive Δ favors CLP-CSGM Ridge. Wins count seeds with strictly lower CLP RMSE.

step	ρ	CLP	AE $[u, b]$	AE mask	Δ AE	Δ mask	CLP>AE	CLP>mask
8	0.05	0.0448	0.0357	0.0346	-0.0092	-0.0102	0/3	0/3
8	0.10	0.0365	0.0320	0.0312	-0.0045	-0.0053	1/3	0/3
8	0.20	0.0227	0.0260	0.0260	0.0033	0.0034	2/3	3/3
16	0.05	0.0334	0.0371	0.0346	0.0037	0.0011	3/3	2/3
16	0.10	0.0269	0.0342	0.0322	0.0073	0.0053	3/3	3/3
16	0.20	0.0177	0.0321	0.0281	0.0144	0.0103	3/3	3/3
32	0.05	0.0322	0.0390	0.0389	0.0068	0.0067	3/3	3/3
32	0.10	0.0264	0.0389	0.0376	0.0125	0.0112	3/3	3/3
32	0.20	0.0170	0.0396	0.0348	0.0225	0.0178	3/3	3/3

and 0.10.

Because the sampling coordinates are fixed within each configuration, the binary mask is constant across windows, and the conventional AE already receives the measurements in a consistent positional order. The experiment therefore assesses an alternative explicit spatial encoding, but does not isolate the causal effect of latent inverse optimization. The placement check is Appendix B.11.

B.11 Sparse-coordinate placement

The main protocol samples one coordinate set Ω for each model seed and measurement ratio and then holds it fixed across windows. To check stability under new sparse-observation realizations, we redrew Ω and the measurement noise five times on the cross-well V_c grid. The model seeds stay at $\{7, 23, 41\}$, so the decoder and the ridge anchor keep the same initialization. The first draw is the published coordinate set and noise realization. The other four draws change only the generator that samples Ω and the noise on $\boldsymbol{b}$, so the positions and the noise change together. This check does not isolate the effect of position from the effect of noise. The direct AE is fit again on each draw, because $\boldsymbol{b}$ is part of its input, and the CLP-CSGM penalty weight is selected again on the validation windows of that draw. The score is the same full-window RMSE as the main grid. Direct MLP, PCA, structural ablations, and the F03-4 grid were not rerun, and these draws are not merged into the submitted three-seed tables.

Table 18 shows that the cell-level winner matches the published ranking in all nine cells. At steps 16 and 32, CLP-CSGM Ridge has lower RMSE in all fifteen comparisons at every ρ. At step 8 and $\rho = 0.05$, AE $[\boldsymbol{u}, \boldsymbol{b}]$ is lower in all fifteen. The two closer step-8 cells are not unanimous: five

Table 18: Cross-well CLP-CSGM Ridge versus AE $[\boldsymbol{u}, \boldsymbol{b}]$ across five sparse-coordinate placements and the three paper seeds ($n = 15$ per cell). RMSE values are means over those comparisons. Positive mean ΔRMSE (AE − CLP) favors CLP-CSGM Ridge. Wins count comparisons with strictly lower RMSE.

step	ρ	n	CLP RMSE	AE RMSE	mean ΔRMSE	CLP wins	AE wins
8	0.05	15	0.04231	0.03586	-0.00645	0	15
8	0.10	15	0.03581	0.03225	-0.00356	5	10
8	0.20	15	0.02383	0.02591	0.00208	11	4
16	0.05	15	0.03072	0.03710	0.00638	15	0
16	0.10	15	0.02615	0.03412	0.00797	15	0
16	0.20	15	0.01840	0.03036	0.01196	15	0
32	0.05	15	0.02895	0.03873	0.00977	15	0
32	0.10	15	0.02470	0.03870	0.01401	15	0
32	0.20	15	0.01795	0.04023	0.02228	15	0

of fifteen comparisons favor CLP-CSGM Ridge at $\rho = 0.10$, and four of fifteen favor the direct AE at $\rho = 0.20$. The low-training-support advantage therefore does not depend on the one published observation draw. The step-8 ranking is the less stable of the two regimes, and a single extra draw can reverse an individual seed there. These fifteen comparisons are three model seeds times five draws on the single held-out well F03-4. They are not fifteen independent wells, and they are not evidence of cross-well generalization. This check does not add seeds beyond $\{7, 23, 41\}$.

B.12 Direct linear maps, depth interpolation, and measurement noise

The main grid does not include a direct linear map from $[\boldsymbol{u}, \boldsymbol{b}]$ or an interpolation of the sparse measurements. This supplement adds both on the cross-well V_c grid, with the same windows, seeds $\{7, 23, 41\}$, measurement ratios, and published coordinate draws. Ordinary least squares and ridge regression map standardized $[\boldsymbol{u}, \boldsymbol{b}]$ to the window. The ridge penalty is chosen on validation full-window RMSE. Interpolation connects the noisy measurements along the depth index of the window and keeps the nearest measured value at the ends. It does not use the dense logs. The score is the same full-window RMSE as the main grid. CLP-CSGM Ridge and AE $[\boldsymbol{u}, \boldsymbol{b}]$ in Table 19 are the published offset-0 runs. These comparators are not merged into the submitted tables.

Table 19 shows that interpolation has the lowest RMSE in all nine cells. Seed by seed, it is also lowest in all 27 comparisons. CLP-CSGM Ridge is lower than both direct linear maps at step 16, $\rho = 0.20$, and at step 32 for every ρ. At step 8, and at step 16 for $\rho = 0.05$ and $\rho = 0.10$, one or both linear maps are lower than CLP-CSGM Ridge. The measurements at this noise level are a

Table 19: Cross-well full-window RMSE at the published noise standard deviation 0.01. Values are means over seeds $\{7, 23, 41\}$. Linear $[u, b]$ is ordinary least squares and Ridge direct is validation-selected ridge, both from $[\boldsymbol{u}, \boldsymbol{b}]$ to the window. Interpolation uses only the sparse measurements. The lowest value in every cell is interpolation.

step	ρ	Linear $[u, b]$	Ridge direct	Interpolation	CLP-CSGM	AE $[u, b]$
8	0.05	0.03122	0.02916	0.01779	0.04483	0.03567
8	0.10	0.02790	0.02384	0.01537	0.03646	0.03195
8	0.20	0.02158	0.01862	0.01100	0.02267	0.02600
16	0.05	0.02903	0.02910	0.01751	0.03344	0.03712
16	0.10	0.02634	0.02661	0.01478	0.02689	0.03420
16	0.20	0.02062	0.02089	0.01115	0.01771	0.03214
32	0.05	0.03382	0.03980	0.01755	0.03217	0.03901
32	0.10	0.03083	0.03878	0.01533	0.02640	0.03894
32	0.20	0.02421	0.02648	0.01125	0.01705	0.03959

slightly perturbed copy of the same smooth curve, so connecting them is enough.

The same windows and the same standard-normal draws are scaled to noise standard deviations 0.01, 0.05, 0.10, and 0.20. Table 20 and Fig. 6 use one refit of that series. The decoder and the dense-log map are fit once per step and seed, without the sparse measurements. For each measurement fraction and each noise level, the penalty weight is selected again on validation full-window RMSE. The reported CLP-CSGM values are that selected-penalty result, not a reconstruction with one fixed weight. The sensitivity sweep uses a different, wider grid, with candidates 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, and 100. It omits the factor-of-three candidates from the published grid. At 0.01 and at 0.05 every selected weight also belongs to the published grid. That membership does not mean the two searches were the same. At noise 0.10, step 16 and $\rho = 0.10$, one seed selects the weight 1, above the published grid and still inside the sweep. At noise 0.20, some cells select 100, the largest candidate that was tried. That selection is not an optimum outside the searched grid. At noise 0.01 the ordering matches Table 19, but the cell means are those of this refit. At step 16 and $\rho = 0.20$, the published CLP-CSGM mean is 0.01771 and the refit mean is 0.0159. In that cell the selected weight is 0.0001, inside the published grid. The held-out curve has standard deviation 0.035. The window predicted from the dense logs alone, with no measurement term, has mean RMSE 0.0564 at step 8, 0.0426 at step 16, and 0.0432 at step 32.

At noise standard deviation 0.05, CLP-CSGM Ridge is lower in every seed at steps 16 and 32 and at step 8 for $\rho = 0.20$. At step 8 for $\rho = 0.05$ and $\rho = 0.10$, interpolation still has the lower mean. At 0.10 and 0.20, CLP-CSGM Ridge is lower in all nine cells and in all three seeds. Figure 6

Table 20: Interpolation versus CLP-CSGM Ridge when only the measurement-noise standard deviation changes. RMSE values are means over the three paper seeds. Wins count seeds in which CLP-CSGM Ridge has strictly lower RMSE. The held-out V_c standard deviation is 0.035, on the same scale as the noise. Noise 0.05 is about 1.43 times that deviation. Noise levels 0.10 and 0.20 are about 2.86 and 5.71 times that deviation. These noise levels were used for sensitivity analysis and were not calibrated to field measurement uncertainties. The decoder and the dense-log map are fit once per step and seed, and the penalty is selected on validation at each noise level. Noise 0.01 in this table is that refit. At 0.20 some selected weights equal 100, the largest candidate on this grid, which is not an optimum outside the grid.

		$\sigma_\eta = 0.01$			$\sigma_\eta = 0.05$			$\sigma_\eta = 0.10$			$\sigma_\eta = 0.20$		
step	ρ	Interp.	CLP	wins	Interp.	CLP	wins	Interp.	CLP	wins	Interp.	CLP	wins
8	0.05	0.0178	0.0456	0	0.0456	0.0508	1	0.0864	0.0547	3	0.1703	0.0557	3
8	0.10	0.0154	0.0365	0	0.0442	0.0456	1	0.0853	0.0539	3	0.1690	0.0555	3
8	0.20	0.0110	0.0212	0	0.0423	0.0372	3	0.0838	0.0513	3	0.1673	0.0548	3
16	0.05	0.0175	0.0334	0	0.0442	0.0386	3	0.0838	0.0416	3	0.1653	0.0424	3
16	0.10	0.0148	0.0269	0	0.0427	0.0352	3	0.0825	0.0416	3	0.1636	0.0424	3
16	0.20	0.0112	0.0159	0	0.0428	0.0286	3	0.0847	0.0387	3	0.1690	0.0424	3
32	0.05	0.0175	0.0317	0	0.0430	0.0382	3	0.0820	0.0423	3	0.1625	0.0430	3
32	0.10	0.0153	0.0264	0	0.0445	0.0365	3	0.0860	0.0414	3	0.1706	0.0431	3
32	0.20	0.0112	0.0170	0	0.0425	0.0279	3	0.0840	0.0381	3	0.1674	0.0409	3

shows this ordering with one panel per step. The smaller change in the CLP-CSGM RMSE is not attributed here to the conditional penalty alone. This supplement does not separate that penalty from the decoder restriction or from the validation choice of the weight. Fig. 23 shows one held-out interval at the published noise and at noise 0.05. The two panels use separate horizontal scales. Panel (b) is not one of the larger sensitivity levels. In this displayed interval, interpolation closely follows the reference at noise standard deviation 0.01. At 0.05, it exhibits larger excursions, while CLP-CSGM Ridge retains more of the broad profile structure. The example is consistent with the cell-level result for the corresponding configuration, step 32 and $\rho = 0.20$, and it does not extend to every configuration at that noise level.

The same observation stream is also scored with a second-difference smoother. For each window it solves $\|M_\Omega \boldsymbol{y} - \boldsymbol{b}\|_2^2 + \alpha \|D^2 \boldsymbol{y}\|_2^2$, where D^2 is the second difference along the window index. The smoother does not use the dense logs. α is chosen only by validation full-window RMSE, from the frozen grid 10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2, 10^4, and 10^6. Equal validation scores keep the smaller value. Test targets are not used, and the grid was not changed after the test scores were known. CLP-CSGM was not refit. The largest grid value is selected in 62 of the 108 seed rows, and the smallest value is never selected. A selection at the ceiling is the best candidate inside the grid, not an optimum beyond it.

Table 21: Interpolation, validation-selected second-difference smoothing, and CLP-CSGM Ridge on the noise sweep. Entries are mean full-window RMSE over seeds $\{7, 23, 41\}$, rounded to four decimals. The smoother does not use the dense logs. These noise levels were used for sensitivity analysis and were not calibrated to field measurement uncertainties.

		$\sigma_\eta = 0.01$			$\sigma_\eta = 0.05$			$\sigma_\eta = 0.10$			$\sigma_\eta = 0.20$		
step	ρ	Interp.	Smooth	CLP	Interp.	Smooth	CLP	Interp.	Smooth	CLP	Interp.	Smooth	CLP
8	0.05	0.0178	0.0251	0.0456	0.0456	0.0646	0.0508	0.0864	0.1232	0.0547	0.1703	0.2437	0.0557
8	0.10	0.0154	0.0181	0.0365	0.0442	0.0394	0.0456	0.0853	0.0721	0.0539	0.1690	0.1404	0.0555
8	0.20	0.0110	0.0110	0.0212	0.0423	0.0225	0.0372	0.0838	0.0389	0.0513	0.1673	0.0743	0.0548
16	0.05	0.0175	0.0246	0.0334	0.0442	0.0589	0.0386	0.0838	0.1302	0.0416	0.1653	0.2548	0.0424
16	0.10	0.0148	0.0181	0.0269	0.0427	0.0385	0.0352	0.0825	0.0706	0.0416	0.1636	0.1375	0.0424
16	0.20	0.0112	0.0120	0.0159	0.0428	0.0237	0.0286	0.0847	0.0418	0.0387	0.1690	0.0801	0.0424
32	0.05	0.0175	0.0429	0.0317	0.0430	0.0759	0.0382	0.0820	0.1177	0.0423	0.1625	0.2326	0.0430
32	0.10	0.0153	0.0183	0.0264	0.0445	0.0442	0.0365	0.0860	0.1104	0.0414	0.1706	0.2157	0.0431
32	0.20	0.0112	0.0114	0.0170	0.0425	0.0224	0.0279	0.0840	0.0395	0.0381	0.1674	0.0751	0.0409

Table 21 and Fig. 21 keep every cell. At noise 0.01, smoothing has lower mean RMSE than CLP-CSGM in eight of nine cells. The exception is step 32 and $\rho = 0.05$, where the smoother mean is 0.0429 and the CLP-CSGM mean is 0.0317. Interpolation remains lower than CLP-CSGM in all nine cells at that noise. At 0.05, CLP-CSGM has the lower mean in five cells and smoothing in four: step 8 at $\rho = 0.10$ and $\rho = 0.20$, and steps 16 and 32 at $\rho = 0.20$. At 0.10, CLP-CSGM is lower in eight cells, and smoothing is lower at step 8 and $\rho = 0.20$. At 0.20, CLP-CSGM is lower in all nine cells. At $\rho = 0.05$, the smoother mean is higher than interpolation at every noise level in the table. The comparison does not establish general superiority, and it does not isolate the conditional penalty, the decoder restriction, or the validation choice as a separate mechanism.

An affine least-squares fit to the same noisy observations is an additional comparator, not a replacement for the selected smoother. It has no penalty weight. It is the second-difference problem as $\alpha \to \infty$. On all 108 rows, the curve at $\alpha = 10^6$ and this affine curve differ by a median mean-window profile RMSE of 5.2×10^{-5} and a maximum of 3.7×10^{-4}. The largest absolute gap in the scored full-window RMSE is 2.0×10^{-5}. On the 62 ceiling rows the profile median is 7.6×10^{-5} and the largest scored-RMSE gap is 1.9×10^{-5}. Those ceiling solutions are already near the affine limit (Fig. 22). The selected smoother can still differ from that limit when validation chooses an interior weight. At noise 0.01 the affine cell mean is lower than CLP-CSGM in all nine cells. In the step-32, $\rho = 0.05$ cell, only one of three seeds selects the ceiling: the affine mean there is 0.0240, against the selected-smoother mean 0.0429 and the CLP-CSGM mean 0.0317.

On the same splits, the retrospective decoder reconstruction $G(E(\boldsymbol{y}))$ has mean test RMSE 0.0073 at step 8, 0.0099 at step 16, and 0.0110 at step 32. That reconstruction is given the full

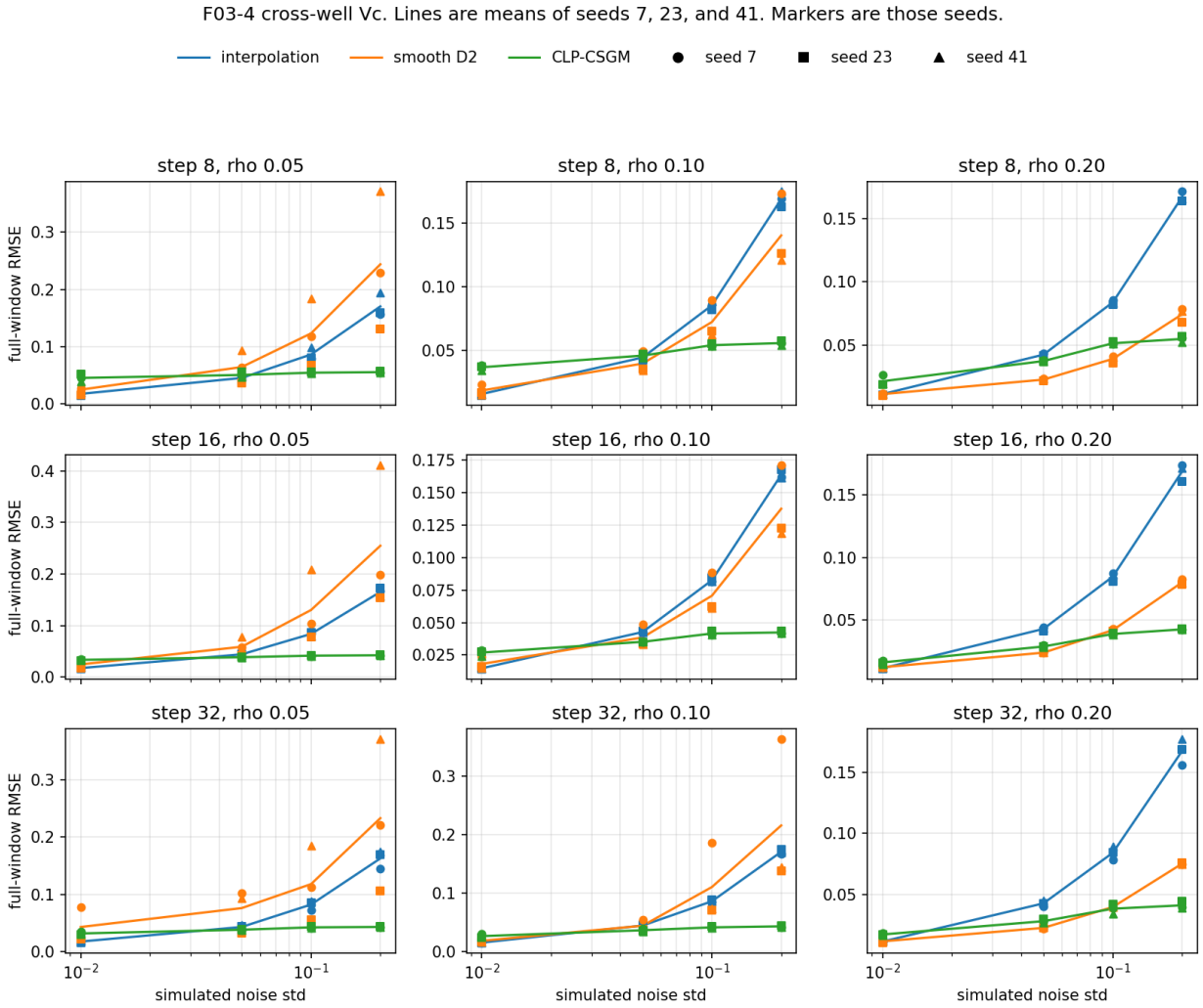


Figure 21: Full-window RMSE of interpolation, validation-selected second-difference smoothing, and CLP-CSGM Ridge. Each panel is one step and one measurement fraction on held-out F03-4. Lines are means of seeds 7, 23, and 41; markers are those seeds. The noise levels are simulated and were not calibrated to field measurement uncertainties.

target window, so it is a check that the decoder can represent the curve. It is not a predictor and it is not ranked with the methods above.

The step-8 pattern at noise standard deviation 0.05 is compatible with a limit of the dense-log map. It is not explained by the decoder check alone, and these comparisons do not isolate that map as the only cause. At step 8 the decoder represents the held-out curve with mean test RMSE 0.0073. The window predicted from the dense logs alone has mean RMSE 0.0564. Interpolation at the same step and noise has mean RMSE 0.0456 for $\rho = 0.05$ and 0.0442 for $\rho = 0.10$. Those ratios observe 4 and 6 coordinates in a window of length 64. CLP-CSGM Ridge then stays between the

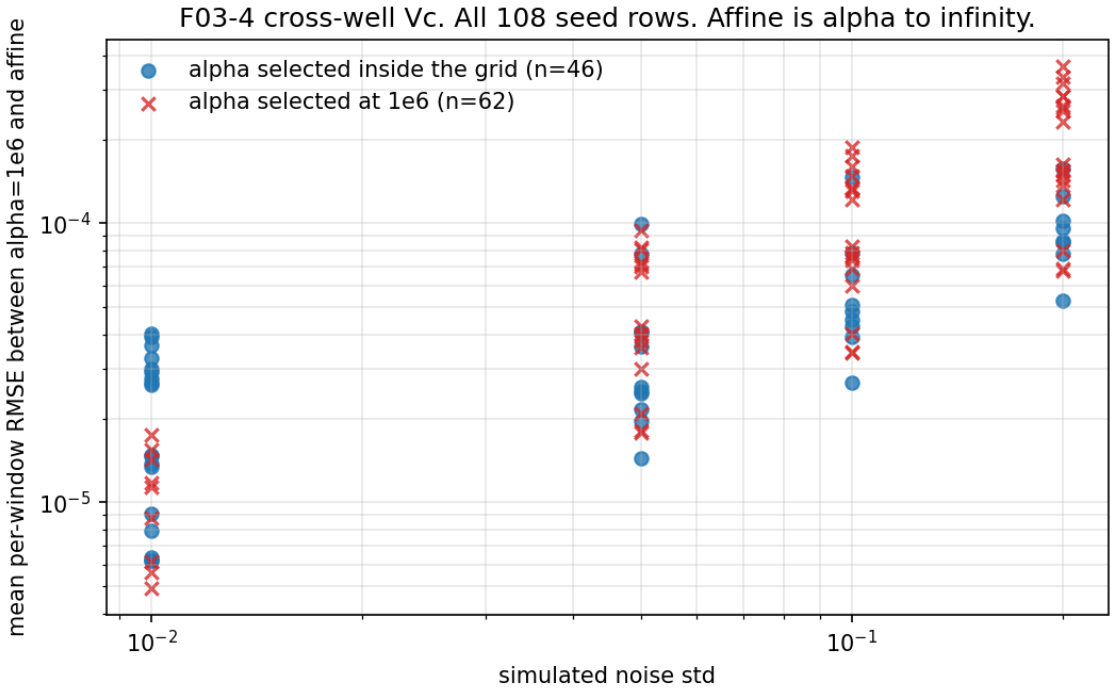


Figure 22: Mean per-window RMSE between the $\alpha = 10^6$ reconstruction and the affine least-squares fit, for all 108 seed rows. Crosses are the 62 rows whose validation choice is the grid ceiling. Circles are the 46 rows with an interior choice. The vertical coordinate is the profile gap at fixed $\alpha = 10^6$, not the gap of the validation-selected curve.

two, at 0.0508 and 0.0456. At $\rho = 0.20$ the same window observes 13 coordinates, and CLP-CSGM Ridge has mean RMSE 0.0372 against an interpolation RMSE of 0.0423. At steps 16 and 32 the dense-log window itself has mean RMSE 0.0426 and 0.0432, and interpolation at this noise has errors of similar magnitude. Those similar magnitudes do not show that the two reconstructions lie close to each other. The pattern agrees with the ablation in Section 6.3, where the conditional code $\boldsymbol{z}_0$ is not helpful at step 8.

B.13 Wide-range depth profiles

Figure 8 omits MLP $[\boldsymbol{u}, \boldsymbol{b}]$ and PCA $[\boldsymbol{u}, \boldsymbol{b}]$ so that the reference, ML only, AE, and CLP-CSGM Ridge can share one V_c scale. Figure 24 uses the same held-out interval, step 16, seed 7, and $\rho = 0.20$, and shows every stored value of those two predictions. Their horizontal range includes negative clay volume. No coordinate was dropped.

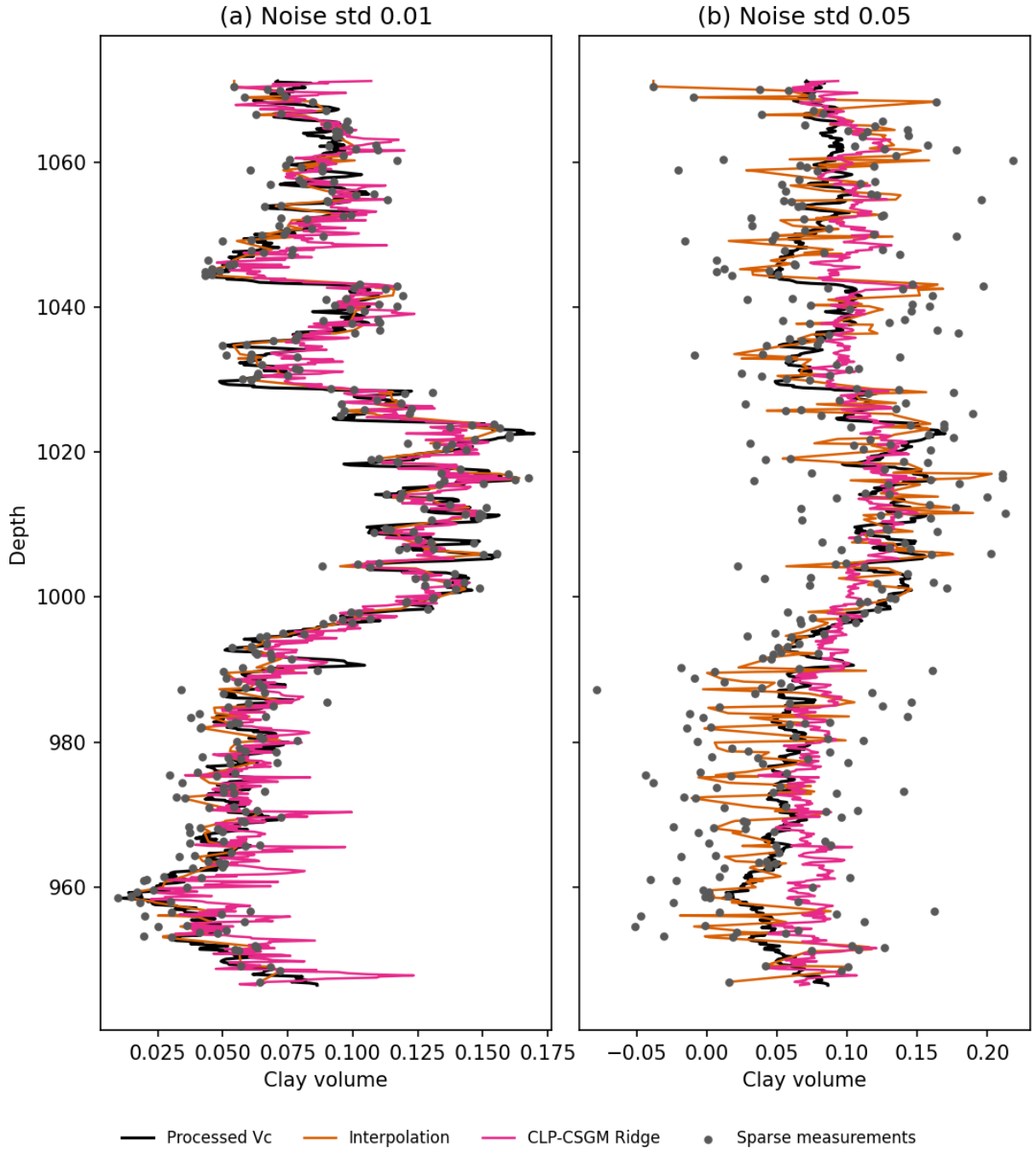


Figure 23: Held-out F03-4 depth tracks, seed 7, step 32, $\rho = 0.20$. Panel (a) uses the published noise standard deviation 0.01. Panel (b) uses 0.05, the same order as the curve's standard deviation 0.035. Each panel has its own clay-volume scale. This interval is one example. Tables 19 and 20 are the comparison.

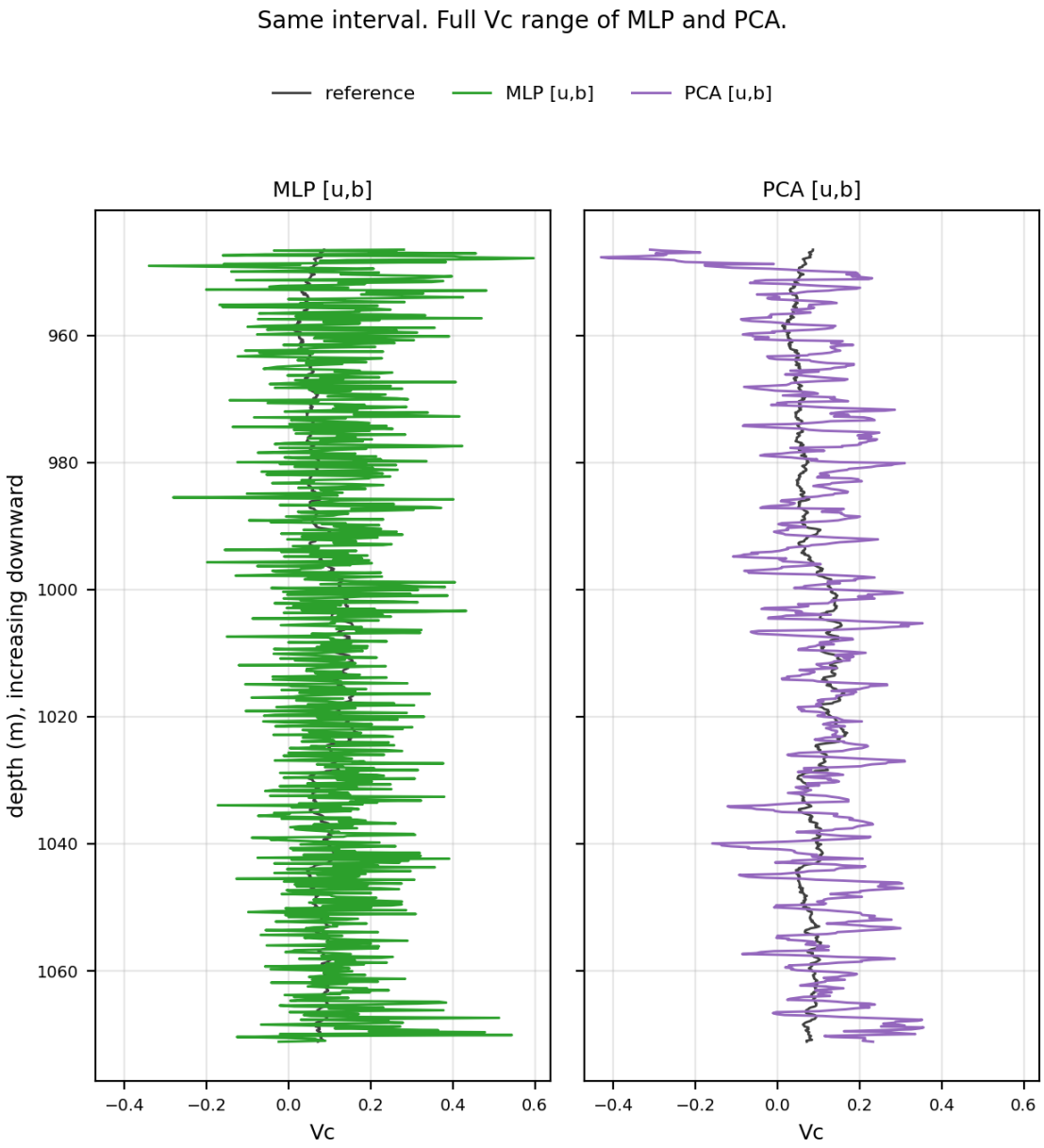


Figure 24: Same F03-4 interval as Fig. 8. MLP $[\boldsymbol{u}, \boldsymbol{b}]$ and PCA $[\boldsymbol{u}, \boldsymbol{b}]$ share the full horizontal range of their stored predictions. Depth increases downward.

B.14 Computational Trade-Offs

Table 22 and Fig. 25 quantify the practical computational trade-off. CLP-CSGM Ridge is more expensive than AE $[u, b]$ because it solves a latent inverse problem at validation and test time instead of producing a single direct forward prediction. Table 22 reports both mean wall-clock time per test window and a separate normalized runtime diagnostic relative to AE; the latter is the diagnostic shown in Fig. 25; it is not the exact quotient of the averaged seconds in the previous column. In the representative cross-well case, the additional cost is substantial relative to AE, but CLP-CSGM Ridge has the lowest mean RMSE among the methods listed in Table 22. That statement refers only to the computational-cost table. It is not a ranking against the full set of comparators in the paper, including depth interpolation. In the F03-4 GR-only case, CLP-CSGM Ridge and AE $[u, b]$ have closer average RMSE, so the runtime gap is less clearly justified. The extra optimization is most readily justified when explicit sparse-measurement consistency produces a meaningful reconstruction gain, especially under stronger ill-posedness or weaker direct supervision.

Table 22: Representative computational-cost summary. Time is reported as average wall-clock seconds per test window across the tested measurement ratios. The normalized runtime diagnostic column is computed relative to AE $[u, b]$ within each dataset and should not be read as the exact quotient of the averaged wall-clock times.

Dataset	Method	Time/window (s)	Normalized runtime diagnostic vs. AE	Mean RMSE
Cross-well V_c	ML only	0.00787	0.32	0.02582
Cross-well V_c	MLP $[u, b]$	0.04348	2.33	0.21399
Cross-well V_c	PCA $[u, b]$	0.07129	3.15	0.14464
Cross-well V_c	AE $[u, b]$	0.03156	1.00	0.01793
Cross-well V_c	CLP-CSGM Ridge	0.23517	9.79	0.01480
F03-4 GR-only	ML only	0.06010	2.13	0.02391
F03-4 GR-only	MLP $[u, b]$	0.04771	1.73	0.03981
F03-4 GR-only	PCA $[u, b]$	0.31793	11.67	0.02149
F03-4 GR-only	AE $[u, b]$	0.02999	1.00	0.01429
F03-4 GR-only	CLP-CSGM Ridge	0.08838	3.23	0.01418

C Supplementary F03-4 GR-only stress tests

The main F03-4 benchmark in Section 6 uses $\rho \in \{0.20, 0.30, \ldots, 0.60\}$. This appendix summarizes controlled supplements on the same well with GR-only u, L=64, depth-first splits before windowing, coordinate subsampling, the same three fixed random-seed realizations, and the same paired measurement protocol as Section 5. To align the grid with the sparse end of the cross-well study,

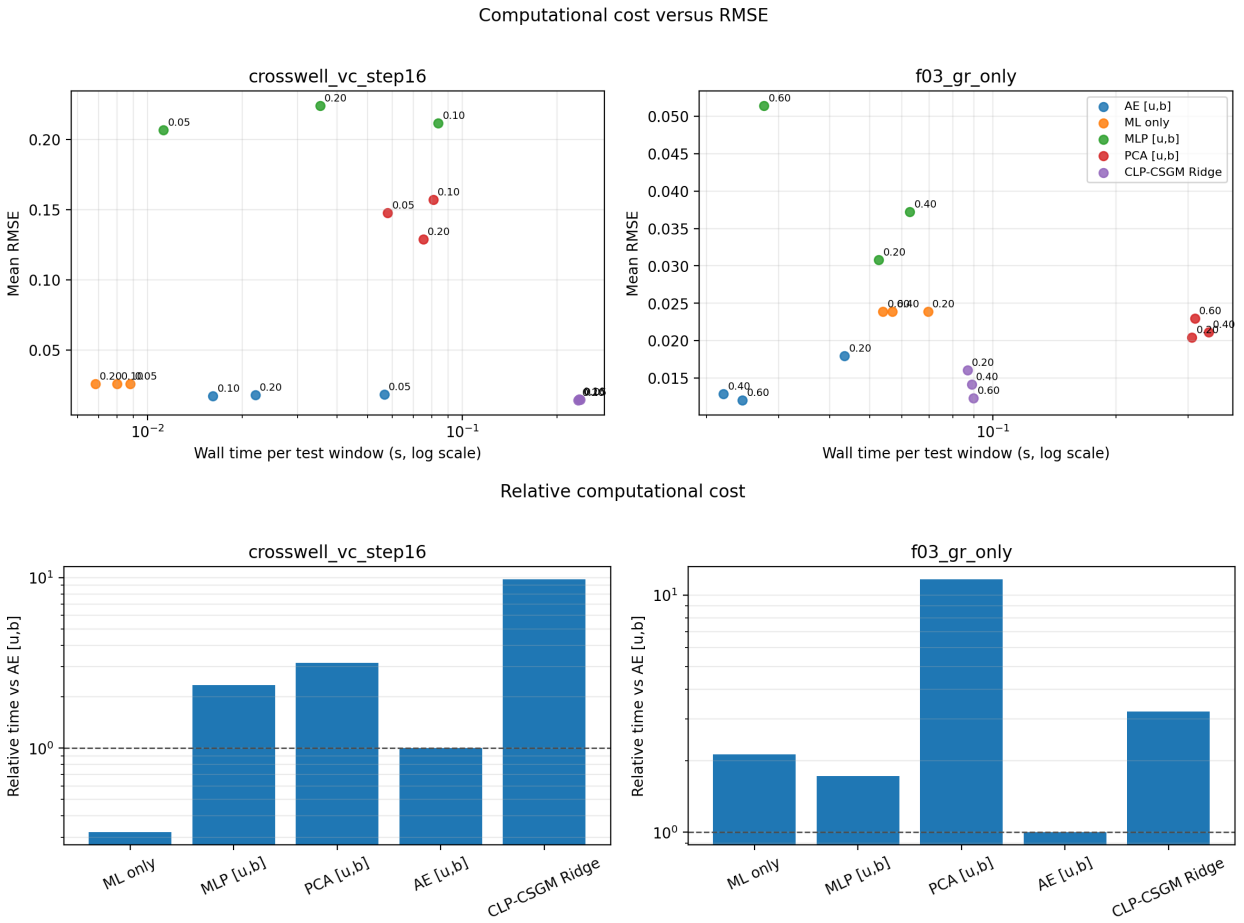


Figure 25: Representative computational-cost diagnostics. The upper panel compares mean RMSE with wall-clock time per test window, and the lower panel reports the normalized runtime diagnostic relative to AE $[u, b]$ within each dataset.

we use $\rho \in \{0.05, 0.10, 0.20\}$ and compare CLP-CSGM Ridge with AE $[u, b]$.

Cases A–D differ only by the stress axes enumerated below:

- **A (baseline noise):** η=0.01 on b=$My + \eta$; standard depth fractions 0.6/0.2/0.2; step 1.
- **B (noisy b):** η=0.05; otherwise as A.
- **C (smaller train pool):** depth fractions 0.45/0.25/0.30; η=0.01; step 1.
- **D (coarse sliding step):** step 4 with standard fractions; η=0.01. This yields far fewer test windows (n_{test}=56 versus 224 in A/B), yielding high-variance paired RMSE summaries at each ρ.

In A–C, CLP-CSGM Ridge has lower mean RMSE than AE $[u, b]$ at every tested ρ, including

Table 23: F03-4 GR-only stress tests: mean RMSE across seeds for AE $[u, b]$ versus CLP-CSGM Ridge. Relative RMSE gap is $(\mathrm{RMSE}_{\mathrm{AE}} - \mathrm{RMSE}_{\mathrm{Ridge}})/\mathrm{RMSE}_{\mathrm{AE}}$ in percent (positive favors Ridge). Test counts n_{test} count windows in the test depth segment after sliding-window construction.

Case	ρ	AE $[u, b]$	CLP-CSGM Ridge	Rel. RMSE gap (%)	n_{test}
A	0.05	0.02011	0.01682	+16.36	224
	0.10	0.01900	0.01596	+15.98	224
	0.20	0.01489	0.01314	+11.74	224
B	0.05	0.02444	0.01919	+21.51	224
	0.10	0.02397	0.01897	+20.87	224
	0.20	0.02125	0.01846	+13.13	224
C	0.05	0.01810	0.01596	+11.84	369
	0.10	0.01712	0.01509	+11.85	369
	0.20	0.01433	0.01243	+13.26	369
D	0.05	0.01676	0.01672	+0.24	56
	0.10	0.01493	0.01547	-3.62	56
	0.20	0.01115	0.01298	-16.33	56

under larger η and smaller training fractions. In D, mean RMSE crosses over at $\rho \in \{0.10, 0.20\}$ and is similar to or slightly lower at ρ=0.05. The small effective test count is consistent with this behavior because aggressive overlap reduction amplifies seed variability. The A–D results position the coarse-step F03 stress as an exploratory supplement to the primary ρ sweep in Table 6.

D Supplementary Lapa Pre-salt Permeability Case Study

This appendix reports a complementary case study built from the well-log and permeability data context described by Alvarez et al. [2025] for the Lapa pre-salt carbonate reservoir in the Santos Basin. Geologically, the Lapa oilfield targets the Aptian Barra Velha Formation, a heterogeneous pre-salt carbonate succession in which laminated microbialites, spherulite, stromatolite, rudstone, and grainstone facies coexist. In such rocks, permeability is controlled by pore structure and diagenesis as much as by porosity. Permeability is therefore the sparse reconstruction target in this supplementary check. Full core–log integration, hydraulic flow units, and nonlinear permeability prediction details are reported in Alvarez et al. [2025]. The benchmark adaptation uses the same CLP-CSGM protocol as the main paper: sliding windows, contiguous train/validation/test split, coordinate subsampling for sparse target observations, paired seeds and measurement ratios, and matched direct $[u, b]$ baselines.

This limited-sample case provides supplementary evidence. The configuration uses $L = 16$, step

= 4, seven dense channels in u, target k_{lab} in $\log_{10}$ scale, $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$, and the same three fixed random-seed realizations used only for reproducible paired averaging.

Table 24: Supplementary Lapa case study by measurement ratio. Values are means over three seeds.

rho	method	rmse	mae	relative_l2
0.20000	CLP-CSGM MLP	1.07615	0.84823	1.00530
0.20000	CLP-CSGM Ridge	1.22639	0.97273	1.08660
0.20000	Measurement-only CSGM	1.03628	0.75013	0.93124
0.20000	AE [u,b]	1.30639	1.11253	1.22956
0.20000	PCA [u,b]	1.73942	1.42881	1.60571
0.20000	ML only	2.06280	1.70838	2.00587
0.20000	MLP [u,b]	2.21106	1.78860	2.13801
0.30000	CLP-CSGM MLP	1.07099	0.81987	0.96733
0.30000	CLP-CSGM Ridge	1.06503	0.79065	0.91318
0.30000	Measurement-only CSGM	0.96152	0.67917	0.86494
0.30000	AE [u,b]	1.39133	1.17212	1.30771
0.30000	PCA [u,b]	1.89848	1.57954	1.72206
0.30000	ML only	2.06280	1.70838	2.00587
0.30000	MLP [u,b]	2.14391	1.73034	2.00236
0.40000	CLP-CSGM MLP	0.96555	0.73670	0.88465
0.40000	CLP-CSGM Ridge	0.97383	0.71050	0.86519
0.40000	Measurement-only CSGM	0.97680	0.65052	0.86968
0.40000	AE [u,b]	1.44311	1.21575	1.35981
0.40000	PCA [u,b]	1.87246	1.54857	1.79426
0.40000	ML only	2.06280	1.70838	2.00587
0.40000	MLP [u,b]	2.27913	1.85405	2.18847
0.50000	CLP-CSGM MLP	0.95447	0.74089	0.84011
0.50000	CLP-CSGM Ridge	0.99413	0.75851	0.85865
0.50000	Measurement-only CSGM	0.91277	0.55221	0.80389
0.50000	AE [u,b]	1.36684	1.18683	1.24828
0.50000	PCA [u,b]	2.03242	1.68385	1.85003
0.50000	ML only	2.06280	1.70838	2.00587
0.50000	MLP [u,b]	2.05267	1.61547	2.03388
0.60000	CLP-CSGM MLP	0.78164	0.59283	0.70811
0.60000	CLP-CSGM Ridge	0.74144	0.54218	0.67252
0.60000	Measurement-only CSGM	0.84417	0.45574	0.74392
0.60000	AE [u,b]	1.31942	1.11830	1.22740
0.60000	PCA [u,b]	1.88084	1.56736	1.77341
0.60000	ML only	2.06280	1.70838	2.00587
0.60000	MLP [u,b]	2.27679	1.86123	2.16631

Table 24 shows that both CLP-CSGM prior variants remain competitive with direct $[u, b]$ baselines across the tested ρ grid. CLP-CSGM MLP and CLP-CSGM Ridge have lower mean RMSE than AE $[u, b]$, PCA $[u, b]$, ML only, and direct MLP $[u, b]$ at all reported ratios. The comparison with measurement-only CSGM is ratio-dependent: measurement-only has lower mean RMSE at some lower or intermediate ratios, while the prior-guided variants have lower mean RMSE at others. This variation is consistent with the expected regime dependence when sparse measurements

are informative but the sample size is limited.

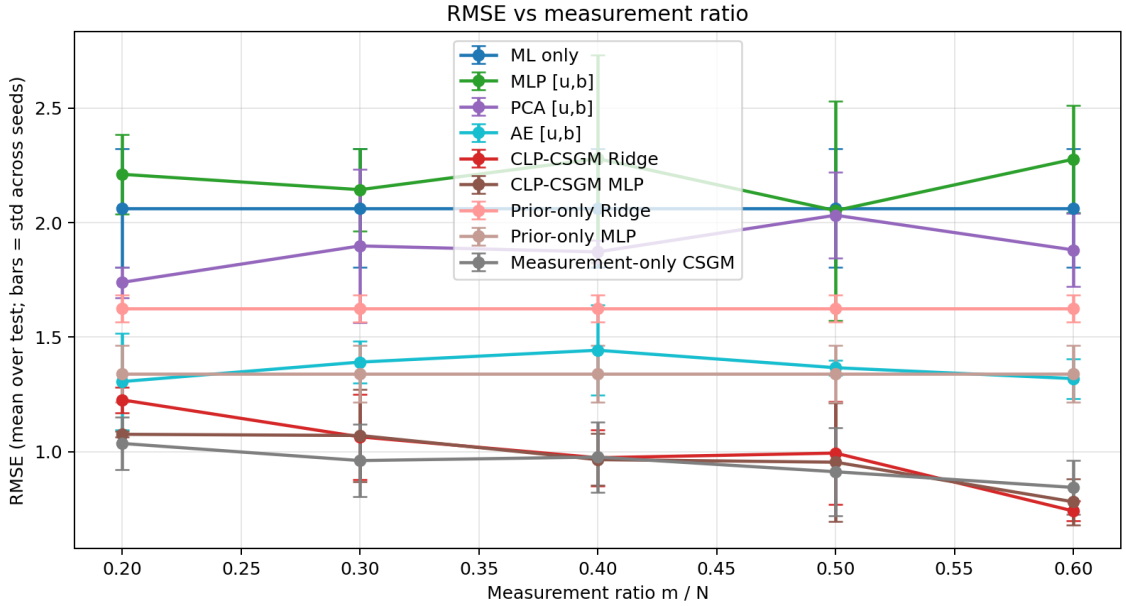


Figure 26: Supplementary Lapa case study RMSE by measurement ratio. Prior-guided CLP-CSGM variants remain competitive across the sparse-measurement grid.

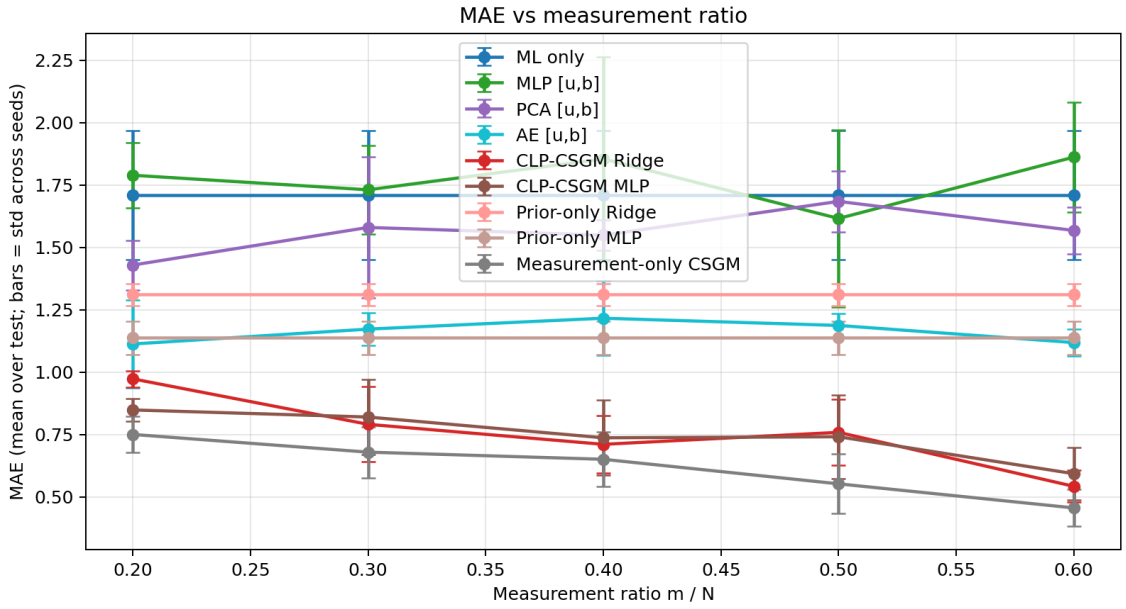


Figure 27: Supplementary Lapa case study MAE by measurement ratio. The ranking pattern is consistent with the RMSE view.

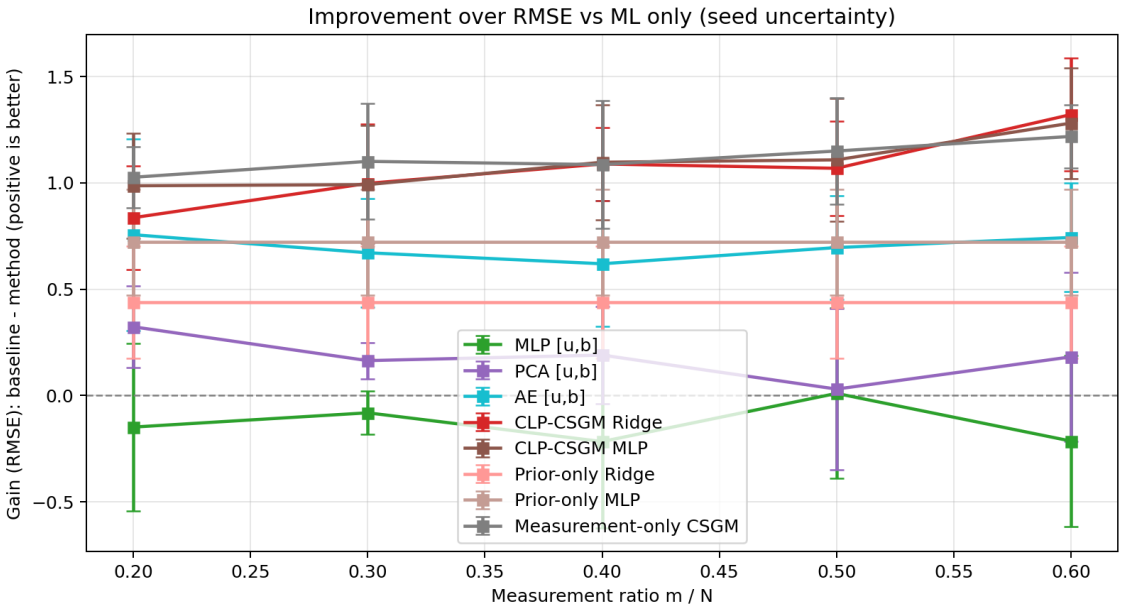


Figure 28: Supplementary Lapa case study RMSE gain over ML-only. Positive values indicate improvement over dense-context-only prediction.

E Laboratory porosity on a cored interval

This supplement uses the 87 laboratory-porosity samples of well 861 over 5205.91–5233.72 m. The target consists of porosity values identified as laboratory measurements in the source table associated with the Lapa field dataset of Alvarez et al. [2025] (Logs table). These values are not deterministically computed from the auxiliary gamma-ray, resistivity, or lithotype channels by our processing pipeline. Original per-plug laboratory assay records were not available for independent verification. Dense context is gamma ray, deep resistivity, shallow resistivity, and lithotype. Density, neutron porosity, sonic porosity, and $\mathrm{Phi}_{\mathrm{ND}}$ are excluded because they track $\mathrm{Phi}_{\mathrm{lab}}$.

Training and test samples are separated into contiguous depth blocks before window construction. Windows have length 8 and step 1. The last 16 rows of the deepest training block form the validation segment. Seeds are $\{7, 23, 41\}$. For each fold, seed, and ratio, one coordinate set is shared by every window. The laboratory value on those coordinates is the measurement, with no added noise. Within each test window, RMSE is computed only at coordinates not supplied as measurements to that particular reconstruction. However, because the length-8 windows overlap with step 1, the same physical depth can be measured in one window and scored as unobserved in another. This protocol evaluates within-window sparse assimilation on depth-held-out test blocks,

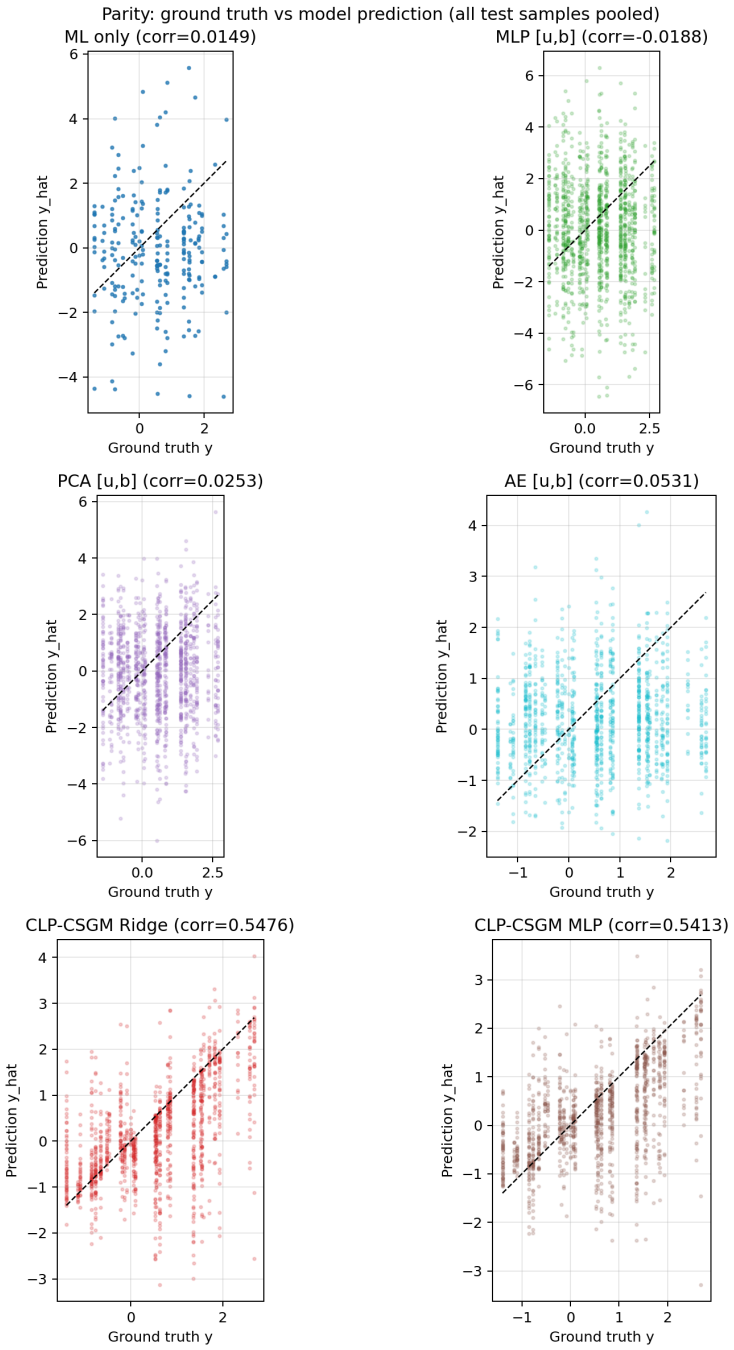


Figure 29: Supplementary Lapa pooled parity plot. The visual agreement is consistent with the quantitative summary, while still reflecting variability expected under small test counts.

not reconstruction at physical depths withheld globally from all test windows. Across the 18 fold–seed–ratio configurations, 21–27 physical depths per test block appear both as measured and scored coordinates across different windows. Between 84% and 100% of scored residual entries correspond to physical depths that were supplied as measurements in at least one other window. Those percentages count residual entries, not RMSE contribution, and they do not indicate training leakage: training and test rows remain disjoint. The nine results per ratio are three depth blocks repeated under three seeds, not nine independent geological intervals. $\rho = 0.25$ uses $m = 2$ measurements and $\rho = 0.50$ uses $m = 4$. Coordinate sets are sampled separately at each ratio. The direct AE receives $[\boldsymbol{u}, \boldsymbol{b}]$ with the same coordinates. The decoded prediction of the conditional prior is $G(h(\boldsymbol{u}))$, with no latent search. It is reported below as the log-only anchor. Neither method is added to the main F3 tables.

Table 25: Well 861 laboratory-porosity supplement. Entries are the mean within-window held-out RMSE across three depth blocks and three seeds ($n = 9$ block–seed pairs, not nine independent intervals). Δ is AE minus CLP-CSGM Ridge; a negative value means the direct AE has the lower mean. CLP>AE counts splits in which CLP-CSGM Ridge has the lower RMSE.

ρ	CLP	AE $[u, b]$	prior-only	Δ AE	CLP>AE
0.25	0.0602	0.0537	0.0729	-0.0065	5/9
0.50	0.0580	0.0523	0.0753	-0.0057	6/9

Table 25 shows that the direct AE has the lower mean held-out RMSE at both ratios. CLP-CSGM Ridge has the lower RMSE in five of nine splits at $\rho = 0.25$ and in six of nine at $\rho = 0.50$, and it has lower RMSE than the log-only anchor in all nine splits at both ratios. The two summaries answer different questions. The mean asks which method is lower after every split is weighted equally. The split count asks how often CLP-CSGM Ridge is the lower of the two. They disagree because one depth block carries errors large enough to set the mean.

Table 26: Well 861 held-out laboratory-porosity RMSE by depth block. Each entry is the mean across the three seeds. Prior-only is $G(h(\boldsymbol{u}))$ with no measurement term.

Block	ρ	CLP	AE $[u, b]$	prior-only
5205.91–5214.48 m	0.25	0.0338	0.0371	0.0486
5205.91–5214.48 m	0.50	0.0228	0.0350	0.0472
5214.67–5223.63 m	0.25	0.0536	0.0548	0.0559
5214.67–5223.63 m	0.50	0.0497	0.0543	0.0554
5224.39–5233.72 m	0.25	0.0931	0.0691	0.1142
5224.39–5233.72 m	0.50	0.1016	0.0676	0.1234

Table 26 locates the block-level pattern. On 5205.91–5214.48 m, CLP-CSGM Ridge has the

lower block-mean RMSE at both ratios. In the shallowest block, mean held-out RMSE decreases from 0.0338 at $\rho = 0.25$ to 0.0228 at $\rho = 0.50$, while the direct AE stays near 0.035. Because the coordinate sets are sampled separately at each measurement ratio, this comparison changes both the number and the placement of observations and does not isolate the effect of measurement count. On 5214.67–5223.63 m the two methods are close, with CLP-CSGM Ridge still slightly lower. On 5224.39–5233.72 m the direct AE is lower by a wide margin (0.0691 and 0.0676 against 0.0931 and 0.1016).

In every block and at both ratios, CLP-CSGM Ridge has lower RMSE than the log-only anchor $G(h(\boldsymbol{u}))$. In the shallowest block, mean $\mathrm{Phi}_{\mathrm{lab}}$ is 0.092 with standard deviation 0.019, mean gamma ray is about 15 API, and the median deep resistivity is about 1900. The direct AE already receives the laboratory coordinates as input features. Each test fold still has 28 to 35 overlapping training windows, so this is not the scarce-window regime of the F3 steps 16 and 32.

The larger share of the pooled AE advantage sits in the deepest block. When that block is held out, mean $\mathrm{Phi}_{\mathrm{lab}}$ rises to 0.145, mean gamma ray rises to about 30 API, and the median deep resistivity falls to about 600. The lithotype mix also changes: codes 2 and 4 account for 25 of the 29 rows, whereas the middle block is mostly code 1. The decoded prediction $G(\boldsymbol{z}_0)$ differs substantially from the target windows there. With only $m = 2$ or $m = 4$ measurements in a length-8 window, CLP-CSGM Ridge still improves on that anchor (0.0931 versus 0.1142, and 0.1016 versus 0.1234) but does not catch the direct AE.

The results are consistent with the hypothesis that sparse-measurement refinement is more effective when the conditional latent prior provides a representation compatible with the target interval. The larger errors in the deepest block coincide with changes in the auxiliary-log characteristics and a less accurate prior-only reconstruction. However, the present experiment does not isolate prior mismatch as the causal factor explaining the relative performance of CLP-CSGM and the direct autoencoder. The result does not extend the F3 low-support advantage to a mean win on this laboratory target. It is one cored interval.